Advanced Quantum Mechanics

Supplementary track, 2013 Fall

Lectures by Leonard Susskind

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ABOUT THESE NOTES

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CHAPTER 1

STATE VECTORS, UNITARY EVOLUTION, AND TRANSLATION SYMMETRY

After a two-quarter gap, a rapid review of quantum mechanics is useful. It cannot occupy the whole quarter. The subject ahead is the application of quantum mechanics to the problems for which it was originally developed: atoms, electrons and related particles, and photons.

Atoms come first. How can atoms come before electrons, when electrons are parts of atoms? Only enough electron mechanics is needed at first to write the basic equations of the hydrogen atom. A more detailed study of electron-like particles can follow. Throughout these applications, symmetry will be a recurring tool: how symmetries are represented in quantum mechanics and what they imply about a system.

1.1 States, observables, and orthogonality

In classical mechanics, a state is represented by a point in phase space. In quantum mechanics, it is represented by a vector in a complex vector space. This is an abstract vector, not an arrow in ordinary three-dimensional space. State vectors can be added and multiplied by complex numbers, and the space containing them need not have any particular finite dimension.

Dirac notation writes a state vector as a ket,

$$|\psi\rangle\,, \tag{1.1}$$

and associates with it a bra,

$$\langle\psi|\,. \tag{1.2}$$

Bras and kets are related by Hermitian conjugation. Putting a bra and a ket together gives an inner product,

$$\langle\phi|\psi\rangle, \tag{1.3}$$

which is generally a complex number.

An observable is something that can be measured: position, momentum, energy, or any other measurable quantity. In quantum mechanics an observable is represented by a linear Hermitian operator A,

$$A^\dagger = A. \tag{1.4}$$

Hermiticity is the operator analogue of reality. In particular, the eigenvalues of a Hermitian operator are real.

An eigenvector of A satisfies

$$A\,|\alpha\rangle = \alpha\,|\alpha\rangle\,. \tag{1.5}$$

The possible eigenvalues are the possible results of measuring A. If, for example, an observable can give only 1, 3, or 7, then these numbers are its eigenvalues. Its eigenvectors are the states in which the result is definite rather than statistical. In the state $|\alpha\rangle$, measuring A gives α with certainty.

The inner product expresses an important logical relation between states. Suppose there is a measurement that distinguishes $|\phi\rangle$ from $|\psi\rangle$ with certainty. The states are then orthogonal:

$$\langle\phi|\psi\rangle = 0. \tag{1.6}$$

Orthogonality means that the states are physically distinguishable without ambiguity. For a normalized state,

$$\langle\psi|\psi\rangle = 1. \tag{1.7}$$

1.2 A particle in the position representation

Consider first a particle moving on a line. Its position x is an observable. For every possible position x_0, there is a state $|x_0\rangle$ in which a position measurement certainly gives x_0.

States associated with different positions are distinguishable and therefore orthogonal:

$$x \neq x' \quad \Longrightarrow \quad \langle x'|x\rangle = 0. \tag{1.8}$$

For an arbitrary state $|\psi\rangle$, its overlap with a position state defines the wavefunction:

$$\psi(x) = \langle x|\psi\rangle. \tag{1.9}$$

This is the position representation of the state vector.

The wavefunction is not itself a probability. It can be positive, negative, imaginary, or fully complex.

The position probability density is

$$P(x) = \psi^*(x)\psi(x) = |\psi(x)|^2. \tag{1.10}$$

The product is nonnegative, as a probability density must be. For a normalized wavefunction,

$$\int_{-\infty}^{\infty} dx\, |\psi(x)|^2 = 1. \tag{1.11}$$

In the position representation, the position operator acts by multiplication:

$$(\hat{x}\psi)(x) = x\psi(x). \tag{1.12}$$

Its eigenfunctions are idealized functions concentrated at one point. They are described formally as Dirac delta functions: very narrow spikes centered at the position eigenvalue. Their precise distributional definition is not needed for the present review.

In three dimensions, position has three components, x, y, and z, and a position state represents a point in three-dimensional space. Two such states are orthogonal whenever the points differ in at least one coordinate. The one-dimensional notation will usually be retained, with the corresponding three-dimensional extension understood.

1.3 Momentum and definite-momentum states

For every component of position there is a corresponding component of momentum. In the position representation, the momentum operator along the x-axis is

$$\hat{p}_x = -i\hbar\frac{d}{dx}. \tag{1.13}$$

Thus an abstract Hermitian operator can also be viewed concretely as an operation on wavefunctions.

A state with definite momentum p_0 satisfies the eigenvalue equation

$$\hat{p}_x \psi_{p_0}(x) = p_0 \psi_{p_0}(x). \tag{1.14}$$

Substituting the differential form of $\hat{p}_x$ gives

$$-i\hbar \frac{d\psi_{p_0}(x)}{dx} = p_0 \psi_{p_0}(x). \tag{1.15}$$

The derivative of the wavefunction is proportional to the wavefunction itself, so the solution has exponential form:

$$\psi_{p_0}(x) \propto \exp\left(\frac{ip_0 x}{\hbar}\right). \tag{1.16}$$

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[a] **Question from the audience.** Does this differential equation hold for any wavefunction?

Response. No. It is the eigenvalue equation for a state of definite momentum. Most wavefunctions do not satisfy it; it selects the momentum eigenfunctions.

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[a]Lecture timestamp: 00:24:26.

The distinction between position and momentum eigenstates is sharp. A position eigenfunction is concentrated at one point. A momentum eigenfunction has

$$\left|\exp\left(\frac{ip_0 x}{\hbar}\right)\right|^2 = 1. \tag{1.17}$$

Its modulus is therefore constant over the entire line. In this idealized sense, definite momentum means complete uncertainty in position; definite position likewise means complete uncertainty in momentum. This is the extreme form of the position–momentum uncertainty relation.

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[a] **Question from the audience.** If the momentum is known to be p_0, why is there still a probability distribution?

Response. The displayed quantity concerns a position measurement, not another momentum measurement. Momentum and position are different observables. For the idealized momentum eigenstate, the position amplitude has constant modulus over the whole line. Amplitudes for momentum belong to the momentum representation and are related to the position wavefunction by a Fourier transform.

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[a]Lecture timestamp: 00:26:36.

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[a] **Question from the audience.** What does the position probability look like when the wavefunction is a Dirac delta function?

Response. It is concentrated at the specified position. The idealized spike expresses certainty of position.

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[a]Lecture timestamp: 00:28:02.

[1]

[1]Editorial clarification: Exact position and momentum eigenstates are generalized states. A Dirac delta and a plane wave are not ordinary square-integrable normalized

A wavefunction can be regarded formally as a vector with one component for every value of x. Unlike an ordinary finite column vector, however, it has a continuous infinity of components.

---

[a] **Question from the audience.** Are state vectors represented by column vectors?

Response. They can be represented that way, but a position wavefunction is awkward to write as an ordinary column because x takes continuously many values. It can be regarded formally as a continuous column vector, with the corresponding bra regarded as a continuous row vector.

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[a]Lecture timestamp: 00:28:53.

1.4 Time evolution and unitarity

How does a state change with time? Time evolution is a transformation on the space of states. Let $U(t)$ denote evolution through an elapsed time t. Then

$$U(t) |\psi(t_1)\rangle = |\psi(t_1 + t)\rangle . \tag{1.18}$$

In particular,

$$|\psi(t)\rangle = U(t) |\psi(0)\rangle , \qquad U(0) = \mathbf{1}. \tag{1.19}$$

The condition on $U(t)$ follows from conservation of information. If two initial states are distinguishable,

wavefunctions on the full line, and a literal square of a delta function is not an ordinary distribution. The statements above concern their idealized localization and constant modulus.

time evolution must not erase that distinction. Orthogonal states must remain orthogonal. More generally, the inner product between any two states is preserved:

$$\langle U(t)\phi|U(t)\psi\rangle = \langle\phi|\psi\rangle. \tag{1.20}$$

Since

$$\langle U(t)\phi|U(t)\psi\rangle = \langle\phi|U^\dagger(t)U(t)|\psi\rangle, \tag{1.21}$$

preservation of all inner products requires

$$U^\dagger(t)U(t) = \mathbf{1}. \tag{1.22}$$

An operator satisfying this equation is unitary.

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[a] **Question from the audience.** Is preserving every inner product a stronger statement than merely preserving orthogonality?

Response. For the linear state transformations under discussion, preservation of orthogonality leads to preservation of inner products. Ordinary rotations provide the familiar analogy: they preserve orthogonality and dot products.

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[a]Lecture timestamp: 00:36:36.

[2]

Unitary operators should not be confused with Hermitian observables. Position and momentum

[2]Editorial clarification: Strictly, preservation of orthogonality alone permits an overall nonzero rescaling. Preservation of normalization, already required for physical time evolution, removes that freedom and gives preservation of inner products.

are Hermitian, but they are not generally unitary. The eigenvalues of a unitary operator are phases, numbers of the form $e^{i\theta}$ lying on the unit circle.

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[a] **Question from the audience.** What is an example of a nonunitary transformation?

Response. The position and momentum operators are examples: they are Hermitian observables, but they are not unitary. A unitary operator has eigenvalues of unit magnitude, so its eigenvalues are phases.

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[a]Lecture timestamp: 00:38:12.

1.5 The Hamiltonian and the Schrödinger equation

Time evolution is assumed to be continuous. Over a very short interval ϵ, the evolution operator must therefore be close to the identity:

$$U(\epsilon) = \mathbf{1} + \epsilon G + \mathcal{O}(\epsilon^2). \tag{1.23}$$

Using unitarity,

$$\begin{aligned} U^\dagger(\epsilon)U(\epsilon) &= \left(\mathbf{1} + \epsilon G^\dagger\right)(\mathbf{1} + \epsilon G) + \mathcal{O}(\epsilon^2) &(1.24)\\ &= \mathbf{1} + \epsilon(G^\dagger + G) + \mathcal{O}(\epsilon^2). &(1.25)\end{aligned}$$

To first order,

$$G^\dagger = -G. \tag{1.26}$$

Thus G is anti-Hermitian. It can be written as

$$G = -\frac{i}{\hbar}H, \tag{1.27}$$

where H is Hermitian:

$$H^\dagger = H. \tag{1.28}$$

The infinitesimal time-evolution operator is therefore

$$U(\epsilon) = \mathbf{1} - \frac{i\epsilon}{\hbar} H + \mathcal{O}(\epsilon^2). \tag{1.29}$$

The Hermitian operator H is the Hamiltonian.

---

[a] **Question from the audience.** What permits the ϵ^2 term to be ignored?

Response. The calculation is being carried out consistently to first order in ϵ. A second-order calculation would require all second-order terms, including the term proportional to $-\epsilon^2 H^2/(2\hbar^2)$. The approximation lies in the truncated expression for $U(\epsilon)$, not in the Hermiticity of H.

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[a] Lecture timestamp: 00:45:48.

A finite time interval can be built from many small intervals. Divide t into N steps of length t/N. Repeating the infinitesimal transformation gives

$$U(t) = \lim_{N\to\infty} \left(\mathbf{1} - \frac{itH}{N\hbar}\right)^N = \exp\left(-\frac{iHt}{\hbar}\right). \tag{1.30}$$

The exponential is the finite version of the infinitesimal transformation.

Now evolve a state from t to $t + \epsilon$:

$$\begin{aligned} |\psi(t+\epsilon)\rangle &= U(\epsilon)\,|\psi(t)\rangle \qquad (1.31)\\ &= \left(\mathbf{1} - \frac{i\epsilon}{\hbar} H\right)|\psi(t)\rangle + \mathcal{O}(\epsilon^2). \qquad (1.32) \end{aligned}$$

Subtracting $|\psi(t)\rangle$, dividing by ϵ, and taking the limit gives

$$\frac{d}{dt}|\psi(t)\rangle = -\frac{i}{\hbar}H|\psi(t)\rangle . \tag{1.33}$$

Equivalently,

$$i\hbar\frac{d}{dt}|\psi(t)\rangle = H|\psi(t)\rangle . \tag{1.34}$$

This is the time-dependent Schrödinger equation. In the position representation it becomes

$$i\hbar\frac{\partial}{\partial t}\psi(x,t) = H\psi(x,t), \tag{1.35}$$

where H acts as an operator on the wavefunction.

The time-independent Schrödinger equation is the eigenvalue equation for the Hamiltonian:

$$H|E\rangle = E|E\rangle . \tag{1.36}$$

The eigenvalues E are the possible measured energies, or energy levels. The same Hamiltonian therefore determines both the allowed energies and the evolution of the state.

If the initial state is an energy eigenstate, its evolution is

$$|E;t\rangle = \exp\left(-\frac{iEt}{\hbar}\right)|E\rangle . \tag{1.37}$$

It remains the same energy eigenvector and acquires only a phase.[3]

[3]The minus sign is fixed by the preceding derivation of $U(t) = \exp(-iHt/\hbar)$. A later spoken rendering of the phase has the opposite sign and is inconsistent with that derivation.

[a] **Question from the audience.** How does an energy eigenvector evolve under the time-dependent Schrödinger equation?

Response. Keep the ket. Since $H|E\rangle = E|E\rangle$, time evolution multiplies the energy eigenvector by a phase determined by its energy: $\exp(-iEt/\hbar)$.

[a] Lecture timestamp: 01:01:05.

1.6 What counts as a symmetry?

Time evolution is one unitary transformation on the state space. Symmetry transformations are another important class.

Rotational symmetry is central to atomic physics. In a hydrogen atom, the electron moves in the central Coulomb potential. Rotating the coordinate system does not change the equations, and physically rotating the atom still leaves a hydrogen atom governed by the same equations. The state may change under the rotation, but the dynamical law does not.

Translation gives another example. An isolated system in empty space should behave the same way after every part of it has been displaced by the same amount. The qualification matters. If a furnace is present and only the system is moved closer to it, the system may behave differently; the furnace breaks that translation symmetry. If both the system and the furnace are translated together, the original relation between them is restored.

Not every operation is a symmetry. Compressing a crystal of rock salt by a factor of two generally changes its behavior. Squeezing is a well-defined transformation, but it is not generally a symmetry.

A symmetry is therefore an operation on states that preserves the form of the dynamics. Denote it by V. It must preserve distinguishability: two orthogonal states must remain orthogonal after the transformation. Accordingly,

$$V^\dagger V = \mathbf{1}, \tag{1.38}$$

so V is unitary.

1.7 Symmetries commute with time evolution

Suppose an initial state $|\psi_1\rangle$ evolves into $|\psi_2\rangle$:

$$|\psi_2\rangle = U(t)\,|\psi_1\rangle\,. \tag{1.39}$$

Apply a proposed symmetry V to both states:

$$|\psi_1'\rangle = V\,|\psi_1\rangle\,, \qquad |\psi_2'\rangle = V\,|\psi_2\rangle\,. \tag{1.40}$$

If V is a symmetry of the dynamics, the transformed initial state must evolve into the transformed final state:

$$|\psi_2'\rangle = U(t)\,|\psi_1'\rangle\,. \tag{1.41}$$

Substituting the definitions gives

$$VU(t)\,|\psi_1\rangle = U(t)V\,|\psi_1\rangle\,. \tag{1.42}$$

This must hold for every initial state. Therefore

$$VU(t) = U(t)V. \tag{1.43}$$

$$\begin{tikzcd}
|\psi_1\rangle \arrow[r, "U(t)"] \arrow[d, "V"'] & |\psi_2\rangle = U(t)\,|\psi_1\rangle \arrow[d, "V"] \\
V\,|\psi_1\rangle \arrow[r, "U(t)"'] & V\,|\psi_2\rangle = U(t)V\,|\psi_1\rangle
\end{tikzcd}$$

Figure 1.1: A symmetry operation and time evolution give the same final state in either order.

The two equivalent routes can be displayed as a commutative diagram.[4]

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[a] **Question from the audience.** Must the symmetry commute with time evolution for all times?

Response. Yes. A symmetry of the dynamics must satisfy the commutation condition for the full time evolution.

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[a]Lecture timestamp: 01:16:38.

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[a] **Question from the audience.** Does this mean that one may rotate and then wait, or wait and then rotate?

Response. Exactly. If rotation is a symmetry, both orders produce the same transformed final state.

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[a]Lecture timestamp: 01:17:13.

[4]The diagram reconstructs the two-route argument developed between 01:10:30 and 01:16:34.

For a small time interval,

$$U(\epsilon) = \mathbf{1} - \frac{i\epsilon}{\hbar} H + \mathcal{O}(\epsilon^2). \tag{1.44}$$

The commutation condition becomes

$$V\left(\mathbf{1} - \frac{i\epsilon}{\hbar} H\right) = \left(\mathbf{1} - \frac{i\epsilon}{\hbar} H\right) V + \mathcal{O}(\epsilon^2). \tag{1.45}$$

The identity terms cancel, leaving

$$VH = HV, \tag{1.46}$$

or

$$[V, H] = 0. \tag{1.47}$$

Thus a unitary operation is a symmetry of the dynamics precisely when it commutes with the Hamiltonian.

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[a] **Question from the audience.** What requires the symmetry operator V to be unitary?

Response. A symmetry must preserve logical distinctions between states. It should not take two mutually exclusive, orthogonal states into states that can no longer be distinguished. Preserving those relations requires a unitary transformation.

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[a] Lecture timestamp: 01:19:29.

An operator with no explicit time dependence that commutes with H is conserved. This connects symmetry with conservation. In classical mechanics the relation is expressed by Noether's theorem. In the present quantum-mechanical form, the central test is the commutator with the Hamiltonian.

1.8 Discrete and continuous symmetries

Symmetries can be discrete or continuous. A mirror reflection is discrete: it exchanges left and right, with no sequence of infinitesimal reflections between the two. Interchanging two identical particles is another discrete operation. Flipping an unmarked two-sided object provides the same kind of example.

Rotations are continuous. A rotation through 90° can be built from many small rotations about the same axis. Spatial translations are continuous for the same reason: a finite displacement can be built from many tiny displacements.

This divisibility into small transformations gives continuous symmetries a special form. Near the identity, write

$$V(\epsilon) = \mathbf{1} - i\epsilon G + \mathcal{O}(\epsilon^2), \tag{1.48}$$

where ϵ parametrizes the transformation. Unitarity implies

$$G^\dagger = G. \tag{1.49}$$

The Hermitian operator G is called the generator of the symmetry.

If the transformation is a symmetry, then

$$[H, V(\epsilon)] = 0. \tag{1.50}$$

Expanding to first order gives

$$[H, G] = 0. \tag{1.51}$$

Continuous symmetries are therefore generated by Hermitian operators that commute with the Hamiltonian. Those generators are conserved quantities.

The word “generated” has a concrete meaning here. A finite transformation is built by applying the infinitesimal one repeatedly, just as finite time evolution is built from many short time steps.

1.9 Translation symmetry and momentum

Consider an active translation of a wavefunction to the right by a distance ϵ. If the original wavefunction is peaked at x_0, the translated wavefunction should be peaked at $x_0 + \epsilon$. Its value at the coordinate x is therefore the old wavefunction evaluated at $x - \epsilon$:

$$(T(\epsilon)\psi)(x) = \psi(x - \epsilon). \tag{1.52}$$

For a small displacement,

$$\psi(x - \epsilon) = \psi(x) - \epsilon \frac{d\psi(x)}{dx} + \mathcal{O}(\epsilon^2). \tag{1.53}$$

Hence

$$T(\epsilon) = \mathbf{1} - \epsilon \frac{d}{dx} + \mathcal{O}(\epsilon^2). \tag{1.54}$$

Using

$$\hat{p}_x = -i\hbar \frac{d}{dx}, \qquad \frac{d}{dx} = \frac{i}{\hbar}\hat{p}_x, \tag{1.55}$$

the infinitesimal translation becomes

$$T(\epsilon) = \mathbf{1} - \frac{i\epsilon}{\hbar}\hat{p}_x + \mathcal{O}(\epsilon^2). \tag{1.56}$$

The corresponding finite translation is

$$T(\epsilon) = \exp\left(-\frac{i\epsilon}{\hbar}\hat{p}_x\right). \tag{1.57}$$

With the convention $V(\epsilon) = \mathbf{1} - i\epsilon G + \cdots$, the dimensionful translation parameter gives

$$G_{\text{trans}} = \frac{\hat{p}_x}{\hbar}. \tag{1.58}$$

Equivalently, momentum is called the generator of spatial translations when the factor of $\hbar$ is displayed in the exponential.

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[a] **Question from the audience.** Has a minus sign been lost in the translation derivation?

Response. The issue is the direction assigned to the translated wavefunction. An active shift to the right is represented by $\psi(x-\epsilon)$, not $\psi(x+\epsilon)$. With that correction, $T(\epsilon) = \mathbf{1} - i\epsilon\hat{p}_x/\hbar + \cdots$.

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[a]Lecture timestamp: 01:34:08.

Whether momentum is conserved depends on the Hamiltonian. For a free particle moving in one dimension,

$$H = \frac{\hat{p}_x^2}{2m}. \tag{1.59}$$

Since every operator commutes with every function of itself,

$$[H, \hat{p}_x] = 0. \tag{1.60}$$

The free-particle Hamiltonian is therefore invariant under translations, and $\hat{p}_x$ is conserved. Physically, the particle behaves the same way wherever its initial position is chosen. Translating the initial condition simply translates the entire motion.

In three dimensions,

$$H = \frac{\hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2}{2m}. \tag{1.61}$$

Each momentum component commutes with H:

$$[H, \hat{p}_x] = [H, \hat{p}_y] = [H, \hat{p}_z] = 0. \tag{1.62}$$

There is a separate translation symmetry in each spatial direction. Momentum is both the conserved quantity and the generator associated with those translations.

The next case is rotational symmetry. Rotations are more involved because rotations about different axes do not generally commute: an x-rotation followed by a y-rotation is not the same as the reverse order. That leads to noncommuting symmetry generators and to the group-theoretic structure of rotations.

CHAPTER 2

SYMMETRY, DEGENERACY, AND ANGULAR-MOMENTUM LADDERS

2.1 Symmetries and degenerate energy levels

A symmetry is an operation that does not change the physical description of a system. It does not change the Hamiltonian, the Schrödinger equation, or the possible energies. If a unitary operator S represents the operation, invariance means

$$SHS^\dagger = H, \tag{2.1}$$

or equivalently

$$[S, H] = 0. \tag{2.2}$$

Translation is one example. Move an atom from one place to another and its energy does not change. This is true for every atomic state, not only the ground state. Rotation is another example and will be the main subject here.

Interchanging two identical particles is also a symmetry. Calling two electrons electron one and electron two, or Harry and Fred, does not make them different. Interchanging the labels cannot change the properties of the atom.

A crystal has an additional kind of translation symmetry. Moving the whole crystal through space is an ordinary translation. For an infinite crystal, however, translation by one lattice spacing also returns the same physical arrangement. Equivalently, the lattice sites can be relabeled by shifting every site index by one. Boundaries spoil this exact statement for a finite crystal, but it becomes a good symmetry far from the ends of a very long crystal.

The homogeneity of space—the fact that every position is like every other position—is translation symmetry. The isotropy of space—the fact that every direction is like every other direction—is rotational symmetry.

Degeneracy is related to symmetry, but it is not the same thing. An energy level is degenerate when more than one state has exactly that energy. For a generic Hamiltonian with a discrete spectrum, an exact equality between two unrelated levels would be a remarkable accident. Agreement to one part in 10^{15} is still not exact equality. Exact degeneracies ordinarily point to symmetry.

Symmetry sometimes implies degeneracy, but not always. The first example shows both why it can and why one symmetry alone may be insufficient.

2.2 A particle on a circle

Consider a bead constrained to move on a circular wire. Its position could be described by Cartesian coordinates x and y, but one angular coordinate is enough:

$$\psi = \psi(\theta). \tag{2.3}$$

The position probability density on the circle is $\psi^*(\theta)\psi(\theta)$.

An active counterclockwise rotation through a small angle ϵ sends

$$\psi(\theta) \longrightarrow \psi(\theta - \epsilon). \tag{2.4}$$

Therefore

$$\delta\psi(\theta) = \psi(\theta - \epsilon) - \psi(\theta) \tag{2.5}$$
$$= -\epsilon \frac{d\psi}{d\theta} + \mathcal{O}(\epsilon^2). \tag{2.6}$$

The operator

$$L_z = -i\hbar \frac{d}{d\theta} \tag{2.7}$$

is Hermitian on periodic wavefunctions. It is the angular analogue of

$$p_x = -i\hbar \frac{d}{dx}. \tag{2.8}$$

The factor of i is essential to Hermiticity. In terms of L_z,

$$\delta\psi = -\frac{i\epsilon}{\hbar} L_z \psi + \mathcal{O}(\epsilon^2), \tag{2.9}$$

so the infinitesimal rotation operator is

$$R(\epsilon) = 1 - \frac{i\epsilon}{\hbar} L_z + \mathcal{O}(\epsilon^2). \tag{2.10}$$

This identifies L_z as the generator of rotations about the axis perpendicular to the circle.

[a] **Question from the audience.** Should one of the factors involving i have the opposite sign?

Response. Checking the signs gives the relation above. With the convention $\psi(\theta) \to \psi(\theta - \epsilon)$, the change is $-\epsilon\, d\psi/d\theta$, which equals $-i\epsilon L_z \psi/\hbar$.

[a] Lecture timestamp: 00:11:19.

Now find the eigenstates of angular momentum:

$$L_z \psi = M\psi. \tag{2.11}$$

Substitution of the differential operator gives

$$-i\hbar \frac{d\psi}{d\theta} = M\psi, \tag{2.12}$$

whose solutions have the form

$$\psi(\theta) = C \exp\left(\frac{iM\theta}{\hbar}\right). \tag{2.13}$$

There is an additional condition that does not occur for motion on an infinite line. The points $\theta = 0$ and $\theta = 2\pi$ are the same point, so a single-valued orbital wavefunction must satisfy

$$\psi(\theta + 2\pi) = \psi(\theta). \tag{2.14}$$

For the eigenfunction above, this requires

$$\exp\left(\frac{2\pi i M}{\hbar}\right) = 1. \tag{2.15}$$

Consequently,

$$M = m\hbar, \qquad m \in \mathbb{Z}. \tag{2.16}$$

The possible orbital angular momenta on the circle are therefore

$$\ldots, -2\hbar, -\hbar, 0, \hbar, 2\hbar, \ldots. \tag{2.17}$$

Must a state with angular momentum $m\hbar$ have the same energy as a state with angular momentum $-m\hbar$? Positive and negative m describe opposite senses of circulation. The question is whether rotational symmetry alone requires those two motions to have equal energy.

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[a] **Question from the audience.** Does using $\psi(\theta - \epsilon)$ mean that the particle is moving clockwise?

Response. No. That convention only fixes which direction is called positive rotation. It defines the signs of angular position and angular momentum, but it does not determine the actual direction in which the particle moves.

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[a]Lecture timestamp: 00:22:51.

A magnetic field perpendicular to the plane of the circle supplies a counterexample. It can shift the energies of positive and negative angular momentum in opposite directions and thereby split the pair. The sign of the shift depends on the charge and orientation conventions, but the splitting is the important point.

Such a field still preserves rotations about the perpendicular axis. It therefore shows that rotational symmetry by itself does not require

$$E(m) = E(-m). \tag{2.18}$$

Without the splitting, $m = 1$ and $m = -1$ may form a degenerate pair, as may $m = 2$ and $m = -2$. The state $m = 0$ has no distinct partner of this kind.

2.3 Reflection symmetry and the $m, -m$ pair

Add a discrete reflection symmetry. Let $\mathcal{M}$ denote reflection across a diameter of the circle:

$$(\mathcal{M}\psi)(\theta) = \psi(-\theta). \tag{2.19}$$

It reverses the sense of circulation:

$$\mathcal{M}\psi_m = \psi_{-m}. \tag{2.20}$$

If reflection is a symmetry, then

$$[\mathcal{M}, H] = 0. \tag{2.21}$$

Suppose

$$H\left|m\right\rangle = E_m\left|m\right\rangle. \tag{2.22}$$

Applying $\mathcal{M}$ gives

$$H\mathcal{M}\left|m\right\rangle = \mathcal{M}H\left|m\right\rangle \tag{2.23}$$
$$= E_m\mathcal{M}\left|m\right\rangle. \tag{2.24}$$

Since $\mathcal{M}\left|m\right\rangle$ is the state with angular momentum $-m\hbar$,

$$E_{-m} = E_m. \tag{2.25}$$

Rotation symmetry conserves angular momentum; reflection symmetry relates the two opposite values. Together they require the pair to be degenerate.

[a] **Question from the audience.** Since the m and $-m$ modes have the same probability density, does that already show that their energies are equal?

Response. No. The wavefunctions $e^{im\theta}$ and $e^{-im\theta}$ are different even though their absolute squares agree. The angular-momentum eigenvalue equation involves the wavefunction, not only $\psi^*\psi$. Quantum mechanics requires amplitudes, not probabilities alone.

[a]Lecture timestamp: 00:29:54.

The magnetic field preserves rotations about its axis but breaks the reflection used above.

[a] **Question from the audience.** Does reflection symmetry reverse the magnetic field?

Response. Yes. A magnetic field is an axial vector. Reflecting a field directed into the board gives one directed out of the board. Equivalently, a current loop and its mirror image carry current in opposite senses and produce oppositely directed fields.

[a]Lecture timestamp: 00:33:05.

The noncommutativity of rotation and reflection can be seen directly. On an angular-momentum eigenfunction,

$$\mathcal{M} L_z \psi_m = m\hbar\, \psi_{-m}, \tag{2.26}$$

$$L_z \mathcal{M} \psi_m = -m\hbar\, \psi_{-m}. \tag{2.27}$$

Thus

$$[\mathcal{M}, L_z]\psi_m = 2m\hbar\, \psi_{-m}, \tag{2.28}$$

which is nonzero for $m \neq 0$. Rotating and then reflecting is not the same operation as reflecting and then rotating.

Both operators commute with H when both transformations are symmetries, but they do not commute with each other. That is the pattern responsible for the degeneracy.

---

[a] **Question from the audience.** Does any noncommuting Heisenberg pair, such as x and p, generate degeneracies?

Response. Not generally. The operators must also be symmetries: they must commute with the Hamiltonian. Position and momentum are typically not both conserved. If both did commute with H, then their noncommutativity would imply degeneracy.

---

[a]Lecture timestamp: 00:40:28.

2.4 Commutator algebras of symmetries

Suppose A and B are Hermitian generators of two continuous symmetries:

$$[A, H] = 0, \qquad [B, H] = 0. \tag{2.29}$$

Nothing in these equations requires A and B to commute with each other.

The commutator $[A, B]$ is anti-Hermitian, so define a Hermitian operator

$$C = i[A, B]. \tag{2.30}$$

Numerical factors and an overall sign are immaterial here. The important point is that C also commutes with the Hamiltonian. Indeed,

$$[AB, H] = A[B, H] + [A, H]B = 0, \tag{2.31}$$

and similarly

$$[BA, H] = 0. \tag{2.32}$$

Therefore

$$[C, H] = i\Big([AB, H] - [BA, H]\Big) = 0. \tag{2.33}$$

The new generator C may be independent of A and B, or it may only be a linear combination of generators already known. If it is new, add it to the list and form further commutators. Repeating this process may continue indefinitely, but often it closes after finitely many generators: every further commutator is a linear combination of those already present. That closed system is a commutator algebra or Lie algebra.

The finite symmetry transformations form a group. Their infinitesimal generators form the associated Lie algebra.

Degeneracies matter because they organize spectra. If two states have equal energy, one cannot decay to the other by emitting a photon carrying their energy difference. In atomic spectroscopy the arrangement and degeneracy of the levels determine the possible spectral lines. The same logic applies to excited hadrons and their transitions.

[a] **Question from the audience.** Are these commutators themselves elements of the symmetry group?

Response. The operators under discussion are generators. The generators and their commutators form a commutator algebra, or Lie algebra. Finite symmetry operations built from the generators are the group elements.

[a]Lecture timestamp: 00:49:07.

If all the symmetry generators commute with one another, the symmetry is called abelian. If some of their commutators are nonzero, it is non-abelian.

[a] **Question from the audience.** What is the fundamental definition of a generator?

Response. Take a unitary symmetry transformation very close to the identity and write it as $1 - i\epsilon G/\hbar + \cdots$. The Hermitian operator G is its generator. Repeating the corresponding infinitesimal transformation builds a finite one.

[a]Lecture timestamp: 00:50:25.

[a] **Question from the audience.** What is meant by the algebra of symmetry generators?

Response. It is a collection of generators that can be added, subtracted, multiplied by ordinary numbers, and commuted. In this algebra the commutator plays the role of multiplication, and closure means that these operations do not produce generators outside the collection.

[a]Lecture timestamp: 00:52:08.

2.5 Rotations about different axes

Rotations provide the most important example of a non-abelian symmetry. A rotation about the x-axis followed by one about the y-axis is not generally the same as performing them in the reverse order. Their mismatch is itself related to rotation about the third axis.

---

[a] **Question from the audience.** Do infinitesimal rotations commute even though finite rotations do not?

Response. They differ at second order in the two small rotation angles. That is enough: after those small parameters are factored out, the generators themselves have a nonzero commutator.

---

[a]Lecture timestamp: 00:54:22.

Before calculating the quantum commutators, consider the corresponding classical picture. Take a circular planetary orbit with angular momentum perpendicular to its plane. Rotating the whole orbit about the y-axis gives another allowed orbit with exactly the same energy. Its angular-momentum vector, however, now has a different x-component.

Thus rotational invariance produces same-energy configurations with different angular-momentum components. Quantum mechanically, this appears as noncommutativity among the generators of rotations about different axes.

2.6 Angular momentum in Cartesian coordinates

First consider a particle moving in the xy-plane, without restricting it to a circle. Under a small counterclockwise rotation,

$$\delta x = -\epsilon y, \qquad \delta y = \epsilon x. \tag{2.34}$$

For the actively rotated state, the transformed wavefunction is evaluated at the inverse-rotated coordinates:

$$\psi_\epsilon(x, y) = \psi(x + \epsilon y, y - \epsilon x). \tag{2.35}$$

Consequently,

$$\begin{aligned} \delta\psi &= \epsilon\left(y\frac{\partial}{\partial x} - x\frac{\partial}{\partial y}\right)\psi + \mathcal{O}(\epsilon^2) && (2.36) \\ &= -\frac{i\epsilon}{\hbar}\left(xp_y - yp_x\right)\psi + \mathcal{O}(\epsilon^2). && (2.37) \end{aligned}$$

The generator of rotations about the z-axis is therefore

$$L_z = xp_y - yp_x. \tag{2.38}$$

This is the z-component of the familiar classical vector

$$\mathbf{L} = \mathbf{r} \times \mathbf{p}. \tag{2.39}$$

It is not only the mechanical angular momentum; in quantum mechanics it is the operator that generates rotations of the wavefunction.[1]

[1] The active-rotation convention is chosen to agree with the earlier transformation $\psi(\theta) \mapsto \psi(\theta - \epsilon)$ and with the final identification $L_z = xp_y - yp_x$ after the intermediate sign uncertainty in the spoken derivation.

Cycling $x \to y \to z \to x$ gives the other components.[2]

$$L_x = yp_z - zp_y, \tag{2.40}$$
$$L_y = zp_x - xp_z, \tag{2.41}$$
$$L_z = xp_y - yp_x. \tag{2.42}$$

For a rotationally invariant system these operators commute with the Hamiltonian:

$$[H, L_x] = [H, L_y] = [H, L_z] = 0. \tag{2.43}$$

A particle in a central force field is the standard example. These equations are angular-momentum conservation in quantum form.

2.7 The angular-momentum Lie algebra

The canonical commutation relations are

$$[x_i, x_j] = 0, \tag{2.44}$$
$$[p_i, p_j] = 0, \tag{2.45}$$
$$[x_i, p_j] = i\hbar\,\delta_{ij}. \tag{2.46}$$

Coordinates commute with momentum components belonging to different directions. Thus x commutes with p_y, for example, while x does not commute with p_x.

Now calculate

$$[L_x, L_y] = [yp_z - zp_y,\ zp_x - xp_z]. \tag{2.47}$$

[2]The L_x and L_y formulas are reconstructed by applying the cyclic substitution $x \to y \to z \to x$, explicitly prescribed in the lecture, to $L_z = xp_y - yp_x$.

Most terms vanish. The two nonzero contributions are

$$[yp_z, zp_x] = -i\hbar\, yp_x, \tag{2.48}$$
$$[zp_y, xp_z] = i\hbar\, xp_y. \tag{2.49}$$

Therefore

$$[L_x, L_y] = i\hbar\,(xp_y - yp_x) \tag{2.50}$$
$$= i\hbar L_z. \tag{2.51}$$

Cycling the coordinate labels gives

$$[L_x, L_y] = i\hbar L_z, \tag{2.52}$$
$$[L_y, L_z] = i\hbar L_x, \tag{2.53}$$
$$[L_z, L_x] = i\hbar L_y. \tag{2.54}$$

[a] **Question from the audience.** When the commutator produces L_z, do we still have to show that L_z is not a linear combination of L_x and L_y?

Response. Yes. There is an obligation to establish that independence rather than assume it. A direct algebraic proof is not pursued at this point.

[a]Lecture timestamp: 01:14:50.

The displayed commutators close on the span of L_x, L_y, and L_z: once all three are included, further commutators produce no operator outside that span. This is the angular-momentum Lie algebra. Together with

$$[H, L_i] = 0, \qquad i = x, y, z, \tag{2.55}$$

these relations contain substantial information about the energy spectrum without requiring the angular Schrödinger equation to be solved directly.

A similar algebraic strategy is used for the quantum harmonic oscillator, where creation and annihilation operators generate its spectrum. Here the analogous operators will change the z-component of angular momentum rather than the energy.

Choosing the z-axis is only a choice of representation. Nothing makes z physically preferable to x or y, but choosing one component gives a convenient basis. Write

$$L_z |m\rangle = m\hbar |m\rangle . \tag{2.56}$$

The task is to determine which values of m can occur and how states with different m are related.

---

[a] **Question from the audience.** When you compare this construction with the harmonic oscillator, do you mean the quantum harmonic oscillator? That was not covered in the preceding class.

Response. Yes, the comparison is with the quantum oscillator. The present angular-momentum argument is self-contained and does not require knowledge of the oscillator. The analogy can later be read in the other direction: the oscillator uses a similar algebraic method.

---

[a]Lecture timestamp: 01:24:02.

2.8 Raising and lowering angular momentum

Define

$$L_+ = L_x + iL_y, \qquad L_- = L_x - iL_y. \tag{2.57}$$

The angular-momentum commutators imply

$$[L_z, L_+] = \hbar L_+, \qquad [L_z, L_-] = -\hbar L_-. \tag{2.58}$$

Apply the first relation to an eigenstate of L_z:

$$L_z L_+ |m\rangle = (L_+ L_z + [L_z, L_+]) |m\rangle \tag{2.59}$$
$$= (m+1)\hbar\, L_+ |m\rangle\,. \tag{2.60}$$

Unless $L_+ |m\rangle = 0$, it is another eigenstate of L_z, now with eigenvalue $(m+1)\hbar$. Similarly,

$$L_z L_- |m\rangle = (m-1)\hbar\, L_- |m\rangle\,. \tag{2.61}$$

Thus L_+ raises m by one and L_- lowers it by one.

Starting from any one eigenvalue, the algebra generates a sequence separated by integers. The sequence may continue indefinitely, or it may terminate because

$$L_+ |m_{\text{top}}\rangle = 0 \tag{2.62}$$

at the upper end and

$$L_- |m_{\text{bottom}}\rangle = 0 \tag{2.63}$$

at the lower end.

---

[a] **Question from the audience.** Was angular momentum not already required to be an integer by the circle argument?

Response. That argument proves integer values for orbital angular momentum represented by single-valued position wavefunctions. The abstract angular-momentum algebra gives a looser result: the ladder may contain either integer or half-integer values.

---

[a]Lecture timestamp: 01:35:11.

If a ladder terminates, rotational invariance relates its two ends. A rotation through 180° about an axis perpendicular to z sends

$$L_z \longrightarrow -L_z. \tag{2.64}$$

The spectrum must therefore be symmetric about zero:

$$m_{\text{bottom}} = -m_{\text{top}}. \tag{2.65}$$

Since neighboring values differ by one, there are two possibilities. The ladder contains either integers,

$$\ldots, -2, -1, 0, 1, 2, \ldots, \tag{2.66}$$

or half-integers,

$$\ldots, -\frac{5}{2}, -\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \ldots, \tag{2.67}$$

with endpoints wherever the raising or lowering operator gives zero.

The single-valued wavefunction argument on the circle selects the integer possibility for orbital angular momentum,

$$\mathbf{L} = \mathbf{r} \times \mathbf{p}. \tag{2.68}$$

The half-integer possibility is also physical and belongs to spin angular momentum, which will be treated separately.

Finally, the ladder states have equal energy when the Hamiltonian is rotationally invariant. Suppose

$$H \left| m \right\rangle = E \left| m \right\rangle. \tag{2.69}$$

Since every component L_i commutes with H, so do L_+ and L_-. Hence

$$HL_+ |m\rangle = L_+H |m\rangle \tag{2.70}$$
$$= EL_+ |m\rangle \, . \tag{2.71}$$

Whenever $L_+ |m\rangle \neq 0$, the raised state has the same energy as the original state. The same calculation gives

$$HL_- |m\rangle = EL_- |m\rangle \, . \tag{2.72}$$

Raising and lowering therefore generate a family of states with different L_z eigenvalues but exactly the same energy. This is the connection between noncommuting symmetries and degeneracy developed throughout the lecture.

---

[a] **Question from the audience.** Are these ladder states associated with choosing different rotation axes?

Response. They are related by rotations. Eigenvectors of L_x or L_y are linear combinations of the chosen L_z eigenvectors. Rotating a state changes its angular-momentum components while leaving its energy unchanged.

---

[a]Lecture timestamp: 01:44:13.

The choice of L_+ and L_- is convenient because these combinations expose the ladder directly. This trick recurs throughout quantum mechanics: when a commutator algebra closes, raising and lowering operators often organize its eigenvalue spectrum.

---

[a] **Question from the audience.** What happens when L_+ acts on the maximum state, and how is the top defined?

Response. It gives zero. That is the definition of the top of a terminating ladder. Likewise L_- gives zero at the bottom. An abstract ladder need not terminate. For the atomic angular-momentum states considered later, one typically encounters terminating ladders.

---

[a]Lecture timestamp: 01:46:33.

The next step is to develop this angular-momentum structure further and apply it to atoms, including hydrogen. The rotational multiplets found here account for part of the degeneracy pattern; further structure remains to be developed.

CHAPTER 3

ANGULAR-MOMENTUM MULTIPLETS, RADIAL MOTION, AND THE OSCILLATOR LADDER

3.1 Angular Dependence of the Wavefunction

A particle in a central force field has a radial position and two angular coordinates. Classically,

$$\mathbf{L} = \mathbf{r} \times \mathbf{p} \tag{3.1}$$

is conserved as a vector. Its direction fixes the plane of the orbit, although the orbit need not be circular.

Quantum mechanically the particle is described by

$$\Psi = \Psi(r, \theta, \phi), \tag{3.2}$$

and angular momentum concerns the dependence of this wavefunction on the angles.

In two dimensions there is only one angular-momentum component, the generator of rotations about the axis perpendicular to the plane:

$$L_z = -i\hbar \frac{\partial}{\partial \theta}. \tag{3.3}$$

An eigenfunction with eigenvalue $\hbar\ell$ obeys

$$-i\hbar \frac{\partial}{\partial \theta} \psi(r, \theta) = \hbar\ell\, \psi(r, \theta). \tag{3.4}$$

The coordinate r is a spectator in this equation. Its solution has the form

$$\psi(r,\theta) = \chi(r)e^{i\ell\theta}. \tag{3.5}$$

For an ordinary single-valued orbital wavefunction, returning to the same point after $\theta \to \theta + 2\pi$ requires ℓ to be an integer. The radial factor $\chi(r)$ may affect the energy, but it does not affect this angular-momentum eigenvalue.

The three-dimensional analogue separates the radial dependence from functions on the sphere:

$$\Psi(r,\theta,\phi) = R(r)Y_{\ell m}(\theta,\phi). \tag{3.6}$$

The functions $Y_{\ell m}$ are spherical harmonics. Their explicit formulas will not be needed here. Rotations act on the angular factor, while the radial function is left alone.

3.2 Finite Angular-Momentum Multiplets

The components of angular momentum satisfy

$$\begin{aligned} [L_x, L_y] &= i\hbar L_z, \\ [L_y, L_z] &= i\hbar L_x, \\ [L_z, L_x] &= i\hbar L_y. \end{aligned} \tag{3.7}$$

Choose eigenstates of L_z,

$$L_z|\ell, m\rangle = \hbar m\,|\ell, m\rangle, \tag{3.8}$$

and define

$$L_\pm = L_x \pm iL_y. \tag{3.9}$$

The commutators

$$[L_z, L_\pm] = \pm\hbar L_\pm \tag{3.10}$$

give

$$L_z\Big(L_\pm|\ell, m\rangle\Big) = \Big(L_\pm L_z + [L_z, L_\pm]\Big)|\ell, m\rangle \tag{3.11}$$
$$= \hbar(m \pm 1)L_\pm|\ell, m\rangle. \tag{3.12}$$

Thus L_+ raises m by one and L_- lowers it by one, unless the resulting vector vanishes.

For a finite multiplet there is a highest state and a lowest state:

$$L_+|\ell, \ell\rangle = 0, \qquad L_-|\ell, -\ell\rangle = 0. \tag{3.13}$$

The $+z$ and $-z$ directions are related by a rotation, so the finite spectrum is symmetric about zero. Its values are therefore

$$m = -\ell, -\ell + 1, \ldots, \ell - 1, \ell, \tag{3.14}$$

and the number of states is

$$2\ell + 1. \tag{3.15}$$

Since neighboring values differ by one, the endpoints can be integers or half-integers. Ordinary orbital spherical harmonics realize the integer case; the abstract angular-momentum algebra also permits half-integer multiplets.

[a] **Question from the audience.** Could an angular-momentum multiplet contain infinitely many states?

Response. That possibility can be contemplated, but it will not be needed here. Angular momentum generally raises the energy when the other features of the state are held fixed, so states of infinite angular momentum are not relevant when discussing finite-energy states.

[a]Lecture timestamp: 00:16:12.

3.3 The Common Value of L^2

The operator measuring the square of the angular-momentum magnitude is

$$L^2 = L_x^2 + L_y^2 + L_z^2. \tag{3.16}$$

The ordered product of ladder operators is

$$\begin{aligned} L_-L_+ &= (L_x - iL_y)(L_x + iL_y) & (3.17)\\ &= L_x^2 + L_y^2 + i[L_x, L_y] & (3.18)\\ &= L_x^2 + L_y^2 - \hbar L_z. & (3.19) \end{aligned}$$

Consequently,

$$L^2 = L_z^2 + \hbar L_z + L_-L_+. \tag{3.20}$$

Apply this identity to the highest state. Since $L_+|\ell, \ell\rangle = 0$,

$$\begin{aligned} L^2|\ell, \ell\rangle &= \left(L_z^2 + \hbar L_z + L_-L_+\right)|\ell, \ell\rangle & (3.21)\\ &= \hbar^2\left(\ell^2 + \ell\right)|\ell, \ell\rangle, & (3.22) \end{aligned}$$

and therefore

$$L^2|\ell,\ell\rangle = \hbar^2\ell(\ell+1)|\ell,\ell\rangle. \tag{3.23}$$

The squared magnitude is not $\hbar^2\ell^2$. The extra term is a consequence of the noncommutativity of the angular-momentum components.

The same value belongs to every state in the multiplet, not only to the highest one.

---

[a] **Question from the audience.** What exactly is the exercise showing that the other states have the same value of L^2?

Response. Show that L^2 commutes with every component of angular momentum and hence with L_-. Then commute L^2 through L_-: if the highest state has eigenvalue $\hbar^2\ell(\ell+1)$, the next state down has the same eigenvalue. Repeating the argument carries that value through the whole ladder.

---

[a] Lecture timestamp: 00:26:08.

Indeed,

$$[L^2, L_i] = 0, \qquad i = x, y, z, \tag{3.24}$$

so

$$[L^2, L_\pm] = 0. \tag{3.25}$$

For example,

$$\begin{aligned} L^2\big(L_-|\ell,\ell\rangle\big) &= L_-L^2|\ell,\ell\rangle \qquad (3.26)\\ &= \hbar^2\ell(\ell+1)L_-|\ell,\ell\rangle. \qquad (3.27)\end{aligned}$$

Continuing downward gives

$$L^2|\ell,m\rangle = \hbar^2\ell(\ell+1)|\ell,m\rangle \tag{3.28}$$

for every $m = -\ell, \ldots, \ell$.

The opposite ordering gives an equivalent factorization:

$$\begin{aligned} L_+L_- &= (L_x + iL_y)(L_x - iL_y) && (3.29)\\ &= L_x^2 + L_y^2 - i[L_x, L_y] && (3.30)\\ &= L_x^2 + L_y^2 + \hbar L_z, && (3.31)\end{aligned}$$

and hence

$$L^2 = L_z^2 - \hbar L_z + L_+L_-. \tag{3.32}$$

---

[a] **Question from the audience.** Could $L_x^2 + L_y^2$ be factored in the opposite order using L_+L_-?

Response. Yes. The sign of the L_z term reverses. With that ordering, apply the identity to the lowest state, which is annihilated by L_-, rather than to the highest state. The same eigenvalue $\hbar^2\ell(\ell+1)$ follows.

---

[a]Lecture timestamp: 00:32:05.

The spherical harmonics therefore satisfy

$$\begin{aligned} L^2 Y_{\ell m} &= \hbar^2\ell(\ell+1)Y_{\ell m}, && (3.33)\\ L_z Y_{\ell m} &= \hbar m\, Y_{\ell m}. && (3.34)\end{aligned}$$

For each orbital value $\ell = 0, 1, 2, \ldots$, there are $2\ell + 1$ functions labeled by $m = -\ell, \ldots, \ell$.

For a central potential,

$$[H, L_i] = 0. \tag{3.35}$$

Once a radial energy eigenstate and ℓ are held fixed, changing m with $L_\pm$ does not change the energy. Each such radial eigenstate therefore generates a $2\ell + 1$-dimensional magnetic multiplet.

The integer orbital ladder is shown below.[1]

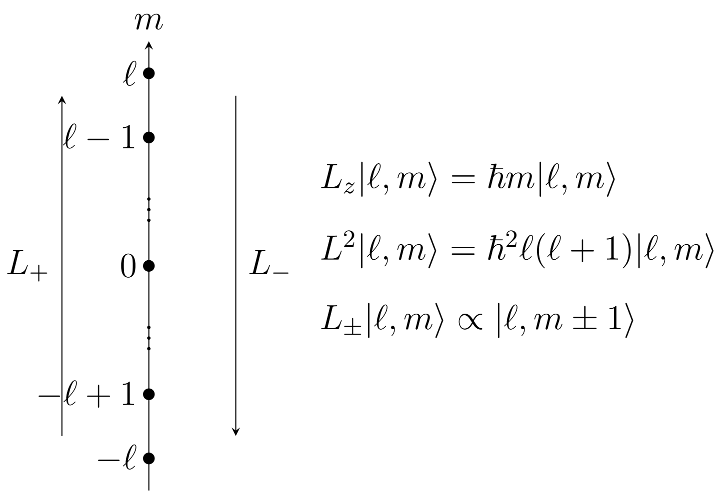

Figure 3.1: An integer orbital angular-momentum multiplet. L_+ and L_- change m by one, while L^2 remains $\hbar^2\ell(\ell + 1)$ throughout the multiplet.

Lecture frame: 00:35:53.

3.4 The Central-Force Problem

Consider

$$H = \frac{\mathbf{p}^2}{2m} + V(r), \tag{3.36}$$

[1] The diagram is reconstructed from the multiplet sketch and operator identities visible at 00:35:53 of the lecture video.

where V depends only on the distance from the origin. Rotational symmetry implies conservation of angular momentum. Classically, $\mathbf{L}$ has fixed magnitude and direction, so the orbit remains in a plane perpendicular to $\mathbf{L}$. The orbit need not be circular, or even closed.

Choose coordinates so that this classical orbital plane is the xy-plane, and resolve the momentum into radial and tangential components:

$$\mathbf{p}^2 = p_r^2 + p_\theta^2. \tag{3.37}$$

Since the magnitude of the angular momentum is

$$L = rp_\theta, \tag{3.38}$$

the tangential contribution may be written

$$p_\theta^2 = \frac{L^2}{r^2}. \tag{3.39}$$

The Hamiltonian becomes

$$H = \frac{p_r^2}{2m} + \frac{L^2}{2mr^2} + V(r). \tag{3.40}$$

For fixed L, this is a one-dimensional problem in the radial coordinate, with effective potential

$$V_{\text{eff}}(r) = \frac{L^2}{2mr^2} + V(r). \tag{3.41}$$

The radial Hamiltonian is shown on the blackboard below.[2]

[2]Source frame: `lecture_03_frame_02.png`, lecture video at 00:47:15.

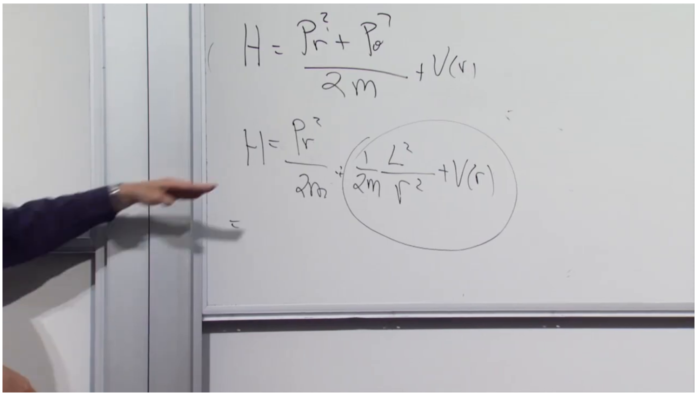

Figure 3.2: For fixed classical angular momentum, the radial Hamiltonian contains radial kinetic energy, the centrifugal term $L^2/(2mr^2)$, and the central potential $V(r)$.

Lecture frame: 00:47:15.

The $1/r^2$ term is the centrifugal barrier. It rises as the particle approaches the origin, so the associated force points outward. For the Coulomb-like attractions under discussion, a trajectory with $L \neq 0$ swings around the center instead of reaching it. When $L = 0$, the barrier is absent.

For the attractive Coulomb potential,

$$V(r) = -\frac{\kappa}{r}, \tag{3.42}$$

the centrifugal term dominates at small r, while the Coulomb term falls more slowly and dominates at large r. For $L \neq 0$, their sum has a minimum. Motion at the minimum has constant radius and is therefore circular. Bound motion above the minimum and below the escape threshold oscillates between a smallest and a largest radius; in the Coulomb problem these are the radial oscillations of an elliptical orbit.

3.5 The Reduced Radial Equation

In quantum mechanics the angular part of a state with definite orbital angular momentum is a spherical harmonic. Write the separated state as

$$\Psi_{n_r\ell m}(r,\theta,\phi) = \frac{u_{n_r\ell}(r)}{r}\,Y_{\ell m}(\theta,\phi). \tag{3.43}$$

The angular operator L^2 has the eigenvalue

$$L^2 Y_{\ell m} = \hbar^2\ell(\ell+1)Y_{\ell m}. \tag{3.44}$$

For a fixed pair n_r, ℓ, abbreviate $u_{n_r\ell}(r)$ to $u(r)$. The equation left for the reduced radial function is

$$\begin{aligned} &-\frac{\hbar^2}{2m}\frac{d^2u}{dr^2} + \frac{\hbar^2\ell(\ell+1)}{2mr^2}u(r) \\ &\qquad + V(r)u(r) = E_{n_r\ell}\,u(r). \end{aligned} \tag{3.45}$$

[3]

The reduced radial equation is shown on the blackboard below.[4]

[3]Editorial clarification: The pure one-dimensional second-derivative equation uses the reduced radial function $u = rR$. Thus the $\psi(r)$ in the displayed lecture equation corresponds to $u(r)$, while the ordinary radial factor in $\Psi = R(r)Y_{\ell m}$ is $R = u/r$. This convention avoids treating $-i\hbar\,d/dr$ as a standalone Hermitian radial-momentum operator on the three-dimensional radial Hilbert space.

[4]Source frame: `lecture_03_frame_03.png`, lecture video at 00:54:32.

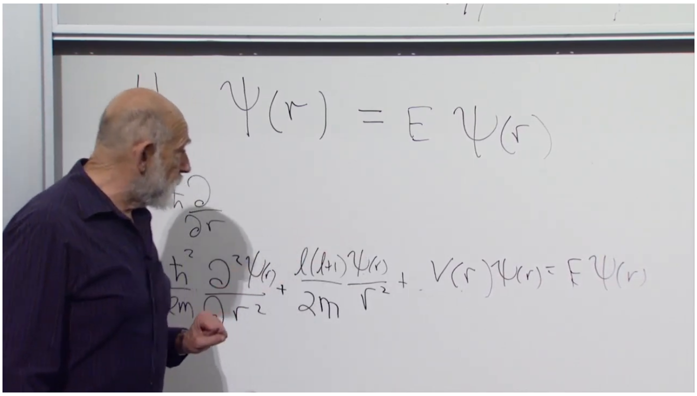

Figure 3.3: For fixed ℓ, the reduced radial wavefunction satisfies a one-dimensional Schrödinger equation with a centrifugal term proportional to $\ell(\ell+1)/r^2$.

Lecture frame: 00:54:32.

The three-dimensional angular structure has been compressed into the parameter ℓ. The magnetic quantum number m does not occur in the radial equation.

---

[a] **Question from the audience.** For a given ℓ, do the $2\ell+1$ magnetic states have the same energy?

Response. Yes, provided the radial eigenstate is also held fixed. For each chosen $u_{n_r\ell}$, all values $m = -\ell, \ldots, \ell$ have the same energy because m does not enter the radial equation. Different radial eigenstates at the same ℓ generally have different energies.

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[a]Lecture timestamp: 00:55:34.

[a] **Question from the audience.** Why may L^2 be replaced by $\hbar^2\ell(\ell+1)$, and how is ℓ related to the allowed values of m?

Response. The number ℓ labels an angular-momentum multiplet and is its largest m-value. Every state in that multiplet has the same L^2-eigenvalue, $\hbar^2\ell(\ell+1)$, so that eigenvalue is inserted into the radial equation.

[a] Lecture timestamp: 00:56:03.

The explicit special-function solutions are not needed for the present argument. What matters is how the solutions are organized.

3.6 Radial Nodes and Energy Levels

For fixed ℓ, the reduced radial equation is an ordinary one-dimensional eigenvalue problem. Its bound states are ordered by their number of interior nodes. The lowest radial state has no interior node, the next has one, and the next has two. More nodes mean more rapid variation of the wavefunction; more rapid variation means larger momentum and therefore larger kinetic energy.

Each value of ℓ defines a different radial operator because the centrifugal term changes with ℓ. Thus $\ell = 0, 1, 2, \ldots$ give separate radial towers. The positive centrifugal term shifts the levels upward as angular momentum is added. Every radial eigenvalue at fixed ℓ carries a magnetic multiplet of dimension

$$2\ell + 1. \tag{3.46}$$

The $\ell = 0$ levels are singlets, the $\ell = 1$ levels are triplets, and the $\ell = 2$ levels are quintuplets. These are the degeneracies guaranteed by rotational symmetry; an additional symmetry may make states with different values of ℓ degenerate as well.

For atomic-type potentials satisfying $V(r) \to 0$ at large r, the escape threshold is conventionally $E = 0$. The negative-energy states form the discrete bound spectrum. At and above $E = 0$, one encounters the scattering continuum. This statement depends on the large-r behavior of the potential and does not apply to a confining central potential.

3.7 The Special Coulomb Degeneracy

The exact nonrelativistic Coulomb potential has an additional alignment of levels. The lowest $\ell = 1$ radial state has the same energy as the first excited $\ell = 0$ state. The next $\ell = 1$ state aligns with the second excited $\ell = 0$ state, and the lowest $\ell = 2$ state joins them. The pattern continues through the higher radial and angular-momentum towers.

---

[a] **Question from the audience.** Does this extra Coulomb degeneracy indicate another symmetry?

Response. Yes. It is an unusual additional symmetry of the exact Coulomb potential. Small changes in the potential destroy the alignment, and its derivation is not pursued here.

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[a]Lecture timestamp: 01:09:57.

The resulting orbital shell degeneracies are

$$1, \qquad 4, \qquad 9, \qquad 16, \qquad \dots \tag{3.47}$$

before electron spin is included.

---

[a] **Question from the audience.** Does this degeneracy describe hydrogen, or only the ideal Coulomb problem?

Response. Hydrogen satisfies it to very good approximation, but not exactly. Relativistic corrections and the finite size of the nucleus disturb the ideal Coulomb pattern.

---

[a]Lecture timestamp: 01:11:46.

A point nucleus produces an exact $1/r$ potential. A nucleus with charge spread over a small volume produces a slightly different potential when the electron penetrates that volume, and even this small departure lifts part of the special degeneracy.

Spin has also been omitted. Including the two electron spin states doubles the shell counts to

$$2, \qquad 8, \qquad 18, \qquad 32, \qquad \ldots \tag{3.48}$$

before the further splittings of a more complete atomic treatment are resolved.

3.8 The Cartesian Equation Revisited

The equation from which the radial problem began is the three-dimensional time-independent Schrödinger equation

$$\begin{aligned} -\frac{\hbar^2}{2m}\nabla^2\Psi + V(r)\Psi = E\Psi, \\ \nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}, \\ r = \sqrt{x^2+y^2+z^2}. \end{aligned} \tag{3.49}$$

Pass to radial and angular coordinates and select an eigenstate of L^2. Replacing L^2 by $\hbar^2\ell(\ell+1)$ then reduces the partial differential equation to the ordinary radial equation above.

---

[a] **Question from the audience.** Could deviations from the ideal Coulomb spectrum be used to determine the size of the nucleus?

Response. Not by this simple comparison. Other corrections are much larger and must first be accounted for. Nuclear size is mentioned here because it gives an immediate example of a departure from the point-Coulomb potential.

---

[a] Lecture timestamp: 01:16:42.

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[a] **Question from the audience.** What is the main difference between the classical and quantum angular-momentum terms?

Response. Classically L^2 is the ordinary square of a conserved vector magnitude. In a quantum orbital multiplet its eigenvalue is $\hbar^2\ell(\ell+1)$, not simply $\hbar^2\ell^2$. The additional term comes from the noncommutativity of the angular-momentum components.

---

[a] Lecture timestamp: 01:17:09.

[a] **Question from the audience.** Do the bound-state wavefunctions have compact support?

Response. No. They extend arbitrarily far from the center. For the atomic bound states under discussion they decrease exponentially at large distance, so the probability becomes negligible but does not vanish beyond any finite radius.

[a]Lecture timestamp: 01:17:58.

[a] **Question from the audience.** When $\ell = 0$ removes the centrifugal well, what becomes of the stable point?

Response. There is no longer a minimum produced by competition between the centrifugal barrier and the attraction. In the point-Coulomb idealization the classical motion heads toward the singular center. Quantum mechanically the lowest state cannot have both an exactly definite position at the center and zero momentum, so the lowest wavefunction has a finite spatial spread.

[a]Lecture timestamp: 01:18:48.

5

The classical reduction placed a single orbit in a plane, but that step cannot be applied to a quantum wavefunction in the same way. A quantum orbital state fills three-dimensional space. The full angular operator is $L^2 = L_x^2 + L_y^2 + L_z^2$, and its eigenvalue

[5]Editorial clarification: For a regular central potential, a classical zero-angular-momentum trajectory may continue through the origin. For the ideal point-Coulomb potential, $r = 0$ is instead a collision singularity, and continuation of the classical trajectory requires an additional prescription.

retains the three-dimensional result $\hbar^2\ell(\ell+1)$.

---

[a] **Question from the audience.** How can the quantum problem remain three-dimensional after the classical orbit was placed in a plane?

Response. Only the classical orbit was put in a fixed plane by choosing coordinates perpendicular to its conserved angular-momentum vector. A quantum wavefunction occupies three-dimensional space. The other angular-momentum components remain present through L^2, which is why the radial equation contains $\ell(\ell+1)$ rather than only a square of one component.

---

[a] Lecture timestamp: 01:21:30.

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[a] **Question from the audience.** Does every quantum problem admit raising and lowering operators?

Response. No. They occur in unusually favorable cases. Angular momentum is one such case, and the harmonic oscillator is another. The oscillator case is especially useful because motion near a stable equilibrium is so often approximately harmonic, including small radial oscillations near the minimum of an effective potential.

---

[a] Lecture timestamp: 01:22:54.

The atomic problem is set aside at this point, and the harmonic oscillator becomes the next example of a spectrum determined by ladder operators.

3.9 Small Oscillations and the Harmonic Oscillator

Near a stable equilibrium, a sufficiently small displacement often produces harmonic motion. The effective radial potential for nonzero angular momentum provides one example: radial motion close to the bottom of its well is approximately harmonic.

Consider a one-dimensional oscillator with unit mass. Let x be the displacement from equilibrium and write the spring constant as $k = \omega^2$. The classical Hamiltonian is

$$H = \frac{p^2}{2} + \frac{\omega^2 x^2}{2}. \tag{3.50}$$

Hamilton's equations give

$$\dot{x} = \frac{\partial H}{\partial p} = p, \qquad \dot{p} = -\frac{\partial H}{\partial x} = -\omega^2 x, \tag{3.51}$$

and therefore

$$\ddot{x} = -\omega^2 x. \tag{3.52}$$

Thus ω is the classical oscillation frequency.

3.10 Factorization and the Zero-Point Energy

Quantum mechanically,

$$[x, p] = i\hbar. \tag{3.53}$$

The Hamiltonian is a sum of squares, suggesting the factorization

$$\frac{1}{2}(p+i\omega x)(p-i\omega x) = \frac{1}{2}\left(p^2+\omega^2x^2+i\omega[x,p]\right) \tag{3.54}$$

$$= \frac{1}{2}\left(p^2+\omega^2x^2-\hbar\omega\right). \tag{3.55}$$

Hence

$$H = \frac{1}{2}(p+i\omega x)(p-i\omega x) + \frac{1}{2}\hbar\omega. \tag{3.56}$$

The additive term $\hbar\omega/2$ is the zero-point energy.

Classically the lowest energy would occur at $x = p = 0$. Quantum mechanically no state can have both position and momentum sharply fixed at zero. Moreover, the two terms $p^2/2$ and $\omega^2x^2/2$ are positive and cannot cancel. A nonzero ground-state energy is therefore unavoidable.

For the ladder calculation the common additive constant may be set aside temporarily and restored at the end.

---

[a] **Question from the audience.** Does the positive sign of the zero-point term tell us anything further?

Response. The oscillator Hamiltonian is a positive operator: it is a sum of positive squares, so its energy cannot be negative.

---

[a]Lecture timestamp: 01:36:10.

3.11 The Oscillator Ladder

Introduce the dimensionless operators[6]

$$A_+ = \frac{p + i\omega x}{\sqrt{2\hbar\omega}}, \qquad A_- = \frac{p - i\omega x}{\sqrt{2\hbar\omega}}. \tag{3.57}$$

They are Hermitian conjugates:

$$A_+^\dagger = A_-. \tag{3.58}$$

Their commutator is

$$[A_-, A_+] = \frac{1}{2\hbar\omega}\left([p, i\omega x] + [-i\omega x, p]\right) \tag{3.59}$$
$$= 1. \tag{3.60}$$

Define

$$N = A_+ A_-. \tag{3.61}$$

The Hamiltonian is then

$$H = \hbar\omega\left(N + \frac{1}{2}\right). \tag{3.62}$$

Thus finding the energy spectrum reduces to finding the eigenvalues of N.

Suppose

$$N|n\rangle = n|n\rangle. \tag{3.63}$$

[6]Around 01:37–01:42, the calculation uses $\hbar = 1$. The factor of $\hbar$ is restored in these normalized definitions so that $A_\pm$ are dimensionless and their commutator is unity.

The commutators with N are[7]

$$[N, A_+] = A_+[A_-, A_+] = A_+, \tag{3.64}$$
$$[N, A_-] = [A_+, A_-]A_- = -A_-. \tag{3.65}$$

It follows that

$$N\Big(A_+|n\rangle\Big) = (n+1)A_+|n\rangle, \tag{3.66}$$
$$N\Big(A_-|n\rangle\Big) = (n-1)A_-|n\rangle. \tag{3.67}$$

Thus A_+ raises an eigenvalue of N by one and A_- lowers it by one whenever the resulting vector is nonzero.

The operator N is nonnegative:

$$\langle\psi|N|\psi\rangle = \langle\psi|A_+A_-|\psi\rangle = \|A_-|\psi\rangle\|^2 \geq 0. \tag{3.68}$$

The lowering process therefore cannot continue into negative eigenvalues. At a bottom state $|b\rangle$, it must terminate:

$$A_-|b\rangle = 0. \tag{3.69}$$

For normalized eigenstates, the coefficients in the ladder are fixed by their norms:

$$\|A_-|n\rangle\|^2 = \langle n|A_+A_-|n\rangle = n, \tag{3.70}$$
$$\|A_+|n\rangle\|^2 = \langle n|A_-A_+|n\rangle = n+1. \tag{3.71}$$

Phases may therefore be chosen so that

$$A_-|n\rangle = \sqrt{n}\,|n-1\rangle, \qquad A_+|n\rangle = \sqrt{n+1}\,|n+1\rangle. \tag{3.72}$$

[7]The intermediate steps at 01:45:37–01:48:10 are reconstructed directly from the stated relations $N = A_+A_-$ and $[A_-, A_+] = 1$.

[a] **Question from the audience.** If A_- lowers a state and A_+ raises it back, how does A_+A_- produce the eigenvalue n?

Response. The ladder action includes normalization factors. Acting first with A_- contributes $\sqrt{n}$, and acting with A_+ on the resulting state contributes another $\sqrt{n}$. Their product is n.

[a] Lecture timestamp: 01:52:42.

Indeed,

$$A_+A_-|n\rangle = \sqrt{n}\,A_+|n-1\rangle = n|n\rangle. \tag{3.73}$$

[a] **Question from the audience.** Must the lowest eigenvalue of N be zero, or could it be positive?

Response. It must be zero, and the spectrum of N consists of the nonnegative integers. A proof of the endpoint has not yet been supplied.

[a] Lecture timestamp: 01:55:52.

[8]

Starting from this lowest state and applying A_+ gives

$$n = 0, 1, 2, \ldots. \tag{3.74}$$

Restoring the additive zero-point term, the oscillator energies are

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right), \qquad n = 0, 1, 2, \ldots. \tag{3.75}$$

[8] Editorial clarification: Let $|b\rangle$ be a lowest state. Since lowering must terminate, $A_-|b\rangle = 0$. It follows immediately that $N|b\rangle = A_+A_-|b\rangle = 0$, so the lowest eigenvalue of N is zero.

The oscillator algebra will be developed further and used to solve other quantum problems.

CHAPTER 4

OSCILLATOR WAVEFUNCTIONS, SPIN-ONE-HALF, AND PARTICLE EXCHANGE

4.1 Oscillators Everywhere

Any system with a stable equilibrium can oscillate when it is displaced from that equilibrium. It may have one frequency or many. A violin string, for example, has a fundamental mode and higher harmonics. Plucking it excites some combination of those modes.

The more important examples here are fields. The equilibrium configuration of the electromagnetic field is the vacuum: no electric field, no magnetic field, and no electromagnetic oscillation. Send radiation into a cavity or a waveguide and the field can oscillate in its allowed modes, each with its own frequency. In the quantum theory those modes describe collections of photons.

The same idea applies to a crystal. It is misleading to picture every atom as an independent oscillator. The atoms exert forces on one another, so a displacement of one atom affects its neighbors and a wave travels through the crystal. The normal modes of that coupled motion are harmonic oscillators, and their quanta are called phonons. Photons and phonons are different physical objects, but both lead back to the same oscillator algebra.

4.2 The Oscillator Algebra

For the oscillator discussion, set the mass and the reduced Planck constant equal to one:

$$m = 1, \qquad \hbar = 1.$$

The Hamiltonian is

$$H = \frac{p^2}{2} + \frac{\omega^2 x^2}{2}. \tag{4.1}$$

The parameter ω fixes the curvature of the potential and is also the oscillation frequency.

Define the raising and lowering operators by

$$A_\pm = \frac{p \pm i\omega x}{\sqrt{2\omega}}. \tag{4.2}$$

The normalization is chosen so that

$$[A_-, A_+] = 1. \tag{4.3}$$

The number operator is

$$N = A_+ A_-. \tag{4.4}$$

Together, these equations imply that A_+ raises an eigenvalue of N by one and A_- lowers it by one. The resulting spectrum is

$$n = 0, 1, 2, 3, \ldots. \tag{4.5}$$

Multiplying the two ladder operators also displays the quantum correction to the classical energy:

$$N = \frac{1}{2\omega}(p + i\omega x)(p - i\omega x) \tag{4.6}$$
$$= \frac{1}{2\omega}\left(p^2 + \omega^2 x^2 + i\omega[x,p]\right) \tag{4.7}$$
$$= \frac{1}{2\omega}\left(p^2 + \omega^2 x^2 - \omega\right), \tag{4.8}$$

where $[x, p] = i$. Consequently,

$$H = \omega\left(N + \frac{1}{2}\right). \tag{4.9}$$

If the additive constant is irrelevant to the problem, it is convenient to use the shifted Hamiltonian

$$H_{\text{shifted}} = H - \frac{\omega}{2} = \omega N. \tag{4.10}$$

The term $\omega/2$ is the oscillator's zero-point, or ground-state, energy. For a field mode the analogous term is called vacuum energy. It reflects the uncertainty principle: no state can have both x and p sharply equal to zero, so the two positive quadratic terms in H cannot both have zero expectation value.

[a] **Question from the audience.** Why is zero included in the spectrum of the number operator?

Response. It can be demonstrated explicitly by constructing the ground-state wavefunction. That construction will exhibit a state with $N = 0$.

[a]Lecture timestamp: 00:09:46.

To begin that construction, replace state vectors by wavefunctions of position. Energy eigenstates become

$$\psi_0(x),\ \psi_1(x),\ \psi_2(x), \ldots,$$

and the basic operators act as

$$X \longmapsto x, \qquad P \longmapsto -i\frac{d}{dx}. \tag{4.11}$$

With $\hbar$ restored, the second replacement would be $P \mapsto -i\hbar\, d/dx$.

Before solving for ψ_0, there is a faster abstract answer to the question. On the discrete ladder of normalizable oscillator eigenstates, $N = A_+A_-$ is nonnegative, so repeated lowering in unit steps cannot continue indefinitely. At the endpoint—whose normalizable wavefunction will be constructed below—there must be a state, denoted $|0\rangle$, for which

$$A_-|0\rangle = \mathbf{0}. \tag{4.12}$$

Otherwise $A_-|0\rangle$ would be another state one step lower. It follows immediately that

$$N|0\rangle = A_+A_-|0\rangle = \mathbf{0} = 0\,|0\rangle. \tag{4.13}$$

Thus the bottom state has number eigenvalue zero.

---

[a] **Question from the audience.** If zero is the eigenvalue, should the right-hand side contain a zero eigenvector?

Response. The ket $|0\rangle$ is a normalized, nonzero vector whose label is zero. In $N|0\rangle = 0\,|0\rangle$, the right-hand side is the zero vector. The ground-state ket and the zero vector are different objects.

---

[a]Lecture timestamp: 00:14:57.

4.3 The Schrödinger Equation and Physical Solutions

Return now to the position representation. The time-independent Schrödinger equation is

$$\left[-\frac{1}{2}\frac{d^2}{dx^2} + \frac{1}{2}\omega^2 x^2\right]\psi(x) = E\psi(x). \tag{4.14}$$

As a second-order differential equation, it has formal solutions for arbitrary values of E. Most of those formal solutions are not physical oscillator states. They grow exponentially at large $|x|$, so their norm diverges.

For a bound state, the wavefunction must be square integrable:

$$\int_{-\infty}^{\infty} \psi^*(x)\psi(x)\,dx < \infty. \tag{4.15}$$

It can then be normalized so that the integral equals one. Only special energy values produce such solutions. Position and momentum states on an infinite line are exceptional in this respect, but exponentially growing oscillator solutions are simply excluded from the state space.

4.4 The Gaussian Ground State

The second-order equation can be solved, but the ladder algebra gives a simpler first-order equation:

$$A_{-}\psi_0(x) = 0. \tag{4.16}$$

The overall factor in A_{-} is irrelevant when the right-hand side is zero, so this is equivalent to

$$(p - i\omega x)\psi_0(x) = 0. \tag{4.17}$$

Using $p = -i\,d/dx$ gives

$$\left(\frac{d}{dx} + \omega x\right)\psi_0(x) = 0. \tag{4.18}$$

[a] **Question from the audience.** Could this be solved by separation of variables?

Response. There is only one independent variable, so there are not two variables to separate. Written as $\psi_0'/\psi_0 = -\omega x$, the equation has the form of a logarithmic derivative. That suggests writing the wavefunction as an exponential.

[a] Lecture timestamp: 00:22:32.

A useful way to solve it is to write

$$\psi_0(x) = e^{f(x)}. \tag{4.19}$$

Then

$$\psi_0'(x) = f'(x)e^{f(x)},$$

and the common exponential factor cancels. The equation reduces to

$$\frac{df}{dx} + \omega x = 0. \tag{4.20}$$

---

[a] **Question from the audience.** Should there be a minus sign in $f(x)$?

Response. Yes. Since $df/dx = -\omega x$, integration gives a negative quadratic term.

---

[a]Lecture timestamp: 00:26:28.

Therefore

$$f(x) = -\frac{1}{2}\omega x^2 + C, \tag{4.21}$$

and

$$\psi_0(x) \propto e^{-\omega x^2/2}. \tag{4.22}$$

The multiplicative normalization constant is left unspecified. The important point is the Gaussian shape: it is smooth, concentrated near the origin, and rapidly decreasing at large $|x|$. It is plainly square integrable.

There is no need to substitute the answer back into the second-order equation to discover its energy. The algebra already gives

$$E_0 = \frac{\omega}{2}. \tag{4.23}$$

4.5 Excited States, Nodes, and Classical Motion

The first excited state follows by raising the ground state:

$$\psi_1(x) \propto A_+\psi_0(x) \propto (p + i\omega x)e^{-\omega x^2/2}. \qquad (4.24)$$

The ground-state equation says

$$p\psi_0 = i\omega x\, \psi_0.$$

Changing the sign in the ladder operator turns the cancellation into addition:

$$(p + i\omega x)\psi_0 = 2i\omega x\, \psi_0. \qquad (4.25)$$

Up to an overall constant,

$$\psi_1(x) \propto x\, e^{-\omega x^2/2}. \qquad (4.26)$$

This wavefunction is odd:

$$\psi_1(-x) = -\psi_1(x). \qquad (4.27)$$

Its probability density is even, but it has a node at the origin,

$$|\psi_1(0)|^2 = 0. \qquad (4.28)$$

Thus the first-excited-state probability density has a node at $x = 0$.

Repeated action of A_+ gives higher-degree polynomials multiplying the same Gaussian envelope.[1] Up

[1] The lecture states the higher-degree polynomial factors, the additional node at each step, and alternating symmetry from 00:34:25 to 00:35:49. The Gaussian-times-polynomial form is corroborated by Leonard Susskind and Art Friedman, *Quantum Mechanics: The Theoretical Minimum* (2014), in its discussion of oscillator eigenfunctions.

to normalization,

$$\psi_n(x) = P_n(x)e^{-\omega x^2/2}. \qquad (4.29)$$

The parity alternates between even and odd, and each step adds a node. As n increases, the wavefunction spreads farther from the origin and develops more rapid oscillations.

The two features have direct physical meanings. Support farther from the origin corresponds to larger potential energy because the potential grows as x^2. Rapid oscillation in position space signals larger momentum. A highly excited oscillator is likely to reach farther from the center; when it passes through the central region, it carries more momentum and its wavefunction oscillates more rapidly there.

---

[a] **Question from the audience.** Does the harmonic oscillator fail to approach classical behavior when n becomes large?

Response. Classical motion is not represented by one stationary energy eigenstate. It appears in a wave packet containing many energy levels. A displaced Gaussian, for example, evolves back and forth like a classical oscillator, and the relative uncertainty becomes smaller as the swing grows larger.

---

[a] Lecture timestamp: 00:37:18.

Start with the ground-state Gaussian and displace it from the origin. The displaced packet is no longer an energy eigenstate; it is a superposition of many of them. Under the time-dependent Schrödinger equation, its components acquire different phases. The packet swings from one side of the potential to the other, developing rapid phase variation near the center and becoming smooth again near a turning point. The larger the excursion compared with the packet's width, the more nearly classical the motion appears.

---

[a] **Question from the audience.** Should the second excited-state wavefunction pass through zero at the origin?

Response. No. Its polynomial factor is not simply x^2; it also contains a constant term, so the wavefunction need not vanish at $x = 0$.

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[a]Lecture timestamp: 00:39:29.

The ladder argument also assumes that A_+ and A_- are Hermitian conjugates on the square-integrable oscillator domain, with the boundary behavior needed for the adjoint relation. Exponentially growing formal solutions do not belong to that domain.

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[a] **Question from the audience.** Is there a general way to recognize when a Hamiltonian can be solved by these algebraic methods?

Response. There is no general rule of that kind. Exactly solvable examples are rare. The harmonic oscillator and the ideal nonrelativistic hydrogen atom are fortunate cases in which algebraic methods work.

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[a]Lecture timestamp: 00:40:45.

4.6 Spin One-Half as Angular Momentum

Spin is angular momentum carried by a particle independently of its center-of-mass orbital motion. A particle at rest has no orbital angular momentum $\mathbf{R} \times \mathbf{P}$, but it may still have spin. One can try to picture a tiny spinning object, or treat spin as an intrinsic quantum property. The decisive statement is mathematical: its components transform under spatial rotations and obey the angular-momentum commutation relations.

Restore $\hbar$ from this point onward. Angular-momentum components satisfy

$$[L_z, L_x] = i\hbar L_y, \tag{4.30}$$

with the other two relations obtained by cyclic permutation. The dimensionless Pauli matrices are

$$\begin{aligned} \sigma_z &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \\ \sigma_x &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \\ \sigma_y &= \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \end{aligned} \tag{4.31}$$

In this basis, σ_z is diagonal. Its eigenvalues are $+1$ and -1.

[a] **Question from the audience.** Is there a minus sign in σ_x?

Response. No. The two off-diagonal entries of σ_x are both positive.

[a]Lecture timestamp: 00:47:35.

The board calculation at 00:49:35 begins the entry-by-entry multiplication. Completing the products gives

$$\sigma_z\sigma_x = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = i\sigma_y, \tag{4.32}$$

$$\sigma_x\sigma_z = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = -i\sigma_y. \tag{4.33}$$

Hence

$$[\sigma_z, \sigma_x] = 2i\sigma_y. \tag{4.34}$$

The extra factor of two is removed by defining the physical spin operators as[2]

$$S_i = \frac{\hbar}{2}\sigma_i. \tag{4.35}$$

Then

$$[S_z, S_x] = i\hbar S_y, \tag{4.36}$$

and similarly for cyclic permutations. Half the Pauli matrices, with the factor of $\hbar$ restored, therefore form an angular-momentum system.

Setting $\hbar = 1$ gives the form shown below.[3]

[2]The board calculation suppresses $\hbar$. It is restored here so that S_i has the units of angular momentum.

[3]The source frame is `lecture_04_frame_02.png`.

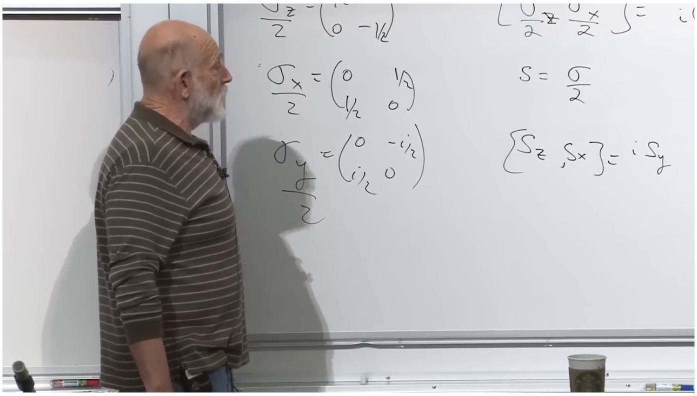

Figure 4.1: In units with $\hbar = 1$, half the Pauli matrices are the spin-one-half angular-momentum operators and satisfy $[S_z, S_x] = iS_y$.

Lecture frame: 00:55:41.

The two S_z eigenvalue equations are

$$S_z|\pm\rangle = \pm\frac{\hbar}{2}|\pm\rangle. \tag{4.37}$$

Equivalently, the dimensionless magnetic quantum number is

$$m_s = \pm\frac{1}{2}, \qquad S_z|m_s\rangle = m_s\hbar\,|m_s\rangle. \tag{4.38}$$

Angular-momentum raising and lowering operators shift m by integer steps, but this does not require every ladder to pass through zero. Integer multiplets contain integer m-values, while half-integer multiplets contain half-integer values. For spin one-half the ladder has only the two members $m_s = +1/2$ and $m_s = -1/2$.

These two states form the spin-one-half doublet. Electrons, muons, neutrinos, and quarks are among the particles described by such half-integer spin multiplets.

4.7 Orbital and Spin Angular Momentum

There are now two kinds of angular momentum. Orbital angular momentum describes motion relative to an origin,

$$\mathbf{L} = \mathbf{R} \times \mathbf{P}, \tag{4.39}$$

while $\mathbf{S}$ is intrinsic spin. Their vector sum is the total angular momentum,

$$\mathbf{J} = \mathbf{L} + \mathbf{S}. \tag{4.40}$$

[a] **Question from the audience.** If a system has total angular momentum $\mathbf{J}$, how is its spin $\mathbf{S}$ identified?

Response. The spin and orbital quantum numbers are normally supplied separately. If $L = 0$, all the angular momentum is spin. Otherwise one needs the quantum rules for adding angular momenta, which are postponed here.

[a] Lecture timestamp: 00:56:49.

The addition rules apply equally to orbital angular momentum plus spin, to two spins, or to two orbital angular momenta. The combined system has its own allowed values of total angular momentum, but those rules are not needed yet.

[a] **Question from the audience.** What experimental evidence led to the introduction of spin?

Response. Spectroscopy supplied evidence, but the periodic table gives a particularly transparent motivation: its shell structure requires electrons to possess an additional two-valued property.

[a]Lecture timestamp: 00:58:18.

4.8 Hydrogenic Degeneracy and Shell Counting

For orbital angular momentum, the L^2 eigenvalue equation is

$$L^2|\ell, m\rangle = \hbar^2 \ell(\ell+1)|\ell, m\rangle, \tag{4.41}$$

and a fixed value of ℓ has

$$2\ell + 1 \tag{4.42}$$

magnetic states, corresponding to $m = -\ell, -\ell+1, \ldots, \ell$.

The ideal nonrelativistic Coulomb problem has an additional degeneracy. Its bound-state energies are negative and approach zero from below, but only the alignment of levels matters for the counting. The first $\ell = 1$ level aligns with the next $\ell = 0$ level; the first $\ell = 2$ level joins the next principal shell; and the pattern continues.[4]

[4]`lecture_04_frame_03.png`, 01:02:01; the completed principal-shell alignment is stated from 01:02:33 through 01:04:10.

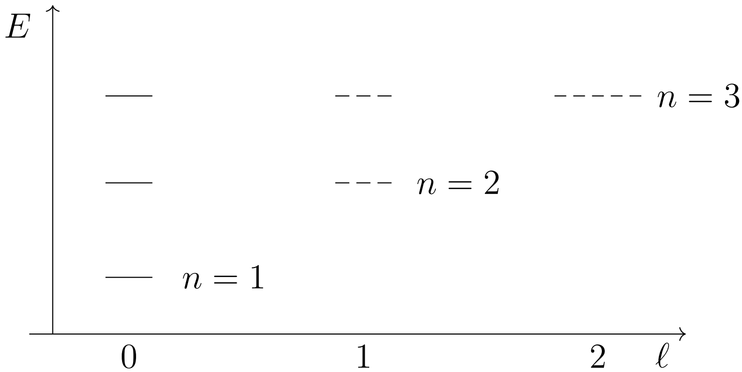

Figure 4.2: The first three ideal Coulomb shells combine orbital multiplets of equal principal energy: one state in the first shell, $1 + 3$ in the second, and $1 + 3 + 5$ in the third.

Lecture frame: 01:02:01.

In the principal shell n, the allowed orbital values are[5]

$$\ell = 0, 1, \ldots, n - 1.$$

The number of spatial states is therefore

$$\sum_{\ell=0}^{n-1} (2\ell + 1) = n^2, \tag{4.43}$$

giving the sequence

$$1,\ 4,\ 9,\ 16, \ldots. \tag{4.44}$$

[5] This compact ℓ-range and the sum below reconstruct the level alignments and the counts $1, 4, 9, 16, \ldots$ stated from 01:02:33 to 01:04:10. The beginning of the spectrum drawing appears in `lecture_04_frame_03.png` at 01:02:01.

This is orbital counting only. Each spatial state permits two spin values, so the corresponding idealized totals are

$$2,\ 8,\ 18,\ 32, \ldots = 2n^2. \tag{4.45}$$

4.9 Exclusion and the Two-Valued Electron Property

Begin with helium in an independent-electron approximation. Its nucleus has twice the proton charge, which rescales the hydrogenic levels and draws the electrons inward. Ignoring electron–electron interactions for the moment, two electrons can occupy the lowest spatial orbital and give a reasonable first approximation to the helium ground state.

Why not put a third electron there as well? That would lower its orbital energy, but it does not happen. The restriction is the Pauli exclusion principle: no two electrons may occupy the same complete one-particle state. Calling this an ordinary repulsive force would be misleading. It is a restriction on the allowed quantum state.

There is an apparent contradiction. The two helium electrons do share the same spatial orbital. The resolution is that a complete electron state contains more than the orbital wavefunction. It also contains a two-valued degree of freedom. Call the values up and down. One electron in the lowest orbital can be spin up and the other spin down; their spatial states agree, but their full quantum states do not.

At this point the counting starts forward toward lithium and then backs up. The backtrack exposes why both exclusion and spin are necessary. Imagine an exclusion rule with no spin at all. Only one electron could occupy the lowest spatial state. A second electron would have to enter one of the four spatial states in the next principal shell, producing several alternatives where the observed helium ground state is unique. That spinless version of exclusion is wrong.

Restore the two spin values. Two electrons fill the lowest orbital with opposite spin. A third electron, as in neutral lithium, must enter the next principal shell. That shell has one $\ell = 0$ spatial state and three $\ell = 1$ spatial states. Before spin is counted, there are four possibilities.

---

[a] **Question from the audience.** How do the s- and p-states give eight possibilities in the first excited shell?

Response. There is one spatial state with $\ell = 0$ and three with $\ell = 1$, giving four spatial states. Each admits two spin values, so the total is $2(1 + 3) = 8$.

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[a] Lecture timestamp: 01:14:58.

This is the simple counting behind the eight entries in the second row of the periodic table. It is deliberately an idealization: the later discussion will make clear that electron–electron interactions soon spoil the independent hydrogenic-shell picture.

The two-valued property can also be tested as angular momentum. The electron's spin is accompanied by a magnetic moment, so a magnetic field gives its two spin projections different energies. A level that was degenerate in zero field splits into two, just as expected for a small magnetic moment aligned or anti-aligned with the field. The splitting tests whether the added two-valued degree of freedom behaves as angular momentum rather than as an arbitrary binary label.

4.10 Photons, Electrons, and the Limits of the Shell Picture

Particles do not all obey the same occupancy rule. Many photons may occupy the same electromagnetic mode; a classical light wave may be understood as a state containing a large number of photons in that mode. Electrons behave differently: exclusion prevents them from piling into one complete quantum state. This contrast motivates the distinction between the two particle statistics.

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[a] **Question from the audience.** What happens for a free stream of electrons rather than electrons bound in atoms?

Response. The exclusion rule still applies: the electrons cannot all occupy one quantum state. A laser can place many photons in the same mode, but an electron stream cannot be prepared in that way.

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[a]Lecture timestamp: 01:20:28.

The restriction is a property of electrons, not a peculiarity of atomic binding.

[a] **Question from the audience.** Is orbital angular momentum specifically rotation about a nucleus?

Response. Angular momentum is defined relative to a chosen origin. The nucleus is the natural origin in an atom, while a free particle has angular momentum $\mathbf{R} \times \mathbf{P}$ relative to whatever origin is selected.

[a] Lecture timestamp: 01:21:12.

Every discussion of rotation chooses a point about which the rotation is made. An atom supplies a natural center because its nucleus is the center of the approximately central potential. In free space no origin is preferred, although angular momentum can still be calculated about any chosen one.

[a] **Question from the audience.** Why does the periodic table contain repeated rows of eight and then repeated rows of eighteen elements?

Response. The simple independent-electron shell picture does not explain the full pattern. It breaks down quickly because it neglects the interactions among the electrons.

[a] Lecture timestamp: 01:22:34.

For a large atom, the outer electrons interact with a substantial inner electron cloud as well as with the nucleus. Treating every electron as if it moved independently in an unscreened Coulomb field is no longer sensible.

[a] **Question from the audience.** Why do atoms not become dramatically larger as more electrons are added?

Response. Adding electrons also adds protons to the nucleus. The increased nuclear charge pulls the inner electrons inward, while those inner electrons screen the nucleus. In this simple picture, an outer valence electron then moves in an effective charge of roughly one.

[a]Lecture timestamp: 01:23:33.

The increased nuclear charge contracts the inner cloud. For an atom with one outer valence electron, the nucleus and the inner electrons nearly cancel at large distance, leaving the valence electron in a field roughly like that of a unit positive charge. Atomic sizes therefore do not simply grow in proportion to the number of electrons.

[a] **Question from the audience.** Is electrical repulsion the reason neighboring atoms remain separated?

Response. Not by itself. Pauli exclusion supplies an important effective repulsion when electron clouds overlap, and for many atoms it is the dominant reason they resist further overlap.

[a]Lecture timestamp: 01:24:49.

Pauli repulsion is an effective consequence of exclusion, not an extra force law.

[a] **Question from the audience.** Is the exclusion principle a separate postulate?

Response. In nonrelativistic quantum mechanics it is imposed as a postulate. In relativistic quantum theory, the spin–statistics connection gives it a deeper basis.

[a]Lecture timestamp: 01:25:23.

4.11 Identical Particles in Classical and Quantum Mechanics

Before formulating that postulate, consider how identical particles are counted classically. Put two particles of the same kind into a set of boxes. If the particles secretly carry names—Harry and Sally—then Harry in the first box and Sally in the second appears different from Sally in the first and Harry in the second.

Classical mechanics permits that picture even when the labels are too faint to detect. One can imagine an arbitrarily weak mark on each particle and still regard their trajectories as distinct. In classical statistical mechanics one usually chooses not to count permutations of identical particles as new configurations. For N particles this convention produces the familiar factor $1/N!$.

Quantum mechanics is stricter. Identical particles do not carry hidden individual labels that can be followed in principle. Two electrons are interchangeable as electrons. Exchanging an electron with a photon would produce a physically different arrangement because they are different species, but exchanging the two electron slots cannot reveal which electron was in fact which.

A one-particle state has a wavefunction

$$\psi(x) = \langle x|\psi\rangle.$$

For two particles the state is a function of two position arguments,

$$\psi(x_1, x_2). \tag{4.46}$$

The subscripts label the two slots in the description, not two spatial directions. Its absolute square gives the joint position probability density.

4.12 The Swap Operator and the Statistics Postulate

Interchange the two particle slots:

$$\psi(x_1, x_2) \longmapsto \psi(x_2, x_1). \tag{4.47}$$

Following the notation used in this discussion, call the swap operator S. Here S denotes exchange, not the spin vector $\mathbf{S}$. Its action is

$$(S\psi)(x_1, x_2) = \psi(x_2, x_1). \tag{4.48}$$

A general function of two variables need not be symmetric, so this operation can produce a different wavefunction.

Swapping twice restores the original ordering:

$$S^2 = I. \tag{4.49}$$

The swap is reversible and preserves inner products, so it is represented by a unitary operator. If ψ is an eigenvector of S with eigenvalue λ, then

$$S^2\psi = \lambda^2\psi = \psi,$$

and therefore

$$\lambda = \pm 1. \tag{4.50}$$

This conclusion identifies two exchange eigenspaces, but it does not say that every arbitrary two-particle wavefunction is already an eigenvector of S. Any such wavefunction can be separated into its two parts,[6]

$$\psi_\pm = \frac{1}{2}(I \pm S)\psi, \tag{4.51}$$

which obey

$$S\psi_\pm = \pm\psi_\pm. \tag{4.52}$$

Thus

$$\psi_+(x_2, x_1) = \psi_+(x_1, x_2), \tag{4.53}$$
$$\psi_-(x_2, x_1) = -\psi_-(x_1, x_2). \tag{4.54}$$

Now comes an additional principle, not a consequence of $S^2 = I$ alone. For identical particles of any fixed species, the allowed states lie entirely in one of these exchange sectors: exchanging a pair always returns the same wavefunction, or always

[6] This decomposition is an editorial algebraic clarification of the two eigenspaces implied by $S^2 = I$. The recorded argument states the two eigenvalues and then introduces the additional statistics principle.

returns its negative. The $+1$ sector is symmetric and the -1 sector is antisymmetric. The physical interpretation of these two possibilities as the two particle statistics is the next step, and is left for the following discussion.

CHAPTER 5

EXCHANGE SYMMETRY, SPIN, AND THE TWO MINUS SIGNS

5.1 Shell Filling and the Spin–Statistics Question

The shell-model review begins with two properties associated with electrons. An electron has spin one-half, with the two dimensionless spin projections

$$m_s = +\frac{1}{2}, \qquad m_s = -\frac{1}{2}, \tag{5.1}$$

and electrons obey the Pauli exclusion principle. These facts are related, but they are not the same statement.

The phrase "the same state" needs care. No two electrons may occupy the same complete one-particle state. Two electrons may nevertheless share one spatial orbital when their spin states differ. In the independent-electron approximation to helium, both electrons occupy the lowest spatial orbital, one with spin up and the other with spin down. More precisely, the alternatives "the first is up and the second is down" and "the first is down and the second is up" belong to an entangled two-electron spin state.

This is only the old shell-model approximation. It ignores the interaction between the electrons and is being used for its simple counting. For lithium, the first two electrons fill the lowest spatial orbital with opposite spin. The third cannot enter either of those complete states and must occupy a higher orbital.

The next step is the familiar orbital-angular-momentum diagram. The sectors with

$$\ell = 0, \qquad \ell = 1, \qquad \ell = 2 \tag{5.2}$$

are called S, P, and D states. A fixed value of ℓ has

$$2\ell + 1 \tag{5.3}$$

orbital states, so

$$\ell = 0 \Rightarrow 1, \qquad \ell = 1 \Rightarrow 3, \qquad \ell = 2 \Rightarrow 5. \tag{5.4}$$

The Coulomb problem has additional degeneracies: radial levels from different ℓ-sectors can have the same energy. For the shell-filling argument, the next shell contains one S orbital and three P orbitals. It therefore has

$$1 + 3 = 4 \tag{5.5}$$

spatial states. Spin doubles the count:

$$2(1 + 3) = 8. \tag{5.6}$$

Those eight complete one-electron states are filled in the simple progression from lithium through neon. Beyond the closed shell, sodium again has one outer electron, which is why this crude picture makes it resemble lithium. The chemistry is not the destination. Its point is that half-integer spin and exclusion appear together.

Photons provide the contrast. They have integer spin and do not obey an exclusion rule. A classical electromagnetic wave corresponds to many photons occupying the same mode. Bosons can accumulate in a common state, whereas identical fermions cannot occupy one complete state more than once. The question is whether those two pairings could have been otherwise: half-integer spin without exclusion, or integer spin with exclusion. In three spatial dimensions the spin–statistics relation says no. The familiar statement requires qualification in one and two spatial dimensions, and no proof of the field-theoretic result is attempted here.

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[a] **Question from the audience.** Is there a maximum number of bosonic quanta that can occupy one state?

Response. No. Bosons have no occupancy limit, and they tend to accumulate in the same state.

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[a]Lecture timestamp: 00:11:48.

5.2 Many-Particle Wavefunctions

The connection will be approached through two minus signs, one from exchange and one from rotation. Begin with exchange, and temporarily suppress spin.

Let x_i denote the position of particle i. The subscript is a particle label, not a Cartesian component; the three spatial coordinates are compressed into the single symbol x_i. A position basis for n particles is

$$|x_1, x_2, \ldots, x_n\rangle. \tag{5.7}$$

For an abstract state $|\Psi\rangle$, the coordinate wavefunction is

$$\psi(x_1, x_2, \ldots, x_n) = \langle x_1, x_2, \ldots, x_n \mid \Psi\rangle. \tag{5.8}$$

Its squared magnitude,

$$\left|\psi(x_1, x_2, \ldots, x_n)\right|^2, \tag{5.9}$$

is the joint probability density for finding particle 1 at x_1, particle 2 at x_2, and so on.

Now suppose the particles are identical. Imagine first that one electron is painted blue and another red. Then "blue at x_1, red at x_2" could be distinguished from "blue at x_2, red at x_1." Actual electrons carry no such colors or individual tags. Interchanging the bookkeeping labels cannot create a new physical arrangement.

An overall phase does not change probabilities or expectation values. For a state in a definite exchange sector, the coordinate wavefunction may therefore obey

$$\psi(x_2, x_1) = \eta\, \psi(x_1, x_2), \qquad |\eta| = 1. \tag{5.10}$$

Applying the same exchange twice restores the original arguments and gives

$$\eta^2 = 1, \tag{5.11}$$

so the two possible exchange eigenvalues are

$$\eta = +1 \qquad \text{or} \qquad \eta = -1. \tag{5.12}$$

1

For bosons the allowed wavefunctions are symmetric,

$$\psi_B(x_1, x_2) = \psi_B(x_2, x_1), \tag{5.13}$$

whereas for fermions they are antisymmetric,

$$\psi_F(x_1, x_2) = -\psi_F(x_2, x_1). \tag{5.14}$$

The minus sign is invisible when it multiplies the whole state, but it matters when amplitudes are added. That is how exchange symmetry acquires observable consequences.

5.3 Products, Sums, and Exclusion

Consider first two particles assigned to the same one-particle orbital ψ_0. Their product is

$$\psi_0(x_1)\psi_0(x_2). \tag{5.15}$$

Exchanging the particle coordinates gives

$$\psi_0(x_2)\psi_0(x_1) = \psi_0(x_1)\psi_0(x_2). \tag{5.16}$$

Wavefunction values are numbers, not operators, so the factors commute. The same-state product

[1] Editorial clarification: The double-exchange algebra determines the possible eigenvalues only after a state has been taken to have definite exchange behavior. The further restriction that the allowed states of a fixed species lie in one exchange sector is a physical rule; it does not follow from $\eta^2 = 1$ alone. The same qualification applies to the exclusion argument below: antisymmetry implies exclusion once the fermionic sector has been selected, but the algebra of exchange by itself does not select that sector.

is symmetric. It is an allowed bosonic form, but it cannot describe two identical fermions.

Next assign the particles to different one-particle orbitals:

$$\psi_0(x_1)\psi_1(x_2). \tag{5.17}$$

Exchange produces

$$\psi_0(x_2)\psi_1(x_1). \tag{5.18}$$

For distinct functions the exchanged product is generally neither the original expression nor its negative. The bare product therefore has neither required exchange symmetry. This does not forbid one particle from occupying ψ_0 while another occupies ψ_1. It means that the two-particle state must combine the two assignments.

The symmetric combination is

$$\psi_B^{(01)}(x_1, x_2) \propto \psi_0(x_1)\psi_1(x_2) + \psi_0(x_2)\psi_1(x_1), \tag{5.19}$$

and the antisymmetric combination is

$$\psi_F^{(01)}(x_1, x_2) \propto \psi_0(x_1)\psi_1(x_2) - \psi_0(x_2)\psi_1(x_1). \tag{5.20}$$

Proportionality signs are sufficient because no normalization is needed for the exchange argument. Both states contain one particle in each orbital; they differ only in how the two indistinguishable assignments are combined.

Now let the orbitals coincide. The bosonic combination becomes

$$\psi_B^{(00)}(x_1, x_2) \propto \psi_0(x_1)\psi_0(x_2) + \psi_0(x_2)\psi_0(x_1) \tag{5.21}$$

$$= 2\psi_0(x_1)\psi_0(x_2), \tag{5.22}$$

which is nonzero. The fermionic combination gives

$$\psi_F^{(00)}(x_1, x_2) \propto \psi_0(x_1)\psi_0(x_2) - \psi_0(x_2)\psi_0(x_1) \tag{5.23}$$

$$= 0. \tag{5.24}$$

With the antisymmetric exchange rule, exclusion follows: there is no nonzero antisymmetric state with both fermions in the same one-particle state.

For a many-particle state the rule applies to any transposition of two arguments. A transposition leaves a bosonic wavefunction unchanged and reverses the sign of a fermionic one. Repeating the same transposition is trivial in both cases. The surprising part is that a single exchange, though it changes no observable particle labels, changes the sign of a fermionic state.

5.4 Spin in the Complete Particle Label

Spin can now be restored with a small change. A one-particle amplitude depends on both position and a discrete spin label,

$$\psi(x, \sigma), \tag{5.25}$$

where $\sigma = +1$ or -1 labels the chosen σ_z result. The squared magnitude is the joint probability density for the stated position and spin label.

For many particles the wavefunction is

$$\psi(x_1, \sigma_1;\, x_2, \sigma_2;\, \ldots). \tag{5.26}$$

Exchange swaps everything belonging to the two particles, not only their positions:

$$\psi_B(x_1, \sigma_1;\, x_2, \sigma_2) = +\psi_B(x_2, \sigma_2;\, x_1, \sigma_1), \tag{5.27}$$
$$\psi_F(x_1, \sigma_1;\, x_2, \sigma_2) = -\psi_F(x_2, \sigma_2;\, x_1, \sigma_1). \tag{5.28}$$

This is the precise form of the opening shell-model statement. Two electrons may share one spatial orbital only when their complete states, including spin, are different.

---

[a] **Question from the audience.** Is exchange symmetry an instance of the permutation group?

Response. Yes. Exchanging particles is a permutation, and the plus or minus transformation law gives a representation of the permutation group.

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[a]Lecture timestamp: 00:34:33.

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[a] **Question from the audience.** Why is the two-particle construction a product rather than a sum of the one-particle wavefunctions?

Response. A sum describes one particle in a superposition. A product belongs to the two-particle tensor-product space and describes one particle assigned to each one-particle state.

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[a]Lecture timestamp: 00:35:12.

The full-label exchange is shown on the board below.[2]

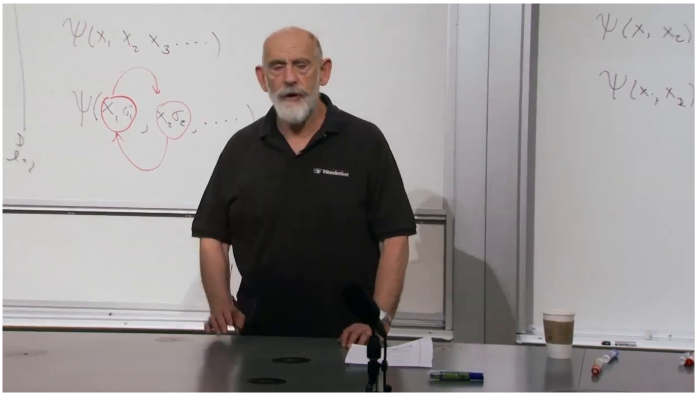

Figure 5.1: Exchanging two identical particles swaps their complete position–spin labels. A bosonic state is unchanged, while a fermionic state changes sign.
Lecture frame: 00:35:25.

These are different uses of linearity. For one particle,

$$\alpha\psi_0(x) + \beta\psi_1(x) \tag{5.29}$$

is a superposition of two alternatives. For two particles,

$$\psi_0(x_1)\psi_1(x_2) \tag{5.30}$$

is a product amplitude on a space with two particle coordinates. Symmetrization or antisymmetrization combines that product with its exchanged partner.

[2] The ordering of the position–spin labels is reconstructed from the legible blackboard frame and the surrounding explanation from 00:33:43 to 00:34:25.

[a] **Question from the audience.** Does exchanging particles produce an interference effect?

Response. Yes. The vanishing same-state fermionic amplitude can be understood as destructive interference between the two exchanged contributions.

[a]Lecture timestamp: 00:36:13.

[a] **Question from the audience.** Did Pauli derive exclusion using this same symmetry argument?

Response. Pauli began from an empirical fact and found a mathematical framework in which it could be expressed. He was not initially deriving exclusion from a more fundamental theory.

[a]Lecture timestamp: 00:37:13.

Antisymmetry supplies the mathematical framework in which the empirical exclusion rule is expressed.

[a] **Question from the audience.** Is particle-exchange interference connected with interference between alternative paths?

Response. Yes. One history has the two particles pass without crossing, while another has them cross into the exchanged destinations. The amplitudes add for bosons and can cancel for fermions.

[a]Lecture timestamp: 00:39:01.

Exchange supplies the first minus sign. Rotation supplies the second.

5.5 The Second Minus Sign: Rotation by 2π

A full rotation appears to do nothing. Every ordinary geometric direction returns to its starting point after an angle 2π. What must return, however, are the probabilities and expectation values. The state vector itself may return with an overall phase, just as it may under exchange.

---

[a] **Question from the audience.** Which axis is used for the rotation, and is the generator orbital angular momentum, spin, or both?

Response. One may choose an external axis; below it is taken to be the z-axis. The generator is total angular momentum, including both orbital and spin contributions. The spin itself need not point along the chosen axis.

---

[a]Lecture timestamp: 00:42:12.

[3]

Let $\psi(\theta)$ denote the state obtained as the rotation parameter advances through an angle θ. In the convention used here,

$$J_z\psi(\theta) = -i\frac{\partial}{\partial\theta}\psi(\theta). \tag{5.31}$$

For a state with definite J_z,

$$J_z\psi(\theta) = m\psi(\theta). \tag{5.32}$$

[3]Editorial clarification: The angular-momentum formulas in this discussion use $\hbar = 1$, so m is the dimensionless eigenvalue of J_z.

Combining the equations gives

$$-i\frac{\partial}{\partial\theta}\psi(\theta) = m\psi(\theta), \tag{5.33}$$

with solution

$$\psi(\theta) = e^{im\theta}\psi(0). \tag{5.34}$$

The allowed values of m are integers or half-integers. Therefore

$$\psi(2\pi) = \begin{cases} +\psi(0), & m \in \mathbb{Z}, \\ -\psi(0), & m \in \mathbb{Z} + \frac{1}{2}. \end{cases} \tag{5.35}$$

After another full turn,

$$\psi(4\pi) = \psi(0) \tag{5.36}$$

in either case. This is the second minus sign: a state in a half-integer angular-momentum multiplet changes sign under a 2π rotation.

The sign does not alter $|\psi|^2$, so it is not visible when it multiplies the entire state. Even so, a 2π rotation is not represented by the identity on a half-integer-spin state. The distinction becomes observable when the sign is made relative to another coherent amplitude.

5.6 The 2π and 4π Paths

The distinction between 2π and 4π can be made visible without further algebra. Hold a cup in one hand and turn it through one full revolution without releasing it. The cup returns to its original orientation, but the arm is twisted. Continue through a second revolution and the arm can be returned to its original position without reversing the cup's motion.

A related construction places a sphere inside a box and joins it to the box with strings. Turning the sphere through 2π leaves a tangle. In ordinary three-dimensional space, the tangle associated with a 4π turn can be unwound. The relevant object is not merely the final orientation of the sphere but the path followed through the space of rotations.

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[a] **Question from the audience.** If the sphere is turned twice about the same axis, can the strings really be untangled? Does that require another spatial dimension?

Response. The 4π path can be unwound in ordinary three-dimensional space. Time parametrizes the motion; it is not another spatial direction. A 2π path cannot be unwound in this way, while a 4π path can.

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[a]Lecture timestamp: 00:54:22.

The demonstrations distinguish a nontrivial 2π path from an unwindable 4π path without developing a formal theory of the rotation group.

5.7 Exchange and Rotation in the Belt Picture

The two minus signs now stand side by side. Exchanging two identical fermions changes the sign of their joint wavefunction. Rotating one half-integer-spin state through 2π also changes its sign.

A belt gives a geometrical relation between the two operations.

Arrange the belt so that its two lower ends have been exchanged. Viewed another way, the same belt carries a 2π twist. Performing the exchange together with the 2π rotation removes the twist and leaves an untwisted belt. The combined process can be continuously reduced to doing nothing.[4]

---

[a] **Question from the audience.** Is this only a statement about how the belt is embedded in space rather than a statement about particle topology?

Response. The belt can be read as a particle history in spacetime. With time running upward, a cross-section through it gives particle-like objects. Pairs are created, one object is rotated through 2π, and two identical objects are exchanged. In that history the rotation and exchange phases cancel.

---

[a]Lecture timestamp: 01:01:12.

The history makes the phase bookkeeping explicit. Begin with no particles. At two separated places, create a particle–antiparticle pair, leaving four particles at an intermediate time. Rotate one particle through 2π while exchanging two identical particles elsewhere in the configuration. The complete history can be deformed into the elementary history in which a pair is created and later annihilated. In the sign

[4]The exchange–rotation observation is attributed to David Finkelstein in the lecture. No specific publication is identified in the source record, so the name is retained here only as that attribution.

bookkeeping under discussion, that trivial history has no additional phase. If η_{rot} and η_{ex} denote the two signs, the spoken cancellation can be written compactly as[5]

$$\eta_{\text{rot}}\eta_{\text{ex}} = 1. \tag{5.37}$$

---

[a] **Question from the audience.** Can exchange be related to rotations by letting each of two particles move through π and then reorienting them?

Response. That construction may reflect related mathematics, but it uses two separate π rotations. The process being analyzed here contains one 2π rotation, so the two arguments are not identical.

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[a]Lecture timestamp: 01:03:34.

The argument depends on the possibility of creating and annihilating particle-like objects. It would not have this form in a theory whose particles were eternal. It is also not offered as the standard proof of the spin–statistics theorem. Its purpose is to display a topological reason that the exchange and rotation signs fit together.

5.8 Lumps, Mirror Images, and Annihilation

The construction can be repeated with localized field configurations. Suppose one or more fields take their vacuum values almost everywhere but have a

[5]This phase equation records the cancellation described from 01:06:50 to 01:07:18; the lecture presents the construction as suggestive rather than as a rigorous proof.

nonzero shape and structure in a small region. That localized configuration is a lump.

Now make a second configuration as the appropriate mirror image of the first. Call one the particle and the other the antiparticle. Bring them together. Corresponding pieces of the field configurations meet and undo one another. At an intermediate stage some parts of the lumps may already have disappeared while other parts remain. Continuing the deformation removes the remaining structure and returns the fields to the vacuum configuration. Two oppositely arranged twists in a belt give the same picture: push the twists together and they unwind.

The process can be run backward. Empty space can produce a lump and its mirror partner. One may create two such pairs, rotate one lump through 2π, exchange two identical lumps, and annihilate the pairs again. The same exchange–rotation cancellation then applies. This remains a heuristic extension of the construction, not a theorem about every localized excitation.

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[a] **Question from the audience.** Are there physical realizations of these lump-like field configurations?

Response. Yes. Examples include nuclei in Skyrmion-type descriptions, magnetic monopoles, and other particle-like excitations of fields.

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[a]Lecture timestamp: 01:11:30.

The mirrored-lump picture is what makes the generalization intelligible. The relevant feature is not that the object began as an elementary point particle, but that a particle-like excitation and its partner can be created, brought together, and annihilated continuously. That feature motivates a broader relation between rotational and exchange behavior.

---

[a] **Question from the audience.** Does spin one-half mean that the state accumulates only half the usual rotation phase?

Response. The phase in this convention is $e^{im\theta}$. For half-integer m, a 2π rotation gives -1, while a 4π rotation gives $+1$.

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[a] Lecture timestamp: 01:16:12.

5.9 A Global Sign and a Null Experiment

Can the minus sign from a 2π rotation be observed? Begin with an experiment in which it cannot. Trap an electron in a magnetic field, let its spin follow the field while the apparatus is turned slowly through 2π, and then release the electron into an interference apparatus. Compare this with the same procedure without the rotation.

The rotation changes the state by

$$\psi \longmapsto -\psi, \tag{5.38}$$

but the minus sign multiplies the whole state. Every probability is unchanged, so the final interference pattern retains no record of whether the preliminary rotation occurred. This is a null experiment: a common phase has nothing with which to interfere.

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[a] **Question from the audience.** Why must the magnetic-field rotation be performed slowly?

Response. The spin must continue to follow the rotating magnetic field. If the field is dragged too quickly, the spin will not remain locked to it.

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[a]Lecture timestamp: 01:20:01.

5.10 One Electron in Two Coherent Branches

The sign becomes observable when it is made relative. Send one electron through a beam splitter. Part of its wavefunction follows one route and part follows another:

$$\psi(x) = \psi_1(x) + \psi_2(x). \tag{5.39}$$

This is one electron, not two electrons. It is a coherent superposition of two alternatives. The distinction must be kept clear: the two packets are not two independent particles.

Guide the two packets into separate regions containing magnetic fields. Rotate the field in one region through 2π, leaving the other unchanged. For a half-integer-spin state,

$$\psi_1(x) + \psi_2(x) \longmapsto \psi_1(x) - \psi_2(x), \tag{5.40}$$

where the label of the rotated branch has been chosen as 2. The sign is now relative. When the two packets later overlap, addition has been changed into subtraction. A point of constructive interference can become a point of destructive interference, or even a node.

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[a] **Question from the audience.** What is being rotated if there is only one electron but two wavefunction branches?

Response. The electron is in a coherent superposition of two localized alternatives. One branch is rotated and the other is not. If the boxes are opened to learn which alternative occurred, that measurement destroys the interference.

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[a]Lecture timestamp: 01:23:46.

Opening the boxes would reveal only one electron, in one box or the other. Without that measurement, both amplitudes remain in the state. Repeating the experiment builds up the interference pattern one detection at a time. At a node produced by the relative minus sign, no electron is detected.

5.11 Magnetic Precession and Two-Slit Interference

A more practical schematic arrangement replaces the very slow box experiment. Split a coherent electron wave into two paths. In one path, let the electron pass through a magnetic region with its spin initially transverse to the field. The spin then precesses. By choosing the field strength and transit time, the spin in that branch can execute one full 2π turn, while the other branch undergoes no such turn.[6]

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[a] **Question from the audience.** How can the path of the single electron be determined?

Response. It cannot be determined without ruining the experiment. Any measurement that records which path the electron took destroys the interference.

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[a]Lecture timestamp: 01:27:05.

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[a] **Question from the audience.** What exactly is a branch of the wavefunction, and how does the splitter produce two branches?

Response. A branch is a wave packet localized along one route. The splitter turns the incoming wave into two spatially separated packets, and electric or magnetic fields can guide those packets along different paths.

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[a]Lecture timestamp: 01:27:50.

Apart from the magnetic region, this is the two-slit experiment. An incoming electron wave separates into two packets. The outgoing waves spread into a common detection region, where their amplitudes overlap. With no relative rotation the central amplitudes may add. A 2π spin rotation in one route reverses one amplitude and can make the same contributions cancel.

[6]The lecture states that a related spinor-phase experiment had been performed and describes its principle, but the source record does not identify a paper or a particular apparatus. The account here is therefore limited to the construction given in the lecture.

[a] **Question from the audience.** How can there be interference if the electron travels through distinct tubes or slits?

Response. The final detection does not reveal the path. The two outgoing wave amplitudes spread and overlap at the detector. Looking to see which path was taken would destroy the interference.

[a]Lecture timestamp: 01:30:46.

[a] **Question from the audience.** Must the tube or slit diameter be comparable to the electron wavelength?

Response. For a literal diffraction-based two-slit realization, the opening must be sufficiently small to produce the needed spreading. The apparatus being described is schematic rather than a practical design.

[a]Lecture timestamp: 01:33:22.

[a] **Question from the audience.** Would a corresponding 2π polarization rotation of photons leave the interference pattern unchanged?

Response. Yes. Photons have integer spin, so a full 2π rotation does not introduce the fermionic minus sign.

[a]Lecture timestamp: 01:33:44.

[a] **Question from the audience.** What is the slight departure of the practical experiment from the idealized version?

Response. The ideal construction rotates the spin adiabatically. In the practical description the electron moves through a magnetic region and its spin precesses more quickly, but the measured phase is still the phase associated with a 2π turn.

[a]Lecture timestamp: 01:34:12.

The slow construction isolates the idea by ensuring that the spin follows the field without additional motion. The precession construction reaches the same relative sign dynamically. In either version, the two routes must cease to be distinguishable before an interference pattern can appear.

[a] **Question from the audience.** Must the two branches be recombined after the splitter?

Response. Yes. The emerging wave packets must overlap, or spread onto a common detection region, so that their amplitudes can form an interference pattern.

[a]Lecture timestamp: 01:36:15.

[a] **Question from the audience.** Is this one-electron rotation experiment a consequence of Pauli exclusion?

Response. No. There is only one electron. The experiment measures the spinor phase under rotation, not the exclusion of two fermions from one state.

[a]Lecture timestamp: 01:38:12.

5.12 Exchange Interference and Relative Rotation

An exchange sign could be exposed by the same general method, although the proposed apparatus is more complicated. Begin with two electrons and split their joint wavefunction into two coherent branches. There are still only two electrons, not four: the pair follows one alternative or the other. Exchange the two electrons in one branch and leave them unexchanged in the other. After the branches are brought back together, the fermionic exchange sign appears as a relative phase and can change the interference.

The final proposal compares the two minus signs directly. One branch acquires the fermionic exchange phase, while the other acquires the corresponding 2π-rotation phase. Both phases are -1, so the relative effect cancels.[7] The lecture presents this as a proposed thought experiment and says that, to the speaker's knowledge, it had not been carried out.

[7] The recording between 01:40:26 and 01:41:15 clearly gives an exchange phase, a 2π-rotation phase, and their cancellation, but the detailed assignment of the rotation is not recoverable from the available speech.

[a] **Question from the audience.** Would observing an exchange phase challenge the indistinguishability of the two electrons?

Response. No. Quantum indistinguishability is subtler than classical sameness. It permits the state vector to change sign under exchange even though no observable label distinguishes the particles.

[a] Lecture timestamp: 01:41:34.

There is one last qualification. Rotating an entire isolated world through 2π would give only a common phase, and no internal experiment could reveal it. The detectable experiment rotates one coherent branch relative to another. What is observed is therefore not an absolute sign but the interference produced by a relative sign.

CHAPTER 6

OSCILLATORS, OCCUPATION NUMBERS, AND THE FIRST QUANTUM FIELD

Quantum field theory is our framework for essentially all known nongravitational physics. That claim should not be confused with a claim that every problem can be solved. A hydrogen atom is manageable; by the time one reaches an atom such as boron, the calculation is already much harder, and a human being is far beyond any realistic field-theory computation. The framework may be fundamental while its equations remain overwhelmingly difficult.

We will therefore enter in the most elementary way possible. The subject is sometimes called *second quantization*, but the name will mean little until we have built the machinery. The shortest route begins not with a relativistic field, but with the harmonic oscillator. Springs are not the point. What matters is the algebra of raising and lowering operators. Once that algebra is repeated for many modes, a raising operator will cease to mean merely “raise an oscillator by one level” and will literally mean “create a particle.”

6.1 The Oscillator as an Algebra

For one oscillator let $a^\dagger$ raise the excitation and let a lower it. The notation a_+ and a_- is equivalent, but the dagger notation anticipates the language used in field theory. The bottom state obeys

$$a|0\rangle = 0, \tag{6.1}$$

and the number operator is

$$N = a^\dagger a, \qquad N|n\rangle = n|n\rangle. \tag{6.2}$$

The two ladder operators are Hermitian conjugates, and their essential commutator is

$$[a, a^\dagger] = 1. \tag{6.3}$$

This small algebra is almost the whole oscillator.

One can recover a and $a^\dagger$ from suitable linear combinations of position and momentum. The precise convention depends on the normalization of x, p, m, and ω; the useful invariant statement here is that $a^\dagger a$ reconstructs the quadratic oscillator Hamiltonian apart from its additive ground-state energy. Thus

$$H = \hbar\omega\left(N + \frac{1}{2}\right). \tag{6.4}$$

For the present discussion we drop the constant term and work with

$$H = \hbar\omega N. \tag{6.5}$$

The frequency tells us the price of one excitation. Raising n by one costs $\hbar\omega$; lowering n by one releases the same amount.

6.2 Many Independent Oscillators

Now take many oscillators and label them by i. The label is only an index, not the imaginary unit. There may be finitely many modes or infinitely many. A collection of independent springs gives one picture; the normal modes of an ideal violin string give another. In the field-theory applications the infinite collection is the important case.

Independence has an operator meaning. Operators belonging to genuinely independent subsystems commute. If they did not, the corresponding quantities could not be measured independently, and the systems would not be independent in the first place. We therefore have

$$[a_i, a_j] = 0, \tag{6.6}$$

$$[a_i^\dagger, a_j^\dagger] = 0, \tag{6.7}$$

$$[a_i, a_j^\dagger] = \delta_{ij}. \tag{6.8}$$

For $i \neq j$, everything belonging to the two modes commutes. For $i = j$, the last line reduces to Equation (6.3).

Each oscillator has its own number operator and, in general, its own frequency:

$$N_i = a_i^\dagger a_i, \qquad H = \sum_i \hbar\omega_i N_i. \tag{6.9}$$

A simultaneous eigenstate is labeled by the complete list

$$|n_1, n_2, n_3, \ldots\rangle, \qquad n_i = 0, 1, 2, \ldots. \tag{6.10}$$

The integer n_i is called the *occupation number* of the i-th oscillator. At this stage that is only terminology. Soon it will count particles occupying a one-particle state, and the name will become literal.

The state with every occupation number equal to zero is

$$|0\rangle \equiv |0, 0, 0, \ldots\rangle, \qquad a_i|0\rangle = 0 \quad \text{for every } i. \tag{6.11}$$

---

[a] **Question from the audience.** Does an infinite family of oscillators correspond to the continuum of modes of a system such as a violin string?

Response. Yes, in the idealized string. A real string is made of atoms, so sufficiently short wavelengths eventually encounter a microscopic cutoff. The continuum model nevertheless motivates the infinite collection of modes used in field theory.

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[a] Lecture timestamp: 00:15:27.

6.3 Why the Ladder Coefficients Are Square Roots

The action of $a^\dagger$ must raise an eigenstate of N to the next eigenstate, but normalization leaves room for a coefficient:

$$a^\dagger|n\rangle = c_n|n+1\rangle. \tag{6.12}$$

Choose the phase convention in which c_n is real. Taking the Hermitian conjugate gives

$$\langle n|a = c_n\langle n+1|. \tag{6.13}$$

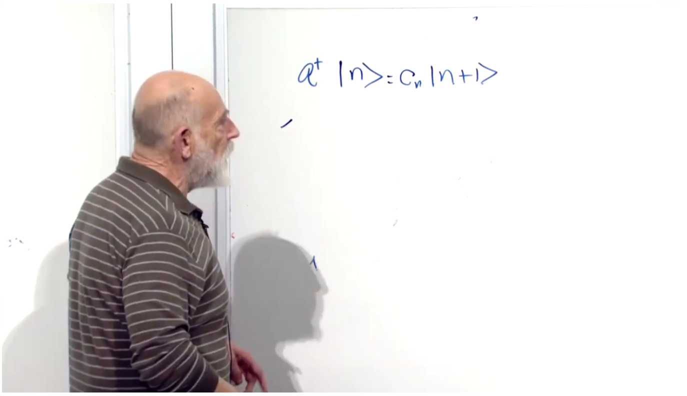

Figure 6.1: The ladder coefficient is introduced before its normalization is derived. The board equation is reproduced in Equation (6.12).

Lecture frame: 00:19:38.

Multiplying the two relations and using normalized basis states,

$$c_n^2 = \langle n|aa^\dagger|n\rangle \tag{6.14}$$
$$= \langle n|(a^\dagger a + 1)|n\rangle \tag{6.15}$$
$$= \langle n|(N + 1)|n\rangle \tag{6.16}$$
$$= n + 1. \tag{6.17}$$

Consequently,

$$a^\dagger|n\rangle = \sqrt{n+1}\,|n+1\rangle. \tag{6.18}$$

[a] **Question from the audience.** Why may one set $\langle n+1|n+1\rangle = 1$ before finding c_n?

Response. The eigenvectors are chosen as a normalized basis from the outset. Their normalization does not determine the normalization of $a^\dagger|n\rangle$; that remaining information is precisely the coefficient c_n.

[a] Lecture timestamp: 00:22:35.

The lowering rule follows in the same way:

$$a|n\rangle = \sqrt{n}\,|n-1\rangle. \tag{6.19}$$

At $n = 0$, the coefficient vanishes automatically, so the rule includes $a|0\rangle = 0$. For many modes, the operator changes only the occupation number in its own slot:

$$a_i^\dagger|n_1,\ldots,n_i,\ldots\rangle = \sqrt{n_i+1}\,|n_1,\ldots,n_i+1,\ldots\rangle, \tag{6.20}$$

$$a_i|n_1,\ldots,n_i,\ldots\rangle = \sqrt{n_i}\,|n_1,\ldots,n_i-1,\ldots\rangle. \tag{6.21}$$

Nothing happens to the other entries.

6.4 A Wavefunction Is Not a Quantum Field

Before defining a field, we must separate it from the familiar wavefunction. Three contrasts organize the transition.

First, a one-particle wavefunction

$$\psi(x) = \langle x|\psi\rangle \tag{6.22}$$

is a representation of a state vector. It is not itself an observable. We measure position, momentum, angular momentum, and other operators; the wavefunction supplies the amplitudes from which the probabilities of those outcomes are calculated.

Second, a many-particle wavefunction depends on the positions of all the particles. For two particles whose position vectors are temporarily called x and y,

$$\psi(x, y) = \langle x, y|\psi\rangle. \tag{6.23}$$

Here x and y label two different particles; they are not the Cartesian x- and y-components of one position vector. With N particles the state is represented by

$$\psi(x_1, x_2, \ldots, x_N). \tag{6.24}$$

Third, this wavefunction describes a fixed number of particles. A fifteen-particle wavefunction has fifteen position arguments and belongs to a fifteen-particle Hilbert space. It does not by itself describe a process in which the number changes from fifteen to fourteen or sixteen.

A quantum field will reverse these features. It is an operator rather than a state, it depends on one spatial point x, and it acts on a state space containing every particle-number sector. The phrase "one point" means one freely variable position vector. It does not mean that x has been assigned one numerical value.

There is one qualification that will matter later. The field $\Psi(x)$ introduced below is not Hermitian, so it is not itself an observable in the strict sense. Its Hermitian combinations are observables. The provisional statement that a field is measurable is therefore sharpened, not discarded: fields are operator-valued functions, and their Hermitian combinations represent measurable quantities.

---

[a] **Question from the audience.** If the field is a function of one position, may that position vary? And what happened to the y in the two-particle wavefunction?

Response. The point x varies over all space. "One position" means that the operator has one spatial argument, not that the argument is fixed. In $\psi(x, y)$, the symbols x and y name the positions of two particles; equivalently they are x_1 and x_2. They are not two Cartesian components of the field argument.

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[a] Lecture timestamp: 00:36:52.

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[a] **Question from the audience.** How are the fixed-particle wavefunction and field descriptions related, and are they describing different kinds of particles?

Response. The connection is the central construction still to come. The particles are of the same kind. For this lecture they are bosons; fermions require a different algebra.

---

[a] Lecture timestamp: 00:38:15.

6.5 One Particle in a Box

Return temporarily to ordinary one-particle quantum mechanics. Put a particle in a one-dimensional box. With only one particle, the distinction between boson and fermion is irrelevant: there is no second identical particle with which to test exchange or exclusion.

Let

$$\psi_1(x),\ \psi_2(x),\ \psi_3(x),\ldots \tag{6.25}$$

be the energy eigenfunctions. The lowest mode is the smoothest wave that vanishes at the walls. The next mode has one interior node, and higher-energy modes oscillate more rapidly. In the ideal infinite square well of width L, a convenient normalized basis is

$$\psi_i(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{i\pi x}{L}\right), \qquad 0 < x < L. \tag{6.26}$$

The explicit sine formula is a standard realization of the qualitative board sketch; the argument below requires only a complete orthonormal set of one-particle states.

The same structure can be redrawn without the occlusion of the live board:

The index i labels a one-particle state, not a particle and not the number of nodes. For the box basis, i happens to be one more than the number of interior nodes. Each state has an energy ε_i. Following the lecture's notation we may write this energy as $\hbar\omega_i$, or simply ω_i after setting $\hbar = 1$.

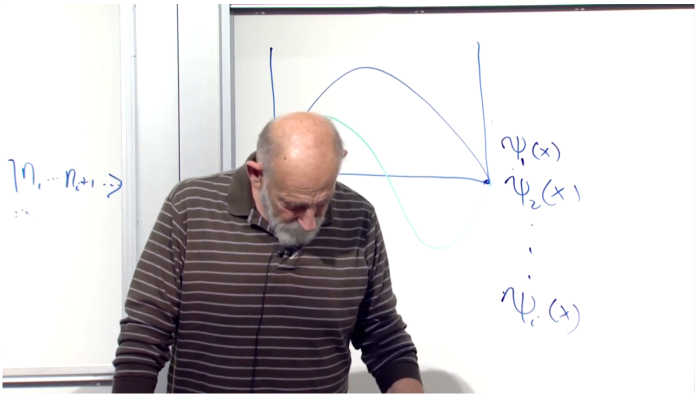

Figure 6.2: One-particle box modes ordered by increasing energy. The first mode has no interior node, the second has one, and the omitted higher modes oscillate progressively faster.

Lecture frame: 00:43:10.

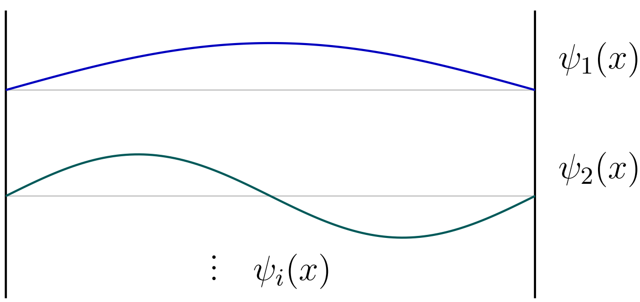

Figure 6.3: A clean reconstruction of the qualitative box-mode sketch. The vertical displacement separates the eigenfunctions for readability and is not an additional physical coordinate.

[a] **Question from the audience.** Was the earlier independence assumption exact, and could the index i label the possible states of an electron in an atom?

Response. For the independent-mode construction the commutation between different modes is exact. The index may label one-particle atomic states. If the particles interact, however, the simple independent-mode Hamiltonian is no longer the whole story.

[a]Lecture timestamp: 00:39:01.

[a] **Question from the audience.** Is the particle in the box a boson?

Response. It may be, but with one particle the statistics do not matter. Exclusion and exchange become meaningful only when there is more than one identical particle.

[a]Lecture timestamp: 00:40:34.

6.6 From Box Modes to Fock Space

Now allow any number of identical bosons in the box. Some number n_1 may occupy ψ_1, some number n_2 may occupy ψ_2, and so on. A basis state is therefore

$$|n_1, n_2, n_3, \ldots\rangle. \tag{6.27}$$

This is exactly the notation already used for many oscillators. The identity is structural: associate one oscillator with every one-particle state, and identify the oscillator excitation number with the number of particles occupying that state.

Bosons are essential here because every n_i may be any nonnegative integer. For fermions each one-particle state can be empty or singly occupied, never doubly occupied, so the ordinary oscillator algebra cannot be used unchanged.

The operators can now be defined by their action on the occupation-number basis. The operator $a_i^\dagger$ creates one particle in one-particle state i; a_i removes one:

$$a_i^\dagger |n_1, \ldots, n_i, \ldots\rangle = \sqrt{n_i + 1}\, |n_1, \ldots, n_i + 1, \ldots\rangle, \tag{6.28}$$

$$a_i |n_1, \ldots, n_i, \ldots\rangle = \sqrt{n_i}\, |n_1, \ldots, n_i - 1, \ldots\rangle. \tag{6.29}$$

This is not an analogy after the fact. It is the definition of the bosonic creation and annihilation operators on this basis. Because the basis spans the space, the definition determines their action on every state by linearity.

---

[a] **Question from the audience.** Are n_1, n_2, n_3 the quantum states?

Response. The labels $1, 2, 3, \ldots$ identify the one-particle states. The symbols $n_1, n_2, n_3, \ldots$ are the numbers of particles occupying those states. Keeping the state label and its occupation number distinct is the main notational burden.

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[a]Lecture timestamp: 00:47:11.

6.6.1 What the square root does

The factor $\sqrt{n_i + 1}$ does not represent an additional particle or a separate physical process. It gives the operator the correct normalization and algebra. In one mode,

$$aa^\dagger|n\rangle = a\Big(\sqrt{n+1}\,|n+1\rangle\Big) \tag{6.30}$$
$$= (n+1)|n\rangle, \tag{6.31}$$

while

$$a^\dagger a|n\rangle = n|n\rangle. \tag{6.32}$$

Their difference is therefore $|n\rangle$, as required by $[a, a^\dagger] = 1$. The two ladder operators are, very roughly, each of order $\sqrt{n}$, since their product gives the number operator of order n.

---

[a] **Question from the audience.** What tangible physical meaning belongs to the square-root factor?

Response. It is primarily mathematical normalization. It ensures that the occupation states are normalized, that $a^\dagger a$ has eigenvalue n, and that the ladder operators obey the oscillator commutator. Sometimes the mathematics is the meaning one needs.

---

[a] Lecture timestamp: 00:47:52.

[a] **Question from the audience.** Have the creation and annihilation operators been defined only through their properties and their action on the basis?

Response. Yes. Definitions need not arise from a more primitive picture. Their value is tested by their consequences. Defining the action on a complete basis defines the operators everywhere, and these definitions reproduce the desired commutation relations.

[a]Lecture timestamp: 00:52:25.

[a] **Question from the audience.** Why does a_i make a particle disappear rather than merely move it to a different state?

Response. Because a_i is defined to lower the occupation of state i. The useful question is not why a definition is true, but why it is useful. It is useful because emission and absorption require operators that change particle number: a photon is created when radiation is emitted and annihilated when it is absorbed.

[a]Lecture timestamp: 00:53:22.

The space spanned by all occupation-number states is *Fock space*. It contains the vacuum, every one-particle state, every two-particle state, and so on without a fixed upper bound:

$$\mathcal{F} = \mathcal{H}_0 \oplus \mathcal{H}_1 \oplus \mathcal{H}_2 \oplus \cdots . \tag{6.33}$$

The direct-sum notation makes explicit what the ordinary N-particle wavefunction did not: all particle-number sectors belong to one state space.

[a] **Question from the audience.** Does defining the operators on the occupation-number eigenvectors determine them on arbitrary vectors, and do the oscillator commutators then follow?

Response. Exactly. The occupation vectors form a basis, so linearity extends the definition to all of Fock space. Acting on that basis verifies the bosonic commutation relations. Each one-particle state is represented by one oscillator, and its oscillator occupation is the number of particles in that state.

[a]Lecture timestamp: 00:54:56.

6.6.2 The vacuum and the free Hamiltonian

The Fock vacuum is the state with no particles in any one-particle state:

$$|0\rangle = |0, 0, 0, \ldots\rangle, \qquad a_i|0\rangle = 0. \tag{6.34}$$

The word *ground state* can be misleading here. The lowest *one-particle* state is the mode $i = 1$; the Fock vacuum has no particle even in that mode. If a particle occupies the lowest mode, a_1 can remove it and return the system to the vacuum.

[a] **Question from the audience.** What is the vacuum in occupation-number language? Is the lowest one-particle state occupied?

Response. The vacuum has every occupation number equal to zero and is annihilated by every a_i. The lowest one-particle mode is unoccupied in the vacuum. If it were occupied, its annihilation operator could remove that particle.

[a]Lecture timestamp: 00:56:19.

For noninteracting particles, the total energy is additive. If one particle in state i has energy $\hbar\omega_i$, then

$$H = \sum_i \hbar\omega_i N_i \tag{6.35}$$

$$= \sum_i \hbar\omega_i a_i^\dagger a_i. \tag{6.36}$$

There are two equivalent readings. It is the sum of oscillator energies, one oscillator per one-particle state. It is also the sum over one-particle energies, each multiplied by the number of particles carrying that energy.

In field-theory notation $a_i^\dagger$ is usually called the creation operator and a_i the annihilation operator. The older a_+ and a_- notation says the same thing.

6.7 Clarifications Before the Field Appears

Several questions after the break separate the new oscillator bookkeeping from an ordinary particle moving in an oscillator potential.

[a] **Question from the audience.** The energy levels of a particle in a box are not equally spaced, so where is the harmonic oscillator?

Response. The one-particle energies ω_i need not be equally spaced. For each fixed mode i, however, the energies of zero, one, two, and three noninteracting particles in that mode are $0, \omega_i, 2\omega_i, 3\omega_i$. That occupation ladder is the oscillator. It is distinct from putting one particle in a harmonic-oscillator potential.

[a] Lecture timestamp: 01:06:09.

[a] **Question from the audience.** Is this construction connected with Fourier decomposition?

Response. Deeply. If the one-particle basis consists of momentum modes, the mode functions are sines, cosines, or complex exponentials, and the a_i are quantum analogues of Fourier coefficients. A general basis need not be trigonometric, but the expansion logic is the same.

[a] Lecture timestamp: 01:08:38.

[a] **Question from the audience.** Are all particles being assigned the same energy, and are interactions included?

Response. Particles occupying the same one-particle state have the same energy, while particles in different states generally do not. For now the particles are free: they do not interact. Interactions would add potential-energy terms, so the Hamiltonian would no longer be only the additive sum in Equation (6.36).

[a] Lecture timestamp: 01:09:07.

[a] **Question from the audience.** What became of the half quantum of zero-point energy in every oscillator?

Response. It is an additive constant in the Hamiltonian. A constant commutes with every operator, so it does not alter Heisenberg equations of motion, and ordinary non-gravitational measurements depend on energy differences. It may matter for special purposes, but it does not affect the construction being developed here.

[a] Lecture timestamp: 01:10:26.

The free theory is already a bookkeeping device of unusual power. It keeps track not only of many particles but of particles that change state, appear, disappear, or exist in superpositions of different total number. The operators are more than bookkeeping symbols because suitable Hermitian combinations can be measured, but bookkeeping is a useful first intuition.

[a] **Question from the audience.** Where did field theory and quantum field theory come from historically?

Response. A brief historical route begins with Faraday's physical picture of electric and magnetic fields and Maxwell's equations for classical fields. Planck introduced energy quanta while studying thermal radiation; Einstein identified the quantization with the electromagnetic field itself and with light quanta. Heisenberg, Pauli, and especially Dirac then helped put the modern operator framework together around the end of the 1920s. The ultraviolet catastrophe and the quantum description of radiation were among the decisive early problems. This is a schematic history rather than a complete assignment of priority.

[a] Lecture timestamp: 01:14:03.

The historical thread also explains why oscillators keep returning. Classical radiation in a cavity decomposes into normal modes. Quantizing those modes turns their excitation numbers into photon numbers. The mathematical quanta occupying the modes become physical particles.

6.8 The Bosonic Field Operator

Let $\{\psi_i(x)\}$ be a complete orthonormal basis of one-particle wavefunctions. Define

$$\boxed{\Psi(x) = \sum_i a_i \, \psi_i(x)} \tag{6.37}$$

and its Hermitian conjugate

$$\boxed{\Psi^\dagger(x) = \sum_i a_i^\dagger \, \psi_i^*(x)}. \tag{6.38}$$

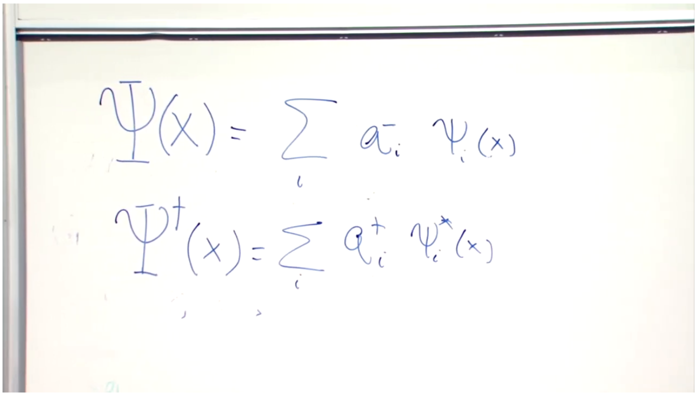

Figure 6.4: The completed mode expansions of the annihilation field and its Hermitian conjugate. Equations (6.37) and (6.38) preserve the same operator and complex-conjugation structure in typeset form.
Lecture frame: 01:25:15.

For any fixed position x, each $\psi_i(x)$ is an ordinary complex number. All operator content resides in a_i and $a_i^\dagger$. As x varies, we obtain a different operator at every point in space. This is the precise sense in which Ψ is an operator-valued function.

If the basis functions are plane waves, Equations (6.37) and (6.38) are quantum Fourier expansions: the classical Fourier coefficients have become operators. In a box, the basis may instead be the sine functions in Equation (6.26). Any complete orthonormal one-particle basis works.

Neither Ψ nor $\Psi^\dagger$ is Hermitian. Observable fields can be built from Hermitian combinations such as

$$\Psi(x) + \Psi^\dagger(x), \qquad i\Big(\Psi^\dagger(x) - \Psi(x)\Big), \tag{6.39}$$

and observable densities will later be built from products such as $\Psi^\dagger(x)\Psi(x)$. The present task is first to understand what the two basic fields do.

---

[a] **Question from the audience.** How can the left side be an operator when the mode functions on the right are ordinary wavefunctions?

Response. The functions $\psi_i(x)$ are numbers once x is chosen. The factors a_i are operators. A number times an operator is an operator, and the sum is therefore an operator. Varying x gives the whole family of operators.

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[a]Lecture timestamp: 01:23:14.

6.9 What the Field Does

The annihilation field kills the vacuum immediately:

$$\Psi(x)|0\rangle = \sum_i \psi_i(x)a_i|0\rangle = 0. \tag{6.40}$$

The creation field does something much more informative.

Let $|i\rangle$ denote the one-particle state with wavefunction

$$\langle x|i\rangle = \psi_i(x). \tag{6.41}$$

Completeness of the one-particle basis gives

$$\sum_i |i\rangle\langle i| = 1. \tag{6.42}$$

Apply this identity to a position eigenstate:

$$|x\rangle = \sum_i |i\rangle\langle i|x\rangle \tag{6.43}$$

$$= \sum_i \psi_i^*(x)|i\rangle. \tag{6.44}$$

In Fock language a one-particle mode state is

$$|i\rangle = a_i^\dagger |0\rangle. \tag{6.45}$$

Substituting this in Equation (6.44),

$$|x\rangle = \sum_i \psi_i^*(x) a_i^\dagger |0\rangle \tag{6.46}$$

$$= \Psi^\dagger(x)|0\rangle. \tag{6.47}$$

Thus $\Psi^\dagger(x)$ creates a particle localized at x, while $\Psi(x)$ removes one there. A position eigenstate is delta-function localized and is a generalized, non-normalizable state; a physical localized particle is obtained by smearing the field with a normalizable wavepacket.[1]

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[a] **Question from the audience.** Where is the information specifying what kind of particle is created? Is the state at x delta-function localized?

Response. A separate field is assigned to each particle species, so the identity of the particle is carried by which field is used. The argument x specifies where it is created. In the ideal position basis, the created state has delta-function localization.

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[a] Lecture timestamp: 01:36:26.

Examples of bosonic species include photons, gravitons, Higgs bosons, and gluons, though realistic fields

[1] Editorial clarification: The lecture speaks in the ideal position-eigenstate language. The wavepacket qualification records the standard mathematical meaning without changing the argument.

also carry internal indices. Even photons require polarization labels. The simplified scalar field notation suppresses those extra labels so that the creation principle remains visible.

6.9.1 The vacuum is not the zero vector

The equation

$$\Psi(x)|0\rangle = 0 \tag{6.48}$$

contains two very different objects denoted by zero. The ket $|0\rangle$ is the vacuum: a normalized physical state with

$$\langle 0|0\rangle = 1. \tag{6.49}$$

The result on the right is the zero vector. Its norm is zero; it cannot be normalized and does not represent a physical state.

The difference becomes sharp under addition. Adding the zero vector to a one-particle state changes nothing. Adding the vacuum to a one-particle state gives a genuine superposition of different particle-number sectors. With normalized coefficients,

$$|\chi\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle + |1\rangle\big) \tag{6.50}$$

has equal probabilities for zero and one particle. It is not the same as $|1\rangle$.

[a] **Question from the audience.** If no particle is present at x, does $\Psi(x)$ simply do nothing? What does it mean for the result to be zero?

Response. It does not return the original state. It returns the zero vector. The vacuum is a normalized state describing empty space; the zero vector has zero length and describes no physical state. Any operator acting on the zero vector returns the zero vector.

[a]Lecture timestamp: 01:37:23.

[a] **Question from the audience.** What happens if the vacuum rather than the zero vector is added to a one-particle state?

Response. One obtains a superposition with amplitudes for different particle numbers. By contrast, adding the zero vector leaves the one-particle state unchanged. The symbol in $|0\rangle$ names zero occupation; it does not mean zero vector.

[a]Lecture timestamp: 01:39:00.

[a] **Question from the audience.** Does the order of a mode function and its creation or annihilation operator matter in the field expansion? Are the mode functions complex?

Response. The mode function is an ordinary number and therefore commutes with every operator, so the order does not matter. In general the mode functions may be complex; whether a chosen basis is real or complex depends on the one-particle Schrödinger problem.

[a]Lecture timestamp: 01:41:56.

6.9.2 Two particles and the emergence of bosonic symmetry

To create one particle at x and another at y, act twice:

$$|x, y\rangle = \Psi^\dagger(y)\Psi^\dagger(x)|0\rangle. \tag{6.51}$$

The points may coincide or differ. Since the underlying creation operators commute,

$$[\Psi^\dagger(x), \Psi^\dagger(y)] = \sum_{i,j} \psi_i^*(x)\psi_j^*(y)[a_i^\dagger, a_j^\dagger] \tag{6.52}$$

$$= 0. \tag{6.53}$$

It follows that

$$\Psi^\dagger(y)\Psi^\dagger(x)|0\rangle = \Psi^\dagger(x)\Psi^\dagger(y)|0\rangle. \tag{6.54}$$

Exchanging the order of the two particles changes nothing. This is precisely bosonic exchange symmetry. We could therefore have started from the oscillator algebra and *derived* that the particles constructed from commuting creation operators are bosons. Fermions will require anticommuting operators instead.

---

[a] **Question from the audience.** How is a second particle added to the one-particle state?

Response. Apply another creation field. To add it at y, act with $\Psi^\dagger(y)$ to the left of $\Psi^\dagger(x)$. The result is the two-particle state in Equation (6.51).

---

[a]Lecture timestamp: 01:42:37.

[a] **Question from the audience.** Is the commutativity of the two creation fields fully equivalent to symmetrizing the two-particle state?

Response. Yes. It makes the state with one particle at x and one at y identical to the state obtained after exchanging the labels. This is the defining exchange property of identical bosons.

[a]Lecture timestamp: 01:46:04.

[a] **Question from the audience.** Does this mean the particles at x and y must be the same kind of boson?

Response. Yes. The arguments x and y are positions, while one field describes one species. Different species, and additional labels such as the two photon polarizations, require distinct field components.

[a]Lecture timestamp: 01:46:32.

6.10 Why Variable Particle Number Matters

If a problem contained exactly thirty-seven particles forever, the field language would be optional bookkeeping. Its real strength appears when particles can be emitted and absorbed, and especially when particle number itself is a quantum variable. Fock space allows states of the form

$$|\Phi\rangle = c_0|0\rangle + c_1|1\rangle + c_2|2\rangle + \cdots, \qquad \sum_{n=0}^{\infty} |c_n|^2 = 1. \tag{6.55}$$

A laser field is an important example: it is not generally a state with one sharp photon number, but a superposition of many number sectors.

The bridge between particles and fields is now explicit. The mode operators count and change particles in energy eigenstates; the field combines those operators with a complete set of wavefunctions so that particles can be created or removed at specified positions. The algebra of harmonic oscillators has become the elementary language of a bosonic quantum field.

CHAPTER 7

FROM PARTICLES TO A SIMPLE QUANTUM FIELD

Let us build the simple quantum field once more, but from a slightly different angle. There are two roads into quantum field theory. We may begin with particles and discover why a field is the natural language for them, or begin with a wave field and discover why its excitations come in quanta. Both roads are useful. Here we stay on the first one.

Ordinary quantum mechanics can describe one particle, two particles, or any other fixed number. Nature does not respect that restriction. The number of photons in a room changes whenever the lights emit or absorb radiation. Unstable particles decay. In beta decay, for example,

$$n \longrightarrow p + e^- + \bar{\nu}_e, \tag{7.1}$$

so the populations of several particle species change at once. We therefore want one quantum mechanics large enough to contain every possible occupation number and, eventually, processes that move among them.

For the moment we simplify severely. There is one species of nonrelativistic particle. We shall first enlarge its state space so that it can hold any particle number, and only afterward discuss more species, relativity, scattering, and decay. None of the mathematics is especially hard, but the change of viewpoint is abstract. It is worth taking the steps slowly enough that the notation does not hide the idea.

7.1 The One-Particle Starting Point

Begin with a single particle in a state $|\psi\rangle$. Its position-space wavefunction is

$$\psi(x) = \langle x|\psi\rangle. \tag{7.2}$$

This lower-case ψ is an ordinary complex-valued function. It is not yet a quantum field and it is not an operator on a many-particle state space.

The Born rule says that

$$\rho_\psi(x) = |\psi(x)|^2 = \psi^*(x)\psi(x) \tag{7.3}$$

is the probability density for finding the particle at x. For a normalized state,

$$\int dx\, \psi^*(x)\psi(x) = 1. \tag{7.4}$$

The right-hand side is the total probability. Since this problem contains exactly one particle, it also happens to equal the number of particles. That coincidence will become a useful clue, although it is trivial here.

[a] **Question from the audience.** Why must the probability density have the form $\psi^*(x)\psi(x)$?

Response. That is the Born rule, one of the postulates of quantum mechanics. The amplitude for the position outcome x is $\langle x|\psi\rangle$; the probability density is its absolute square. It is not derived from the field construction used later.

[a] Lecture timestamp: 00:09:13.

7.2 Modes, Orthogonality, and Completeness

There are many useful bases for the one-particle Hilbert space. Position eigenstates form one basis. The eigenstates of any Hermitian operator form another, with the usual qualifications for degeneracies and continuous spectra. Since energy will soon matter, choose a complete orthonormal set of one-particle energy eigenstates

$$|1\rangle,\ |2\rangle, \ldots, \qquad \psi_i(x) = \langle x|i\rangle. \tag{7.5}$$

A particle in a box makes the index i concrete. The ground-state wavefunction has no interior node, the next state has one, and the higher states oscillate more rapidly. They are essentially sine waves for an ideal infinite square well.

The same mode structure can be separated from the live-board occlusion:

Orthonormality is

$$\int dx\, \psi_i^*(x)\psi_j(x) = \delta_{ij}. \tag{7.6}$$

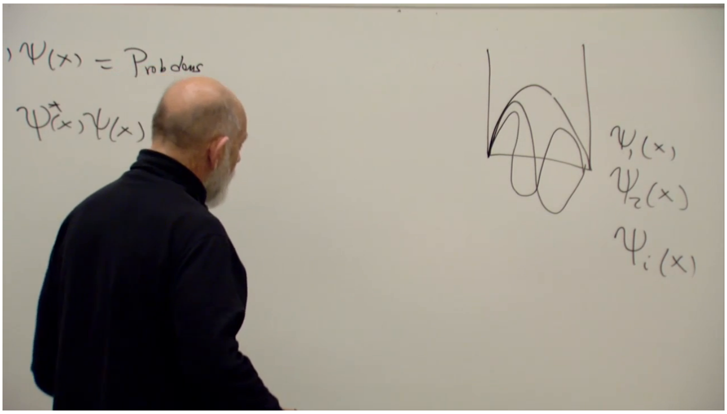

Figure 7.1: The box modes on the lecture board. The remnants at left are the one-particle Born density and normalization; the well at right shows $\psi_1(x)$, $\psi_2(x)$, and the general label $\psi_i(x)$.

Lecture frame: 00:08:25.

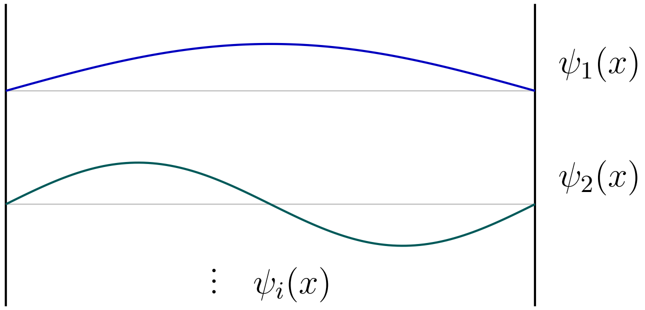

Figure 7.2: A clean reconstruction of the qualitative box sketch. The vertical offsets merely separate the two wavefunctions for display.

If $i = j$, the state is normalized and the integral is one. If $i \neq j$, the two states are orthogonal and the integral vanishes.

Completeness is the equally important statement

$$\sum_i |i\rangle\langle i| = \mathbf{1}. \tag{7.7}$$

Each $|i\rangle\langle i|$ projects onto one basis direction; the sum over the complete basis is the identity. Acting on an arbitrary vector merely decomposes that vector into its components and puts it back together.

Now sandwich Equation (7.7) between two position states:

$$\begin{aligned}
\langle y|x\rangle &= \sum_i \langle y|i\rangle\langle i|x\rangle && (7.8)\\
&= \sum_i \psi_i(y)\psi_i^*(x) && (7.9)\\
&= \delta(y - x). && (7.10)
\end{aligned}$$

The star belongs on the factor whose bra-ket order is $\langle i|x\rangle$. Reversing x and y gives the complex-conjugate but equivalent form. Thus every complete orthonormal basis supplies a representation of the delta function. Sines, cosines, oscillator wavefunctions, and energy eigenfunctions all do the same job.

[a] **Question from the audience.** Should the second wavefunction in the completeness sum carry the complex conjugate?

Response. Yes. With the order $\langle y|i\rangle\langle i|x\rangle$, the factors are $\psi_i(y)\psi_i^*(x)$. Keeping the bra-ket order visible prevents the star from being guessed or misplaced.

[a] Lecture timestamp: 00:13:43.

This is the one-particle mathematics we shall need. We now change the state space.

7.3 Fock Space as the Larger State Space

For many identical particles, a convenient basis records how many particles occupy each one-particle mode:

$$|n_1, n_2, n_3, \ldots\rangle. \tag{7.11}$$

The integer n_i is the occupation number of mode i. This ket is not a one-particle state written in fancy notation. It belongs to a larger Hilbert space containing the zero-particle sector, every one-particle state, every two-particle state, and so on. That larger space is called Fock space.

To change occupations, introduce for every mode a creation operator and an annihilation operator,

$$a_i^\dagger, \qquad a_i. \tag{7.12}$$

The first adds one particle to mode i, while the second removes one. At this stage the harmonic-oscillator language is a bookkeeping device: there need not be a literal spring oscillating. The oscillator algebra gives us exactly the machinery needed to raise and lower occupation numbers.

The mode number operator is

$$\begin{gathered} N_i = a_i^\dagger a_i, \\ N_i|n_1, \ldots, n_i, \ldots\rangle = n_i|n_1, \ldots, n_i, \ldots\rangle. \end{gathered} \tag{7.13}$$

The vacuum is

$$|0\rangle \equiv |0, 0, 0, \ldots\rangle, \qquad a_i|0\rangle = 0 \quad \text{for every } i. \tag{7.14}$$

A one-particle state in mode i is therefore

$$|i\rangle = a_i^\dagger|0\rangle. \tag{7.15}$$

The same symbol $|i\rangle$ can now be read either as a one-particle basis vector or as that vector embedded in the one-particle sector of Fock space. The embedding is Equation (7.15); the surrounding context tells us which space is being discussed.

7.4 Packaging the Modes into a Field

Define the annihilation and creation fields by

$$\Psi(x) = \sum_i \psi_i(x) a_i, \qquad \Psi^\dagger(x) = \sum_i \psi_i^*(x) a_i^\dagger. \tag{7.16}$$

The capital letter matters. The lower-case functions $\psi_i(x)$ are complex numbers; the capital $\Psi(x)$ is an operator on Fock space. At every spatial point x, Equation (7.16) gives a different linear combination of the mode operators. That is what an operator-valued function of position means.

For free particles the mode functions may be plane waves e^{ikx}. Then Equation (7.16) is a Fourier expansion, except that its Fourier coefficients are operators rather than numbers. The old terminology calls ordinary complex numbers c-numbers and quantum operators q-numbers. The mode functions are c-numbers and commute with everything; the a_i and $a_i^\dagger$ carry the operator character.

Neither Ψ nor $\Psi^\dagger$ is Hermitian. Their adjoints are each other. The two combinations

$$\Psi + \Psi^\dagger, \qquad i(\Psi^\dagger - \Psi) \tag{7.17}$$

are Hermitian and can represent observables. For the moment, however, the fields are most useful as devices that create, annihilate, and count particles locally.

---

[a] **Question from the audience.** What does it mean for an operator to depend on x?

Response. It means that each value of x selects a different operator. The spatial dependence sits in the ordinary mode functions, while each resulting linear combination acts on the same Fock space.

---

[a]Lecture timestamp: 00:21:05.

7.5 What the Creation Field Does

A definition becomes useful when we ask what it does. Start with a position eigenstate in the one-particle sector and insert the resolution of identity:

$$|x\rangle = \sum_i |i\rangle\langle i|x\rangle \tag{7.18}$$
$$= \sum_i a_i^\dagger |0\rangle\, \psi_i^*(x) \tag{7.19}$$
$$= \Psi^\dagger(x)|0\rangle. \tag{7.20}$$

Thus $a_i^\dagger$ creates a particle in the definite mode i, whereas $\Psi^\dagger(x)$ creates a particle localized at x. In the idealized position basis that localization is distributional; a physical preparation would use a narrow wavepacket, but the algebraic meaning is the same.

---

[a] **Question from the audience.** Is $\sum_i |i\rangle\langle i|$ a projection?

Response. Each term is a projection onto one basis state. The complete sum is the identity itself and is called a resolution of the identity. Inserting it changes no state, but exposes the state's components in the chosen basis.

---

[a]Lecture timestamp: 00:24:48.

Equation (7.20) is the first direct particle interpretation of the field. We have not merely attached a spatial label to an abstract operator. We have built the operation that makes a particle at that spatial point.

[a] **Question from the audience.** Should the position-space creation field be marked with a star or a dagger?

Response. Use a dagger for an operator adjoint. Stars denote complex conjugation of the ordinary wavefunctions $\psi_i(x)$; daggers belong on $a_i^\dagger$ and $\Psi^\dagger(x)$.

[a]Lecture timestamp: 00:28:20.

[a] **Question from the audience.** May the creation operators be moved past the wavefunctions in the mode expansion?

Response. Yes. The $\psi_i(x)$ are c-numbers, so they commute with every operator. One must be careful only when changing the order of noncommuting operators such as a_i and $a_j^\dagger$.

[a]Lecture timestamp: 00:28:36.

7.6 From Probability Density to Particle Density

Let us now play with the field. The first natural expression is

$$\int dx\, \Psi^\dagger(x)\Psi(x). \tag{7.21}$$

It looks like the one-particle normalization integral, but its factors are now operators. Insert the mode expansions without changing their order:

$$\int dx\, \Psi^\dagger(x)\Psi(x) = \int dx \left(\sum_i a_i^\dagger \psi_i^*(x)\right) \tag{7.22}$$

$$\times \left(\sum_j a_j \psi_j(x)\right) \tag{7.23}$$

$$= \sum_{i,j} a_i^\dagger a_j \int dx\, \psi_i^*(x)\psi_j(x) \quad (7.24)$$

$$= \sum_{i,j} a_i^\dagger a_j\, \delta_{ij} \quad (7.25)$$

$$= \sum_i a_i^\dagger a_i. \quad (7.26)$$

Hence the total-number operator is

$$N = \sum_i N_i = \int dx\, \Psi^\dagger(x)\Psi(x). \quad (7.27)$$

For the states under discussion, finite-energy excitations have occupations that die away at sufficiently high mode energy, so this total is ordinarily finite even though the list of possible modes is infinite.

The integrand is the local particle-density operator:

$$n(x) = \Psi^\dagger(x)\Psi(x). \quad (7.28)$$

Strictly speaking one does not measure the density at a mathematical point. For a small region R, one measures

$$N_R = \int_R dx\, n(x), \quad (7.29)$$

the number of particles in that region. Like any quantum observable, N_R need not have a definite value in an arbitrary state. Repeated preparations can yield different outcomes.

[a] **Question from the audience.** Is $\Psi^\dagger(x)\Psi(x)$ the many-particle probability density?

Response. Its direct meaning is particle density. A many-particle probability amplitude generally depends on all particle coordinates. The local bilinear instead counts particles in a small region when integrated over that region.

[a]Lecture timestamp: 00:36:27.

[a] **Question from the audience.** Do the operators a_i and $a_i^\dagger$ act only on state vectors and not on the one-particle wavefunctions?

Response. Yes. They act on Fock-space states such as $|n_1, n_2, \ldots\rangle$. The one-particle mode functions multiplying them are ordinary numbers and are not the objects on which these ladder operators act.

[a]Lecture timestamp: 00:38:28.

The occupation kets form a basis, not the most general state. They may be superposed, including superpositions in which the same total number is distributed differently among the modes. For neutral species such as photons, one may also superpose different total particle numbers. The operator in Equation (7.29) retains the same operational meaning in every such state.

[a] **Question from the audience.** If the particle density is divided by the total number, does it become the probability density for finding a particle at x?

Response. In a fixed-N, large-population setting, N_R/N is a relative frequency and usually tracks a one-particle probability very well. It is not an operator identity equating one measurement outcome with a probability. The measured count fluctuates, typically by an amount of order $\sqrt{N}$ in an independent-particle picture, and an exceptional run can differ greatly from the probability even though such a run is unlikely.

[a]Lecture timestamp: 00:40:01.

[a] **Question from the audience.** What happens to charge conservation in a superposition of a one-electron state and a zero-electron state?

Response. That is not the kind of superposition used for a closed sector of fixed electric charge. One may superpose different distributions with the same total charge. Photons carry no electric charge, so different photon-number sectors cause no such problem. Electron and positron numbers may also change together while the net charge remains fixed.

[a]Lecture timestamp: 00:42:25.

7.7 The Energy of Independent Particles

The next observable is energy. Restrict first to particles that do not interact with one another. This is often a good approximation in a dilute system. Even photons interact in principle through higher-order processes, but ordinary laboratory radiation is dilute enough that the independent-particle description is excellent.

Let ω_i denote the energy of one particle in mode i, with $\hbar = 1$. A mode means one one-particle state. The total Hamiltonian is then

$$H = \sum_i \omega_i N_i = \sum_i \omega_i a_i^\dagger a_i. \tag{7.30}$$

Each occupied mode contributes its one-particle energy once per particle.

---

[a] **Question from the audience.** Is ω_i a function of position?

Response. No. It is the energy eigenvalue associated with mode i. The function $\psi_i(x)$ depends on position; ω_i depends on which energy level has been selected.

---

[a]Lecture timestamp: 00:48:16.

The values ω_i are determined by the one-particle time-independent Schrodinger equation. Define

$$h = -\frac{\nabla^2}{2m} + V(x). \tag{7.31}$$

Then

$$h\psi_i(x) = \omega_i \psi_i(x). \tag{7.32}$$

In one dimension the momentum operator is $p = -i\partial_x$, and $p^2/(2m)$ becomes $-\partial_x^2/(2m)$. In three dimensions the three second derivatives combine into $-\nabla^2/(2m)$.

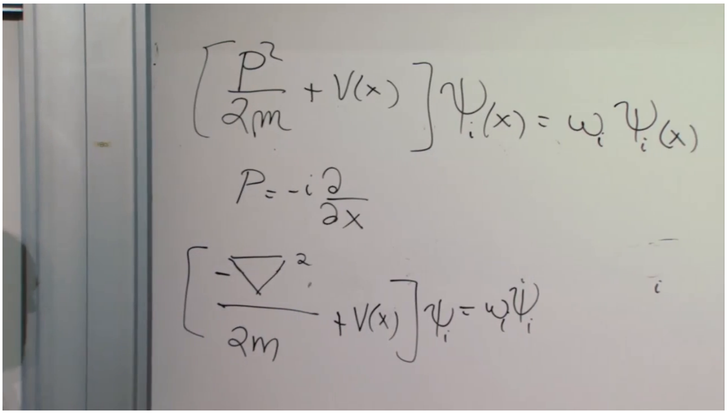

Figure 7.3: The board passage from $p^2/(2m) + V(x)$ to the differential form of the one-particle Schrodinger eigenvalue equation. Equation (7.32) is its cleaned reconstruction.

Lecture frame: 00:54:10.

For an arbitrary normalized one-particle wavefunction, the familiar expectation value is

$$\langle h\rangle_\psi = \int dx\, \psi^*(x) h \psi(x). \tag{7.33}$$

The many-particle formula is suggested by replacing the lower-case wavefunction with the operator-valued field. This is not yet a proof; it is the guess whose correctness we now test:

$$H = \int dx\, \Psi^\dagger(x) h \Psi(x). \tag{7.34}$$

7.8 Writing the Answer First, Then Proving It

Insert Equation (7.16) into the proposed Hamiltonian:

$$\int dx\, \Psi^\dagger h \Psi = \int dx \left(\sum_i a_i^\dagger \psi_i^*(x) \right) h \left(\sum_j a_j \psi_j(x) \right) \tag{7.35}$$

$$= \sum_{i,j} a_i^\dagger a_j \int dx\, \psi_i^*(x) h \psi_j(x). \tag{7.36}$$

The differential operator h acts on the x-dependent wavefunction $\psi_j(x)$, not on the mode operator a_j. By the Schrodinger equation,

$$h\psi_j(x) = \omega_j \psi_j(x). \tag{7.37}$$

Consequently,

$$\int dx\, \Psi^\dagger h \Psi = \sum_{i,j} a_i^\dagger a_j \omega_j \int dx\, \psi_i^*(x) \psi_j(x) \tag{7.38}$$

$$= \sum_{i,j} a_i^\dagger a_j \omega_j \delta_{ij} \tag{7.39}$$

$$= \sum_j \omega_j a_j^\dagger a_j. \tag{7.40}$$

This is precisely Equation (7.30). The local field expression and the sum of all particle energies are the same operator.

No creation operator has been commuted past an annihilation operator in this derivation. The $a_i^\dagger a_j$ order is maintained from beginning to end. The only freely moved objects are the c-number wavefunctions and their eigenvalues.

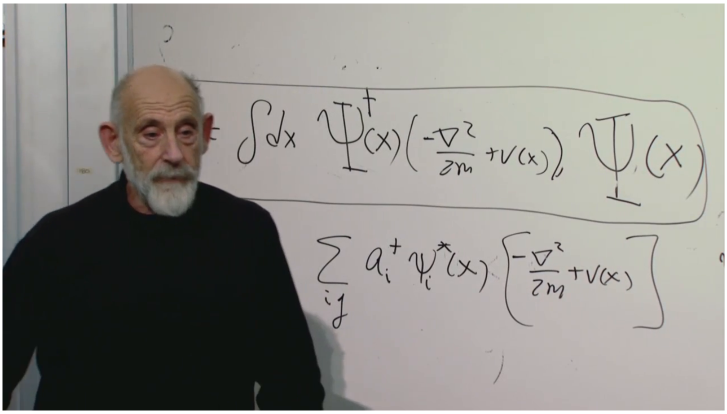

Figure 7.4: The field Hamiltonian emphasized on the board, with the mode-expansion proof beginning below it. The live frame is retained as evidence; Equations (7.34)–(7.40) restore the complete algebra.

Lecture frame: 01:05:34.

7.9 Why This Is a Field

Why call $\Psi(x)$ a field? Because it is an operator-valued quantity defined at every point of space, and Hermitian combinations or bilinears constructed from it are local observables. The density $n(x)$ and the local energy expression below are two examples. There is no additional mathematical test hidden behind the word.

This field is not the electromagnetic field. It describes whichever nonrelativistic species supplied the one-particle modes and the Hamiltonian $p^2/(2m) + V$: slowly moving atoms, neutrons, protons, or electrons, for example. Those particles are the quanta of their corresponding field, in the same structural sense that photons are the quanta of the electromagnetic field.

---

[a] **Question from the audience.** Where, in passing from the many-particle system to the formula above, did a field enter?

Response. It entered in Equation (7.16): $\Psi(x)$ is an operator at every spatial point. The subsequent derivations show that its local products measure particle and energy densities. Position dependence plus operator content is exactly the field structure being constructed.

---

[a] Lecture timestamp: 01:02:45.

7.10 Local Energy Density

Equation (7.34) writes the total energy as an integral over space,

$$
\begin{aligned}
H &= \int dx\, \mathcal{H}(x), \\
\mathcal{H}(x) &= -\frac{1}{2m}\Psi^\dagger(x)\nabla^2\Psi(x) \\
&\quad + V(x)\Psi^\dagger(x)\Psi(x).
\end{aligned}
\tag{7.41}
$$

The integrand is a natural local energy density for this Hamiltonian. Its local form is not unique. For example, integration by parts replaces the kinetic term by

$$
\mathcal{H}_{\text{kin}}(x) = \frac{1}{2m}\nabla\Psi^\dagger(x)\cdot\nabla\Psi(x) \tag{7.42}
$$

up to a boundary term. Both expressions give the same total Hamiltonian under the usual boundary conditions. This is the small ambiguity mentioned in the lecture.

[a] **Question from the audience.** Must the energy density be integrated over all space?

Response. Integrating over all space gives the total energy. Integrating over a smaller region gives the energy assigned to that region, subject to the ordinary ambiguity in choosing a local density whose full integral is the same Hamiltonian.

[a]Lecture timestamp: 01:05:57.

[a] **Question from the audience.** How does the capital N_i in the energy formula differ from the lower-case n_i?

Response. $N_i = a_i^\dagger a_i$ is an operator. The integer n_i is one of its eigenvalues on an occupation-number state. Operators act on states; their eigenvalues are ordinary numbers.

[a]Lecture timestamp: 01:06:25.

[a] **Question from the audience.** Can the field Hamiltonian itself be written as a bra or ket in Dirac notation?

Response. No. Bras and kets denote state vectors and their duals. The field and Hamiltonian are operators acting on those states. An expectation value such as $\langle\Phi|H|\Phi\rangle$ uses both kinds of object and must not identify them.

[a]Lecture timestamp: 01:06:46.

7.11 Recovering One Particle and Approaching a Classical Field

The enlarged formalism contains ordinary one-particle quantum mechanics. Restrict Fock space to states satisfying

$$N|\Phi\rangle = |\Phi\rangle. \tag{7.43}$$

Within that sector, the field construction reproduces every one-particle state and every one-particle prediction. We have not replaced the old theory; we have embedded it in a space that can also describe zero, two, or arbitrarily many particles.

At the other extreme, a mode occupied by a very large number of bosons can behave approximately like a classical wave. In suitable coherent or semiclassical states, the expectation value

$$\Phi_{\rm cl}(x,t) = \langle \Psi(x,t) \rangle \tag{7.44}$$

obeys the classical-looking equation

$$i\partial_t \Phi_{\rm cl}(x,t) = \left(-\frac{\nabla^2}{2m} + V(x) \right) \Phi_{\rm cl}(x,t) \tag{7.45}$$

for the free nonrelativistic theory. The lecture states this connection rather than deriving it. It is the bridge between a collection of quanta and a wave field: large occupations suppress relative fluctuations and make suitable expectation values behave classically.

The formalism is therefore both bookkeeping and physics. It organizes all particle-number sectors, but it also captures interference, entanglement, and the emergence of classical waves from quantum particles.

[a] **Question from the audience.** Why can the creation and annihilation operators be manipulated almost as if they were ordinary numbers inside the integral?

Response. They were never interchanged with each other. The ordinary mode functions commute with them, and the derivatives act only on those functions. Moving an annihilation operator through a creation operator would require the commutator and could add a term; nothing in the proof performs that move.

[a] Lecture timestamp: 01:10:43.

7.12 A Dirac-Notation Interlude

Some of the confusion is historical. A one-particle wavefunction satisfies a wave equation and looks superficially like a classical field, yet it is also a probability amplitude. Early quantum theory had to separate those roles and then explain how genuinely measurable wave fields, such as electromagnetism, are made of quanta. Fock space and operator-valued fields resolve the apparent conflict.

Dirac notation makes the separation unusually compact. A ket belongs to a vector space, a bra to its dual, and their juxtaposition forms an inner product. The resolution

$$\sum_i |i\rangle\langle i| = \mathbf{1} \tag{7.46}$$

then becomes nearly self-explanatory. The lecture illustrates this with an anecdote from teaching quantum mechanics to mathematicians: the mathematical content was already familiar, but the notation made routine manipulations so transparent that the structure looked new. The point is not that one discipline lacked the concept. It is that a well-chosen notation can make the legal operations hard to confuse.

That lesson matters here. There are now several distinct operators:

- h acts on one-particle wavefunctions as a differential operator;
- a_i and $a_i^\dagger$ act on occupation-number states in

Fock space;

- $\Psi(x)$, $\Psi^\dagger(x)$, N, and H also act on Fock space.

The expansion in Equation (7.16) connects the first two settings without merging them.

---

[a] **Question from the audience.** Are fields only a formalism for describing collections of particles?

Response. They are a formalism, but not an empty one. Their local observables can be measured, and the particles are quantum particles with interference and entanglement rather than classical pellets. The field operators encode precisely those measurable collective properties.

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[a] Lecture timestamp: 01:18:16.

7.13 Relativistic Caution and the Electromagnetic Aside

The construction so far is nonrelativistic. Momentum remains a good particle label in relativistic physics, but sharply localized particle position is much more delicate. Consequently, a local photon-number density is not generally as clean an object as the nonrelativistic $n(x)$. Energy density remains well defined and is the safer local observable.

The lecture gives a useful, explicitly heuristic electromagnetic analogy. Combine the electric and magnetic fields into the complex vector

$$\mathbf{F} = \mathbf{E} + i\mathbf{B}, \qquad \mathbf{F}^* = \mathbf{E} - i\mathbf{B}. \tag{7.47}$$

Then

$$\mathbf{F}^* \cdot \mathbf{F} = \mathbf{E}^2 + \mathbf{B}^2, \tag{7.48}$$

which is proportional to the electromagnetic energy density in conventional units. This is not literally a photon probability density. It is better viewed as a compact field variable whose squared magnitude measures energy density, or informally as photon density weighted by photon energy.

---

[a] **Question from the audience.** Can photon density be identified with electric-field strength?

Response. Only qualitatively and with care. For relativistic photons, local particle number is not the preferred concept. The robust statement is that $E^2 + B^2$ is proportional to energy density, and the complex combination $\mathbf{E} + i\mathbf{B}$ packages Maxwell's equations in a Schrodinger-like form.

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[a]Lecture timestamp: 01:19:10.

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[a] **Question from the audience.** Why can a relativistic particle have definite momentum while position is problematic?

Response. Momentum labels relativistic one-particle representations cleanly, whereas strict localization conflicts with particle creation and the relativistic field description. This lecture does not develop that theory; the warning marks the limit of the present nonrelativistic density interpretation.

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[a]Lecture timestamp: 01:20:09.

7.14 Ordering and the Ground-State Constant

Operator order cannot be changed casually. For bosonic modes,

$$a_i a_i^\dagger = a_i^\dagger a_i + 1. \tag{7.49}$$

Accordingly, replacing the normal order $\Psi^\dagger h\Psi$ by the reversed order $(h\Psi)\Psi^\dagger$, or its properly integrated counterpart, adds the commutator contribution. In a mode basis this is an additive sum of one quantum of energy per mode. It is the ground-state or zero-point constant, and in a continuum it requires regulation. The lecture does not pursue that issue here; it only warns that the two orderings are not equal.

---

[a] **Question from the audience.** What happens if Ψ and $\Psi^\dagger$ are interchanged in the Hamiltonian?

Response. Their commutator must be included. Reversing the order therefore adds a state-independent ground-state contribution rather than reproducing the same expression. The displayed Hamiltonian is kept in normal order, $\Psi^\dagger h\Psi$.

---

[a] Lecture timestamp: 01:22:04.

7.15 Momentum as the Final Test

The final minutes repeat the same promotion one more time. For one particle,

$$\langle p\rangle_\psi = \int dx\, \psi^*(x)\,(-i\partial_x)\,\psi(x). \tag{7.50}$$

Replacing the wavefunction by the field gives the total momentum operator

$$P = \int dx\, \Psi^\dagger(x)\, (-i\partial_x)\, \Psi(x), \tag{7.51}$$

or, in three dimensions,

$$\mathbf{P} = \int d^3x\, \Psi^\dagger(\mathbf{x})\, (-i\boldsymbol{\nabla})\, \Psi(\mathbf{x}). \tag{7.52}$$

Fields carry momentum. A sufficiently powerful laser transfers momentum to a wall and exerts a force; in the quantum description that field momentum is the sum of the momenta of the photons.

Equation (7.51) is an operator, not an expectation value. Given a Fock-space state $|\Phi\rangle$, one may then calculate

$$\langle P\rangle_\Phi = \langle\Phi|P|\Phi\rangle. \tag{7.53}$$

An operator and a state are both required to form an expectation value. Confusing the observable with its average would erase exactly the distinction the lecture has worked to establish.

---

[a] **Question from the audience.** Is the displayed field momentum already the average momentum?

Response. No. It is the Hermitian operator representing the observable. Supply a state and sandwich the operator between its bra and ket to obtain an expectation value; a measurement can still fluctuate around that average.

---

[a] Lecture timestamp: 01:26:02.

This is the natural stopping point. Starting from particles, we have constructed local creation and annihilation fields, used them to count particles, and rewritten the sum of one-particle energies and momenta as field operators. The next step is not another change of notation. It is to let field operators represent physical processes: particles decaying, particles scattering, and species transforming into one another.

CHAPTER 8

MIXING, OSCILLATION, ELECTRIC DIPOLE MOMENTS, AND FIELD OPERATORS

We begin a little off the advertised subject, with a question about neutrinos. There is evidence that neutrinos have mass and that the neutrino types mix. If the masses are not all the same, how can one type turn into another without violating energy conservation?

The question points to a pattern that occurs throughout quantum mechanics: the states in which a system is prepared or detected need not be the states that diagonalize its Hamiltonian. A state with definite energy only acquires a phase. A prepared state that is a superposition of several energies changes because its components acquire different phases. That distinction will carry us from a double well to neutrino oscillation and spin precession. After an interlude on the electron electric dipole moment, we shall return to second quantization and see the same position–momentum Fourier structure at the operator level.

8.1 Mixing as the Real Issue

Neutrinos need not all have the same mass. The energy eigenstates do not mix with one another; each evolves independently. What we call the *type*, or flavor, of a neutrino is associated with how it is produced and detected, and a flavor state can be a linear combination of mass-energy eigenstates. Schematically,

$$|\text{type}\rangle = \sum_n c_n |E_n\rangle. \tag{8.1}$$

Thus the right statement is not that energy eigenstates change into one another, but that the states called "electron neutrino" or "muon neutrino" need not themselves be energy eigenstates.

---

[a] **Question from the audience.** If neutrinos have different masses, how can they mix while energy remains conserved?

Response. The mass-energy eigenstates do not change into one another. Electron-, muon-, and tau-flavor states are different linear combinations of those eigenstates. Each mass component evolves with its own phase, and interference changes the flavor probabilities without violating energy conservation.

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[a] Lecture timestamp: 00:00:09.

There are many examples of this structure. The most visible one begins with a particle in a symmetric double well.

8.2 Symmetric Double Well and Approximate Localized States

Consider a particle moving in a symmetric double-well potential with a high barrier between the two minima. If the barrier were taken to be infinitely

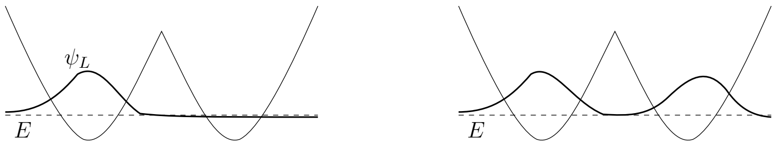

Figure 8.1: A clean redraw of the opening board sketches. On the left is the one-sided trial state ψ_L, obtained by pretending the barrier is effectively infinite. On the right the same wavefunction has been continued symmetrically through the barrier, illustrating the small but nonzero leakage that eventually produces the true even ground state.

high, then the left and right wells would decouple completely. One could solve the left-hand well as though the right-hand well did not exist, and likewise for the right-hand well. The approximate ground-state wavefunctions would then be a left-localized state ψ_L and a right-localized state ψ_R, both with the same energy E.

In each isolated well the ground state has the familiar qualitative properties: it is smooth, has no nodes, and sits at the lowest possible energy consistent with confinement. Think of the high central barrier as a brick wall. Treating it as infinite is only an approximation, but it gives us a useful pair of localized states from which to begin.

Quantum mechanically, however, the barrier is not truly infinite. The wavefunction in the barrier region is not exactly zero; it is only very small. This tiny leakage is the whole story. It is what converts the left-right approximation into a genuine mixing problem.

8.3 Parity Eigenstates and Exponential Level Splitting

The localized picture is now interrupted by a theorem: if the potential is symmetric, then the exact energy eigenfunctions can be chosen to be either symmetric or antisymmetric under reflection.

Theorem 8.1. *If $V(x) = V(-x)$ in one dimension, then the eigenfunctions of the Hamiltonian may be chosen to satisfy either*

$$\psi(x) = \psi(-x) \qquad \text{or} \qquad \psi(x) = -\psi(-x).$$

Now comes the correction. The one-sided states ψ_L and ψ_R are neither symmetric nor antisymmetric, so they cannot be exact eigenstates of the full Hamiltonian. They are only good approximate states when the barrier is high. The true eigenstates are the symmetric and antisymmetric combinations,

$$\psi_+ = \frac{\psi_L + \psi_R}{\sqrt{2}}, \tag{8.2}$$

$$\psi_- = \frac{\psi_L - \psi_R}{\sqrt{2}}. \tag{8.3}$$

Using the notation that will remain on the board, label their energies by

$$H\psi_+ = (E_1 - \epsilon)\psi_+, \tag{8.4}$$

$$H\psi_- = (E_1 + \epsilon)\psi_-. \tag{8.5}$$

The total splitting is therefore

$$\Delta E = (E_1 + \epsilon) - (E_1 - \epsilon) = 2\epsilon. \tag{8.6}$$

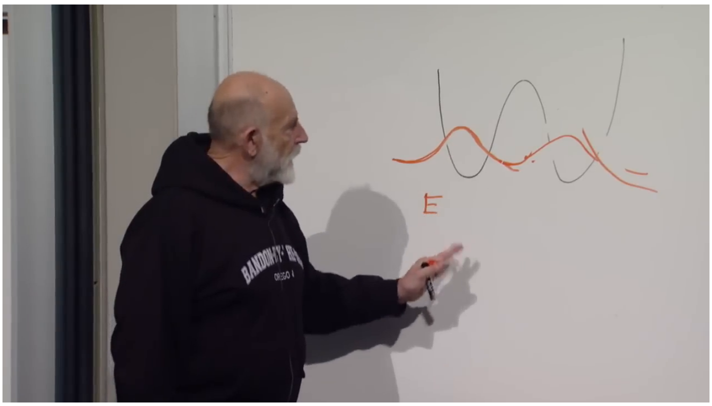

Figure 8.2: The symmetrically continued wavefunction in the double-well potential. Its amplitude is small but nonzero under the barrier, which is the origin of the level splitting.

Lecture frame: 00:06:19.

The ordering of the levels carries the physical point. The symmetric state is slightly lower in energy because it is smoother through the barrier region; allowing the wavefunction to leak and continue symmetrically lowers the energy by a tiny amount. The antisymmetric state is slightly higher because the first excited state in one dimension has one node, and in this symmetric problem that node sits at the midpoint. Extra curvature costs kinetic energy.

The effect is very small, exponentially small in the barrier width and height. We do not need an explicit

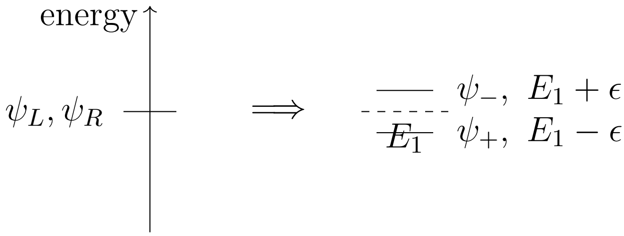

Figure 8.3: The board logic of the tunnel splitting. The approximate left and right states are nearly degenerate, but the exact symmetric and antisymmetric eigenstates are split by a small amount 2ϵ.

WKB formula here; the important point is that the splitting is real but tiny. Also remember that the overall sign of a wavefunction carries no physical information. Replacing ψ_- by $-\psi_-$ does not change the state.

8.4 Time Evolution and Left-Right Oscillation

Now ask the dynamical question: what happens if we place a particle in the left-hand well? This is the decisive step, because oscillation is the time-dependent face of the same mixing phenomenon.

The inverse relations are

$$\psi_L = \frac{\psi_+ + \psi_-}{\sqrt{2}}, \tag{8.7}$$

$$\psi_R = \frac{\psi_+ - \psi_-}{\sqrt{2}}. \tag{8.8}$$

So an initially left-localized state is

$$\psi(0) = \psi_L = \frac{\psi_+ + \psi_-}{\sqrt{2}}. \tag{8.9}$$

It is not an energy eigenstate, and therefore it must evolve nontrivially.

We use the standard Schrödinger convention e^{-iEt}. A common phase has no observable effect; only the relative phase between the two energy components can move probability from one well to the other.

With that convention,

$$\psi(t) = \frac{1}{\sqrt{2}} \left(e^{-i(E_1-\epsilon)t} \psi_+ + e^{-i(E_1+\epsilon)t} \psi_- \right). \tag{8.10}$$

This is the cleaned version of the board formula. Now factor out the common phase:

$$\psi(t) = \frac{e^{-iE_1 t}}{\sqrt{2}} \left(e^{i\epsilon t} \psi_+ + e^{-i\epsilon t} \psi_- \right). \tag{8.11}$$

One phase now runs one way around the unit circle while the other runs the opposite way. The state changes only because these two phases drift out of step.

To see the left-right oscillation explicitly, substitute the definitions of $\psi_\pm$:

$$\psi(t) = \frac{e^{-iE_1 t}}{2} \left[e^{i\epsilon t}(\psi_L + \psi_R) + e^{-i\epsilon t}(\psi_L - \psi_R) \right] \tag{8.12}$$

$$= \frac{e^{-iE_1 t}}{2} \left[(e^{i\epsilon t} + e^{-i\epsilon t})\psi_L + (e^{i\epsilon t} - e^{-i\epsilon t})\psi_R \right] \tag{8.13}$$

$$= e^{-iE_1 t}\left[\psi_L \cos(\epsilon t) + i\,\psi_R \sin(\epsilon t)\right]. \tag{8.14}$$

This is the key worked derivation of the chapter. In the two-state approximation, with ψ_L and ψ_R orthonormal, the probabilities are

$$P_L(t) = \cos^2(\epsilon t), \qquad P_R(t) = \sin^2(\epsilon t). \tag{8.15}$$

So the system oscillates. At intermediate times it is genuinely in superposition; for example, when $\epsilon t = \pi/4$, the probabilities on the two sides are equal. The phases evolve continuously, and the localization changes continuously with them.

The blackboard derivation isolates the first time at which the relative sign flips. One asks when

$$e^{-i\epsilon t} = -\,e^{i\epsilon t}, \tag{8.16}$$

which is the visible milestone on the board. Multiplying through gives

$$e^{2i\epsilon t} = -1. \tag{8.17}$$

For the first occurrence,

$$2\epsilon t = \pi, \qquad \text{so} \qquad t_{L\to R} = \frac{\pi}{2\epsilon}. \tag{8.18}$$

At that instant,

$$\psi(t_{L\to R}) \propto \frac{\psi_+ - \psi_-}{\sqrt{2}} = \psi_R. \tag{8.19}$$

The overall phase is irrelevant. What matters is that the particle initially placed on the left has evolved

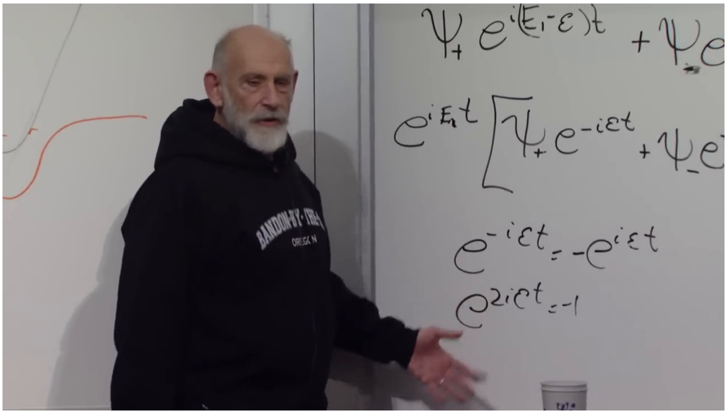

Figure 8.4: The relative-phase calculation on the board. The two energy components become opposite when $e^{2i\epsilon t} = -1$, giving the first transfer time $2\epsilon t = \pi$. The typeset derivation uses one consistent Schrödinger-phase convention.

Lecture frame: 00:17:52.

into the right-localized state. After another equal interval it swings back again. Mixing and oscillation go together.

The equation is unambiguous: small ϵ means slow oscillation and a long transfer time. Neither localized state has a definite energy, but the two exact energy components conserve energy and evolve exactly as they should.

8.5 Neutrino, Ammonia, and Spin Analogies

The double well gives us a template for the original neutrino question. A neutrino created together with an electron is called an electron neutrino; one created together with a muon is called a muon neutrino. Those flavor labels describe production and detection channels. The states with definite masses, and hence definite free-particle energies at fixed momentum, form another basis.

In the simplest two-state, maximally mixed caricature,

$$|\nu_e\rangle \sim \frac{|m_1\rangle + |m_2\rangle}{\sqrt{2}}, \qquad |\nu_\mu\rangle \sim \frac{|m_1\rangle - |m_2\rangle}{\sqrt{2}}, \tag{8.20}$$

where the mass eigenstates play the role of ψ_+ and ψ_-, and the flavor states play the role of ψ_L and ψ_R. Real neutrino mixing uses a nontrivial mixing matrix rather than these equal coefficients, but the mechanism is already visible here.

An electron neutrino is produced in a process involving an electron, just as a particle can be plunked into the left well. For a short flight its two mass components have accumulated almost the same phase, so an interaction still identifies it as electron flavor. Over a long baseline their phases separate. A later charged-current interaction can then produce a muon rather than an electron: the state has acquired a muon-flavor component. Repeating the experiment at different distances or energies reveals the oscillation.

There is no conflict with energy conservation. The mass-energy eigenstates do exactly what eigenstates are supposed to do. It is the flavor superposition that evolves. In the relativistic limit the observable relative phase is governed by mass-squared differences, schematically

$$\Delta\phi_{ij} \simeq \frac{\Delta m_{ij}^2 L}{2E}. \tag{8.21}$$

Oscillation therefore shows that not all neutrino masses can be identical; at least two mass eigenstates differ in mass.

---

[a] **Question from the audience.** What determines the small splitting ϵ in the double-well example?

Response. It depends on the particle mass and on the barrier's height and width. Tunneling is exponentially sensitive to those details, so a modest change in the barrier can produce a large fractional change in ϵ.

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[a]Lecture timestamp: 00:24:06.

[a] **Question from the audience.** Has this oscillation phenomenon actually been observed?

Response. Yes. Tunneling splittings are observed in ordinary quantum systems, and flavor oscillations are observed for neutrinos. The mechanisms differ microscopically, but both realize the same two-state phase evolution.

[a] Lecture timestamp: 00:24:26.

[a] **Question from the audience.** How is neutrino oscillation connected to neutrino mass?

Response. The propagating eigenstates have different masses and therefore accumulate different phases. Their interference changes the flavor composition. In a realistic relativistic treatment it is the mass-squared differences, not a literal time-varying mass of one particle, that control the oscillation.

[a] Lecture timestamp: 00:24:34.

[a] **Question from the audience.** Is neutrino mass merely prevailing theory, or is it experimental fact?

Response. Oscillation is experimental fact, and it requires nonzero mass differences. The absolute mass scale is a separate and more difficult measurement, but the minimal picture in which every neutrino is massless is experimentally excluded.

[a] Lecture timestamp: 00:25:02.

8.5.1 Ammonia inversion: a literal barrier

An ammonia molecule, NH_3, is trigonal pyramidal: the three hydrogen atoms form a plane and the nitrogen lies on one side of it. There is an equivalent configuration with the nitrogen on the other side. The two localized configurations are not exact stationary states. The nitrogen can tunnel through the hydrogen plane, producing symmetric and antisymmetric energy eigenstates and a small inversion splitting. This is the physical origin of the same mathematics in the ammonia maser.

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[a] **Question from the audience.** In ammonia, is the barrier the configuration in which the nitrogen passes through the hydrogen plane?

Response. Yes. The molecular energy is minimized in either pyramidal configuration, while the planar configuration lies higher. Tunneling through that configuration couples the two sides.

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[a]Lecture timestamp: 00:25:11.

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[a] **Question from the audience.** What is the corresponding physical barrier between electron and muon neutrinos?

Response. There is no literal spatial barrier. The analogy is algebraic: an off-diagonal coupling in the neutrino Hamiltonian mixes the flavor basis. In ammonia that coupling arises from tunneling; for neutrinos it is part of the mass and mixing structure.

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[a]Lecture timestamp: 00:27:10.

8.5.2 Spin precession: mixing without tunneling

We have already met the same algebra in spin. Take the two states $|\uparrow_z\rangle$ and $|\downarrow_z\rangle$. With no magnetic field they are degenerate. Now turn on a magnetic field in the x-direction. The energy eigenstates are no longer the z-eigenstates but the x-polarized states. Using the temporary left–right names from the board,

$$|L_x\rangle = \frac{|\uparrow_z\rangle + |\downarrow_z\rangle}{\sqrt{2}}, \tag{8.22}$$

$$|R_x\rangle = \frac{|\uparrow_z\rangle - |\downarrow_z\rangle}{\sqrt{2}}. \tag{8.23}$$

These are the direct analogues of ψ_+ and ψ_-. The magnetic moment aligned and antialigned with the field has slightly different energy. If those energies are $E \mp \epsilon$, then an initially up spin evolves as

$$|\uparrow_z, t\rangle = e^{-iEt}\left[\cos(\epsilon t)\,|\uparrow_z\rangle + i\sin(\epsilon t)\,|\downarrow_z\rangle\right]. \tag{8.24}$$

This is the same formula as for the double well, only now the continuous evolution is spin precession about the x-axis. Starting from $+z$, the spin passes through transverse directions, reaches $-z$ after half a precession period, and returns after another half period. A measurement of S_z still yields only up or down; the continuously changing spin state determines their probabilities. The mechanism does not require tunneling. It requires only that the Hamiltonian be diagonal in a basis different from the preparation basis.

[a] **Question from the audience.** What happens to the tau neutrino in this picture?

Response. It mixes as well. The real problem has three flavor states and three mass states, so a three-by-three mixing matrix replaces the two-state rotation. The phase-interference principle is unchanged.

[a]Lecture timestamp: 00:32:25.

[a] **Question from the audience.** Immediately after the transverse field is applied, is the spin still up, and can it point toward or away from us?

Response. At a very short time it is still almost certainly up, with a small down probability. The state precesses continuously around the field axis and passes through the transverse $\pm y$ directions, while remaining perpendicular to the x-directed field if it began perpendicular. A measurement along any chosen axis still gives one of that axis's discrete eigenvalues.

[a]Lecture timestamp: 00:32:57.

[a] **Question from the audience.** Is the field chosen weak so that the spin never points along the field?

Response. No. The geometry follows from precession about the field axis, not from the field's weakness. A weak field merely makes the energy splitting and precession frequency small, giving a slow oscillation analogous to the neutrino example.

[a]Lecture timestamp: 00:34:30.

The strength of neutrino mixing and the small neutrino masses are parameters whose deeper origin is not explained by the minimal Standard Model as originally formulated. Neutrino mass therefore points beyond that minimal model, though the observed scale is very small and may reflect physics at a remote energy scale.

---

[a] **Question from the audience.** Is oscillation the reason we know neutrinos have mass?

Response. It is the decisive direct evidence that the propagating components have different masses. If all relevant components were exactly massless and degenerate, they would accumulate no mass-dependent relative phase and there would be no vacuum flavor oscillation of this kind.

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[a]Lecture timestamp: 00:36:27.

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[a] **Question from the audience.** At half a period, should a neutrino act half like electron flavor and half like muon flavor?

Response. In the ideal symmetric two-state model, equal probabilities occur at a quarter of the full probability period, while complete conversion occurs at the first maximum. Real three-flavor oscillations have unequal mixing angles, several frequencies, spin-helicity constraints, and matter effects, so complete symmetric conversion is not generic.

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[a]Lecture timestamp: 00:37:57.

[a] **Question from the audience.** If the mass components differ, must a solar neutrino's velocity change in order to conserve momentum?

Response. A flavor state is a coherent superposition of components with fixed masses; its mass is not a classical quantity oscillating in time. In the common equal-momentum description the components have slightly different energies and group velocities, while in an equal-energy description they have slightly different momenta. Both conventions give the same observable oscillation phase when treated consistently.

[a] Lecture timestamp: 00:38:36.

This phase evolution resolved the old solar-neutrino deficit. Early detectors were primarily sensitive to electron neutrinos, so neutrinos that left the Sun as electron flavor could arrive partly as muon or tau flavor and escape that channel. The detailed solar calculation includes three-flavor mixing and matter effects, but the conceptual lesson is already present in the double well: preparation and propagation use different bases.

8.6 Interlude: Electric Dipole Moment, Symmetry, and the "Electron Sphere" Claim

Since we are already talking about particle phenomenology, let us take a break from formal mathematics. A popular article described a new electron electric-dipole-moment bound by saying that the electron had been shown to be a sphere. That slogan confuses three different questions: what an electric dipole moment measures, what rotational symmetry means for a quantum state, and what the experiment actually constrained.

8.6.1 What was measured?

What was measured, or rather bounded, is the electric dipole moment of the electron. For separated charges $\pm q$, the dipole moment is

$$\mathbf{d} = q\,\boldsymbol{\ell}, \tag{8.25}$$

with $\boldsymbol{\ell}$ the separation vector. A nonzero dipole moment means that the charge distribution is displaced relative to some center.

For the electron, however, the relevant question is not merely whether the charge is off-center in some arbitrary direction. The electron has spin and therefore an axis. It also has a magnetic moment. The physically meaningful question is whether there is a correlation between the displacement of charge and that axis:

$$\mathbf{d}_e \parallel \boldsymbol{\mu}_e \qquad \text{or equivalently} \qquad \mathbf{d}_e \parallel \mathbf{S}. \tag{8.26}$$

That correlation is what is meant by the electron's electric dipole moment. Conceptually, one asks whether placing a spin-polarized electron in an electric field splits the two spin orientations:

$$H_{\mathrm{EDM}} = -\mathbf{d}_e \cdot \mathbf{E}, \qquad \mathbf{d}_e = d_e \frac{\mathbf{S}}{|\mathbf{S}|}. \tag{8.27}$$

The actual tabletop experiment is much subtler than a bare electron sitting between capacitor plates, but this is the symmetry and energy-shift idea it exploits.

8.6.2 Why no EDM does not mean sphere

The first correction is elementary. A body can have no dipole moment and still not be spherical. A centered ellipsoid has as much charge on one side as on the other and therefore no dipole, yet it can have a quadrupole moment. A still more complicated distribution may have neither dipole nor quadrupole while retaining a higher multipole. Measuring one multipole coefficient does not determine an entire shape.

Quantum mechanics makes the word "shape" more delicate. Imagine a two-atom, dumbbell-like molecule with an approximately fixed bond length. Ignore radial vibration and keep only rotations. Its rotational states are labeled by angular momentum. If the molecule is bosonic, the lowest rotational state has $J = 0$. Because angular momentum generates rotations,

$$J = 0 \quad \Longrightarrow \quad \text{rotationally invariant ground state.} \tag{8.28}$$

The quantum state is rotationally invariant even though every definite-orientation picture looks like a dumbbell. One may describe it as a coherent superposition of all orientations. Calling the state "spherical" is useful for some measurements and misleading for others.

Take a quick snapshot. A snapshot is a measurement of orientation, and a sufficiently sharp orientation measurement leaves the system in a state localized in angle. Such a state cannot also have definite angular momentum. This is the rotational version of the position–momentum uncertainty principle: localizing angle necessarily introduces a spread of angular momenta and therefore rotational excitation.

For a rigid rotor,

$$E_J = \frac{J(J+1)}{2I}, \tag{8.29}$$

so the resolving energy is controlled by the moment of inertia I. A gymnasium dumbbell has an enormous I and fantastically close rotational levels; ordinary light can easily disturb it enough to reveal an orientation. A molecule has wider level spacing. A compact hadron has much wider spacing still. If the probe lacks enough energy to excite the first resolvable rotational structure, repeated measurements see only the rotationally averaged ground state.

A meson can be pictured loosely as a quark–antiquark two-body system. Its first orbital excitations cost hundreds of millions of electron volts or more, far beyond an optical photon. A low-energy probe therefore cannot take a snapshot of the underlying orientation. "Sphere or not" is partly a question about which state is prepared and partly a question about the resolution of the apparatus.

The electron sharpens the issue. It is a spin-$\frac{1}{2}$

particle, not a rotational scalar. A spin-polarized state carries an axis and a magnetic moment. The familiar picture of circulating charge around that axis is only a heuristic—a pointlike Dirac particle is not literally a tiny spinning ball—but it correctly reminds us that the magnetic moment reverses with spin. The EDM question asks whether the electric charge displacement is correlated with that same axis. A null answer does not turn the electron into a classical sphere.

---

[a] **Question from the audience.** Would displacing part of the electron's charge violate charge quantization?

Response. No. Charge quantization fixes the total charge. A dipole moment concerns the first spatial moment of that charge distribution, or equivalently an allowed point-particle coupling to the electric field; it does not require a fractional change in total charge.

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[a]Lecture timestamp: 00:57:30.

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[a] **Question from the audience.** Are electric dipole moments forbidden by quantum field theory?

Response. Not by quantum field theory itself. They are forbidden when particular discrete symmetries are exact. Once those symmetries are violated, an EDM operator is allowed, although its coefficient may be extremely small.

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[a]Lecture timestamp: 00:58:00.

8.6.3 Parity and time reversal

The real content of the EDM discussion is symmetry. Under parity, an electric dipole moment behaves as an ordinary polar vector, while a magnetic moment behaves as an axial vector:

$$P: \qquad \mathbf{d}_e \to -\mathbf{d}_e, \qquad \boldsymbol{\mu}_e \to \boldsymbol{\mu}_e. \tag{8.30}$$

Under time reversal, the circulating current that defines the magnetic moment reverses, but a static electric displacement does not:

$$T: \qquad \mathbf{d}_e \to \mathbf{d}_e, \qquad \boldsymbol{\mu}_e \to -\boldsymbol{\mu}_e. \tag{8.31}$$

Therefore a nonzero EDM aligned with the electron's spin axis is forbidden by exact parity symmetry and also by exact time-reversal symmetry:

$$(P \text{ exact}) \Rightarrow \mathbf{d}_e = 0, \qquad (T \text{ exact}) \Rightarrow \mathbf{d}_e = 0. \tag{8.32}$$

The geometry is worth spelling out. A magnetic moment is produced by a loop-like current. Mirror reflection reverses ordinary polar vectors but leaves the axial vector normal to that loop unchanged. A charge displacement along the axis does reverse. Thus an electron with $\mathbf{d}_e$ parallel to $\boldsymbol{\mu}_e$ would be mirrored into one with the same magnetic orientation but the opposite relation between electric and magnetic moments. If parity were exact and there were only one electron species, the two descriptions would have to represent the same particle; only $d_e = 0$ is consistent.

Time reversal gives the same conclusion in a different way. Running the film backward reverses the current and therefore $\boldsymbol{\mu}_e$, but it does not move a static charge displacement. A state with parallel moments becomes one with antiparallel moments. Exact time-reversal symmetry would again require $d_e = 0$.

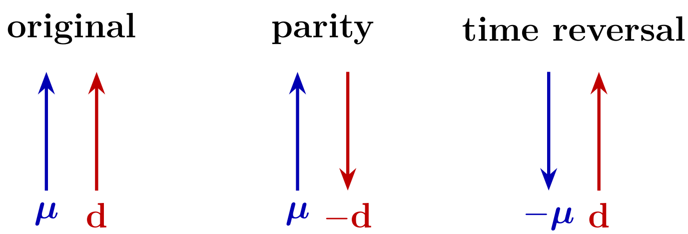

Figure 8.5: The symmetry test reconstructed from the board argument. Parity reverses the electric dipole but not the axial magnetic moment; time reversal reverses the magnetic moment but not the static electric displacement. Either exact symmetry forbids a permanent EDM parallel to spin.

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[a] **Question from the audience.** Under time reversal, are both currents and charges reversed?

Response. Currents reverse because velocities reverse; electric charge itself does not. Charge conjugation C, not time reversal T, reverses the sign of charge.

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[a]Lecture timestamp: 01:06:38.

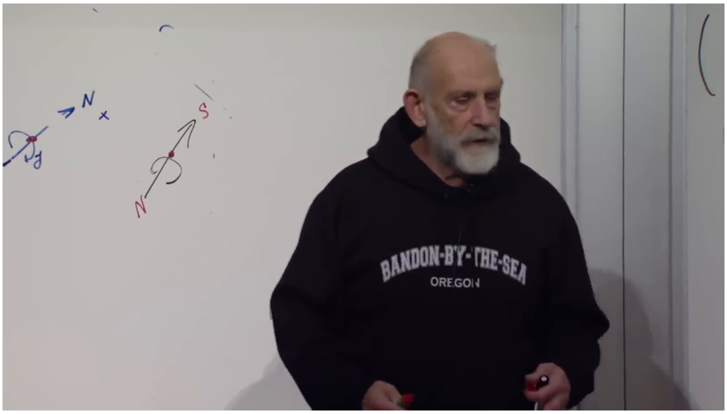

Figure 8.6: The paired magnetic-axis and charge-displacement sketches during the time-reversal argument. The compact TikZ diagram above isolates the transformation content from the partially erased live board.

Lecture frame: 01:06:10.

[a] **Question from the audience.** Does the antiunitary character of time reversal mean that nonrelativistic quantum mechanics is not time-reversal invariant?

Response. No. Antiunitarity is precisely how time reversal is represented consistently in quantum mechanics: it complex-conjugates amplitudes and reverses momenta and spins. Whether a particular Hamiltonian is invariant is a separate question.

[a] Lecture timestamp: 01:07:01.

[a] **Question from the audience.** Does an EDM discussion require the electron to have a finite volume rather than be pointlike?

Response. No. A point-particle effective Hamiltonian can contain $-d_e\,\widehat{\mathbf{S}}\cdot\mathbf{E}$. Experiment asks whether opposite spin orientations acquire different electric-field energies; it need not resolve a literal extended charge cloud.

[a] Lecture timestamp: 01:07:19.

Parity is not a symmetry of the weak interactions; it is violated as strongly as possible in their chiral couplings. Time reversal is also not exact, but the known violation in the Standard Model feeds into the electron EDM only at an extraordinarily small level. One can calculate a nonzero Standard-Model contribution, yet it lies far below the sensitivity discussed here. A null result is therefore consistent with the Standard Model while cutting into extensions—supersymmetry and other new interactions, for example—that can generate much larger time-reversal violation.

The basic experimental thought is simple. A magnetic field reveals a magnetic dipole by splitting spin orientations. An electric field would reveal an EDM by an additional spin-correlated energy shift. Real precision experiments use atoms or molecules, long coherent precession times, reversals of applied fields, and careful control of systematics rather than an isolated electron in a naive apparatus.

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[a] **Question from the audience.** Was the EDM search a high-energy collision experiment or an experiment with material objects?

Response. It was precision tabletop physics using electrons bound in a controlled atomic or molecular environment together with electric and magnetic fields. Its energy is low, but its sensitivity to tiny symmetry-violating shifts probes high-energy new physics indirectly.

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[a]Lecture timestamp: 01:11:02.

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[a] **Question from the audience.** Does the null result constrain which modifications of the Standard Model remain viable?

Response. Yes in principle: every model predicts a coefficient and therefore occupies some parameter region. The detailed numerical exclusion requires a specialist calculation, but the general logic is firm—agreement with the Standard Model at higher precision leaves less room for alternatives that predict a large EDM.

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[a]Lecture timestamp: 01:12:21.

[a] **Question from the audience.** Is time-reversal symmetry required for conservation of quantum information?

Response. No. Unitary time evolution conserves information even when a Hamiltonian is not invariant under T. A particle in a fixed magnetic field is a familiar example: the field selects an orientation in time-reversed movies unless the field is reversed too, yet the quantum evolution remains unitary. Under the usual assumptions of local Lorentz-invariant quantum field theory, the combined CPT transformation is a symmetry, but that is not what guarantees unitarity.

[a]Lecture timestamp: 01:13:09.

The break can now end. The null measurement says that the electron has no detected EDM at the stated sensitivity. It does not say that the electron is a sphere, and it does not overthrow the Standard Model; it makes some proposed departures from that model harder to sustain.

8.7 Return to Second Quantization: Fourier Transforms, Momentum Modes, and Locality

We were doing second quantization. Let us restart gently, with a familiar fact about ordinary single-particle wavefunctions, and then lift it to operators. The connection with Fourier transforms is exact and simple.

8.7.1 From Position Space to Momentum Space

Lower-case ψ, not capital Ψ, is the wavefunction of a single particle. In position space,

$$P(x) = |\psi(x)|^2 \tag{8.33}$$

is the probability density to find the particle at position x. The momentum-space wavefunction is its Fourier transform,

$$\tilde{\psi}(p) = \int dx\, \psi(x)\, \frac{e^{-ipx}}{\sqrt{2\pi}}, \tag{8.34}$$

and the inverse relation is

$$\psi(x) = \int dp\, \tilde{\psi}(p)\, \frac{e^{ipx}}{\sqrt{2\pi}}. \tag{8.35}$$

Then

$$P(p) = |\tilde{\psi}(p)|^2 \tag{8.36}$$

is the probability density for momentum p.

We use $\hbar = 1$. The two transforms are inverses: knowing either wavefunction determines the other, while their absolute squares answer different measurement questions.

For a free particle,

$$H = \frac{p^2}{2m}, \tag{8.37}$$

so momentum eigenstates are also energy eigenstates, and their position-space wavefunctions are plane waves,

$$\psi_p(x) \propto e^{ipx}. \tag{8.38}$$

This is the point of departure for the second-quantized version.

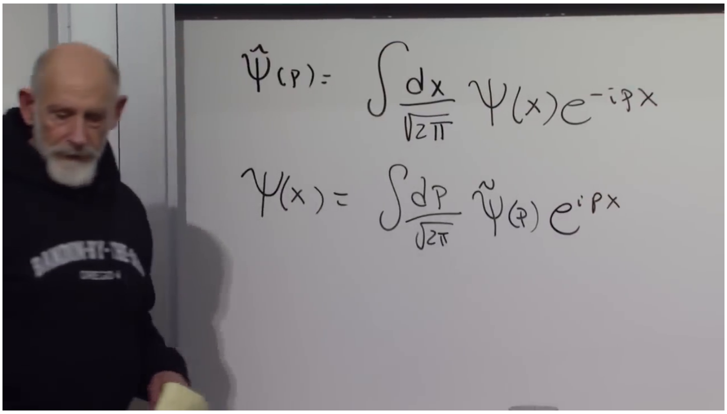

Figure 8.7: The position–momentum Fourier pair for a one-particle wavefunction. The distinction between lower-case ψ and the capital field operator introduced next is essential.

Lecture frame: 01:20:00.

8.7.2 Field operators in position and momentum space

Now return to the field operator. On the blackboard its annihilation and creation pieces are written as $\Psi_-(x)$ and $\Psi_+(x)$. Keeping that notation for a moment,

$$\Psi_-(x) = \sum_i a_i\, \psi_i(x), \tag{8.39}$$

$$\Psi_+(x) = \sum_i a_i^\dagger\, \psi_i^*(x). \tag{8.40}$$

Here $\psi_i(x)$ is the ordinary one-particle wavefunction of the i-th mode, while a_i and $a_i^\dagger$ annihilate and create particles in that mode. From this point on it is convenient to rename

$$\Psi(x) \equiv \Psi_-(x), \qquad \Psi^\dagger(x) \equiv \Psi_+(x).$$

For a free particle the discrete mode label i is replaced by the continuous momentum label p. Using the plane waves e^{ipx}, one obtains the standard reconstructed formulas

$$\Psi(x) = \int \frac{dp}{\sqrt{2\pi}}\, a(p)\, e^{ipx}, \tag{8.41}$$

$$\Psi^\dagger(x) = \int \frac{dp}{\sqrt{2\pi}}\, a^\dagger(p)\, e^{-ipx}. \tag{8.42}$$

These are the field-operator analogues of the ordinary Fourier transform. They can be inverted exactly as before:

$$a(p) = \int \frac{dx}{\sqrt{2\pi}}\, \Psi(x)\, e^{-ipx}, \tag{8.43}$$

$$a^\dagger(p) = \int \frac{dx}{\sqrt{2\pi}}\, \Psi^\dagger(x)\, e^{ipx}. \tag{8.44}$$

The factors of $1/\sqrt{2\pi}$ are a convention chosen so that the forward and inverse transforms have the same normalization.

The conceptual content is straightforward. $\Psi^\dagger(x)$ creates a particle at position x, while $a^\dagger(p)$ creates a particle with momentum p. Position-space and momentum-space creation operators are Fourier conjugates of one another, just as $\psi(x)$ and $\tilde{\psi}(p)$ are Fourier conjugates in ordinary quantum mechanics.

If no particle of momentum p is present, $a(p)$ annihilates the state. If particles of that momentum are present, it removes one with the usual occupation-number coefficient. Likewise, $\Psi(x)$ removes a particle at x, understood distributionally or after smearing into a wavepacket. Each particle species requires its own field operator; one field cannot create every kind of particle.

8.7.3 Bosonic commutators and the independence of separated measurements

The next fact can be derived directly from the definitions. For bosons,

$$[a_i, a_j^\dagger] = \delta_{ij}, \qquad [a_i, a_j] = 0, \qquad [a_i^\dagger, a_j^\dagger] = 0. \tag{8.45}$$

The continuum version is

$$[a(p), a^\dagger(p')] = \delta(p - p'). \tag{8.46}$$

Now compute the equal-time field commutator from the discrete expansion:

$$[\Psi(x), \Psi^\dagger(y)] = \sum_{i,j} \psi_i(x)\psi_j^*(y)\,[a_i, a_j^\dagger] \tag{8.47}$$

$$= \sum_i \psi_i(x)\psi_i^*(y) \tag{8.48}$$

$$= \delta(x - y), \tag{8.49}$$

where the last step uses completeness of the one-particle basis. Likewise,

$$[\Psi(x), \Psi(y)] = 0, \qquad [\Psi^\dagger(x), \Psi^\dagger(y)] = 0. \tag{8.50}$$

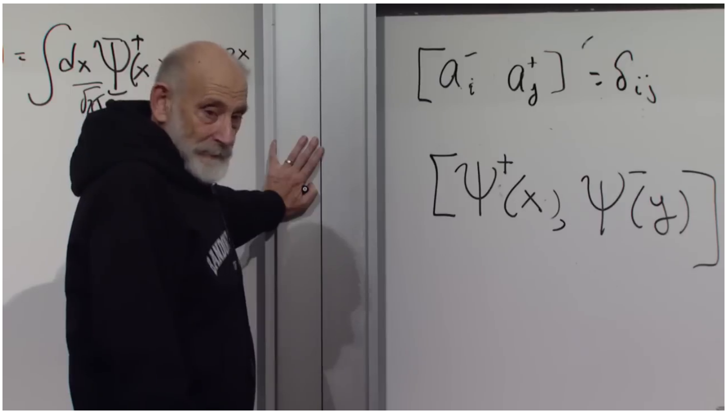

Figure 8.8: The mode commutator and its position-space counterpart on the board. Completeness turns δ_{ij} into the spatial delta distribution $\delta(x-y)$.

Lecture frame: 01:31:00.

The important point is not merely that Ψ is non-Hermitian. Define Hermitian quadratures

$$Q(x) = \frac{\Psi(x) + \Psi^\dagger(x)}{\sqrt{2}}, \qquad P(x) = \frac{\Psi(x) - \Psi^\dagger(x)}{i\sqrt{2}}. \tag{8.51}$$

Their same-point commutator inherits the delta distribution. Thus the two local quadratures are conjugate in the same sense that position and momentum are conjugate. Trying to sharpen one produces uncertainty in the other.

[a] **Question from the audience.** What does the delta function mean when $x = y$? Is the commutator large everywhere?

Response. The delta is a distribution, not an ordinary function with a finite point value. It vanishes for separated arguments and gives a finite result only after integration against a test function. Smearing the fields over a finite detector region turns the formal same-point singularity into the usual finite canonical commutator of two modes.

[a]Lecture timestamp: 01:32:03.

For $x \neq y$, however, the equal-time commutator vanishes. Operations localized in two disjoint regions can be performed independently: measuring the field here does not give the field there a quantum kick. This is the nonrelativistic equal-time precursor of relativistic microcausality. In a relativistic theory, physical local observables commute at spacelike separation, where no light signal can connect the two events.

The contrast with noncommuting observables is familiar. Since $[x, p] = i$, a precise position measurement disturbs momentum, and a precise momentum state spreads in position. Since different spin components fail to commute, preparing one component sharply makes another uncertain. A nonzero commutator between separated local observables would analogously mean that an operation in one region disturbs the other, contradicting the required causal independence.

[a] **Question from the audience.** Do EPR and entanglement experiments show that a measurement here has an effect on a distant measurement?

Response. They show correlations, not a controllable causal influence. Entangled outcomes violate classical factorization, but the local statistics at the distant detector cannot be changed by choosing or performing a measurement here. Commutation of spacelike-separated observables is the operator statement behind that no-signalling independence.

[a] Lecture timestamp: 01:37:51.

Next time the algebra changes. Fermion creation and annihilation operators satisfy anticommutation relations. Taken naively, fermion fields at separated points do not commute, so the fields themselves cannot be interpreted as directly measurable bosonic observables. Physical fermionic observables are even combinations—densities, currents, stress tensors—and those local observables recover the required spacelike commutativity. That is the entry point to the fermion version of second quantization.

CHAPTER 9

LOCAL QUANTUM FIELDS AND THE ONE-DIMENSIONAL DIRAC EQUATION

We have already met quantum fields through their quanta, their creation and annihilation operators, and the simplest Hamiltonian for nonrelativistic particles. We shall now make that machinery do some work. First we will see, rather than merely assert, how a spatially local Hamiltonian conserves momentum. Then we will let products of fields describe scattering and decay, and watch repeated action of the Hamiltonian generate the beginnings of diagrammatic perturbation theory. Only after that will we turn to the promised fermion problem. The cleanest entrance is not yet a fermion field, but the one-dimensional Dirac equation.

9.1 The Field Hamiltonian Once More

Write the Hamiltonian for one species of nonrelativistic particle as

$$H_0 = \int dx\, \psi^\dagger(x) \left[-\frac{\partial_x^2}{2m} + V(x) \right] \psi(x). \qquad (9.1)$$

The notation looks one-dimensional, but nothing in the opening argument depends on that restriction: in three dimensions x can stand for the vector $\mathbf{x}$, dx for d^3x, and $-\partial_x^2$ for $-\nabla^2$.

The operator $\psi^\dagger(x)$ creates a particle at x, while $\psi(x)$ annihilates one there. Their normally ordered product

$$n(x) = \psi^\dagger(x)\psi(x) \tag{9.2}$$

is the local number density. The potential term therefore reads

$$H_V = \int dx\, V(x) n(x). \tag{9.3}$$

This is precisely what we would write for a classical gas: count the particles near x, assign each one the potential energy $V(x)$, and add over space. At this stage the particles do not interact with one another. They may respond to an externally imposed potential, but the Hamiltonian is otherwise a sum of independent one-particle energies.

It is useful to simplify and then reverse course. Take the potential to be a constant, $V(x) = V_0$. A constant potential exerts no force, yet its operator meaning is transparent:

$$\begin{aligned} H_V &= V_0 \int dx\, \psi^\dagger(x)\psi(x) = V_0 N, \\ N &= \int dx\, n(x). \end{aligned} \tag{9.4}$$

It gives each particle the same additive energy. To retain rest energy in a nonrelativistic description, choose $V_0 = mc^2$. The term then counts one rest energy per particle. We keep this otherwise dull constant because it preserves translation invariance and makes the number-counting operation visible.

9.2 The Hamiltonian as an Update Rule

What does it mean to say that momentum is conserved? We should phrase the answer in terms of what the Hamiltonian does. To avoid using ψ for a field operator, a one-particle wavefunction, and a many-body state all at once, call the many-body state $|\Phi(t)\rangle$. Over a short interval ϵ,

$$|\Phi(t+\epsilon)\rangle = \Big(1 - i\epsilon H + O(\epsilon^2)\Big)\,|\Phi(t)\rangle, \tag{9.5}$$

where $\hbar = 1$. The unchanged state is the first term; the first correction is generated by H. Iterating this update is the Schrödinger equation.

Momentum conservation now has a concrete meaning. If the initial state has definite total momentum P, then $H|\Phi\rangle$ must have that same total momentum. Other properties may change, and individual particles may exchange momentum, but the Hamiltonian may not carry the state into a sector with a different total P. We can test this directly by rewriting the Hamiltonian in the momentum basis.

9.3 The Spatial Integral as a Momentum Delta Function

Choose the Fourier convention

$$\psi(x) = \int \frac{dp}{\sqrt{2\pi}}\, a(p) e^{ipx}, \tag{9.6}$$

$$\psi^\dagger(x) = \int \frac{dq}{\sqrt{2\pi}}\, a^\dagger(q) e^{-iqx}. \tag{9.7}$$

The operator $a(p)$ annihilates a particle of momentum p. The operator $a^\dagger(q)$ creates one of momentum

q. Different letters are essential: the two momenta are integrated independently until the spatial integral relates them.

Begin with the number term:

$$N = \int dx\, \psi^\dagger(x)\psi(x) \tag{9.8}$$

$$= \int \frac{dp\, dq}{2\pi}\, a^\dagger(q)a(p) \int dx\, e^{i(p-q)x}. \tag{9.9}$$

The remaining integral is

$$\int dx\, e^{i(p-q)x} = 2\pi\delta(p-q). \tag{9.10}$$

For $p \neq q$, the phase oscillates and cancels between positive and negative contributions. For $p = q$, it does not oscillate at all. The Dirac delta distribution is the precise object that combines those two facts. Consequently,

$$N = \int dp\, dq\, a^\dagger(q)a(p)\delta(p-q) \tag{9.11}$$

$$= \int dp\, a^\dagger(p)a(p). \tag{9.12}$$

The operator removes a particle of momentum p and puts back one of exactly the same momentum. The decisive ingredient was the integral over x: it produced the delta function that enforces equality of momentum in and momentum out.

[a] **Question from the audience.** What happened to the infinity that appears when $p = q$?

Response. A delta function is not used by evaluating its height. It is a distribution defined by integration: $\int dk\, \delta(k) f(k) = f(0)$. The phrase "infinitely high and infinitely narrow" only reminds us that its integrated weight is one. Here it instructs us to set $p - q = 0$, and no infinite answer remains.

[a]Lecture timestamp: 00:17:02.

The same mechanism works for a much more general local monomial. Suppose a term contains several annihilation fields and several creation fields, perhaps belonging to different species, but all evaluated at one point and integrated over that point:

$$\int dx \, \prod_{r=1}^{n} \psi^\dagger_{i_r}(x) \prod_{s=1}^{m} \psi_{j_s}(x). \tag{9.13}$$

Let the annihilated particles carry momenta p_s and the created particles carry momenta q_r. Fourier expansion produces

$$\int dx \, e^{i(\sum_s p_s - \sum_r q_r)x} = 2\pi \, \delta\left(\sum_s p_s - \sum_r q_r \right). \tag{9.14}$$

The result does not pair p_1 with q_1, p_2 with q_2, and so on. It imposes only

$$\sum_{\text{annihilated}} p = \sum_{\text{created}} q. \tag{9.15}$$

That is exactly the rule required by an interaction: individual momenta may change, but total momentum does not. The deeper statement is translation invariance. A local density integrated with the same weight at every point has no preferred origin, and its Fourier transform carries the corresponding momentum-conserving delta function.

---

[a] **Question from the audience.** Is the proposed Hamiltonian term really a product of all those field operators? Why would we write such a product?

Response. Yes. The product specifies which particles are removed and which are created at one point. Its physical motivation will come from observed scattering or decay processes; the momentum argument is established first so that those interactions can be read immediately.

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[a]Lecture timestamp: 00:23:04.

9.4 Kinetic Energy in Momentum and Position Space

Now apply the same machinery to the derivative term. The derivative acts on the plane wave:

$$-\partial_x^2 e^{ipx} = p^2 e^{ipx}. \tag{9.16}$$

Therefore

$$H_{\text{kin}} = \int dx\, \psi^\dagger(x) \left(-\frac{\partial_x^2}{2m}\right) \psi(x) \tag{9.17}$$

$$= \int dp\, \frac{p^2}{2m}\, a^\dagger(p) a(p). \tag{9.18}$$

Again the spatial integral enforces equal incoming and outgoing momentum. The new factor $p^2/2m$ simply weights the number of particles in each momentum mode by the kinetic energy of that mode.

[a] **Question from the audience.** Should differentiating the plane wave twice give a minus sign?

Response. Yes, $\partial_x^2 e^{ipx} = -p^2 e^{ipx}$. The Hamiltonian already contains the compensating minus sign, so $-\partial_x^2$ has the positive eigenvalue p^2.

[a]Lecture timestamp: 00:24:46.

[a] **Question from the audience.** Is the kinetic energy conserved only when the potential is constant?

Response. Yes. With no position-dependent potential, there is no force and the kinetic energy is separately conserved. In a nonconstant external potential, kinetic and potential energy can exchange while the total Hamiltonian remains conserved if it has no explicit time dependence.

[a]Lecture timestamp: 00:27:20.

[a] **Question from the audience.** What is the role of $\psi^\dagger\psi$ in the momentum-space kinetic Hamiltonian?

Response. It is the counting operator. The mode product $a^\dagger(p)a(p)$ counts particles of momentum p, and the factor $p^2/2m$ assigns the appropriate kinetic energy to each one.

[a]Lecture timestamp: 00:27:42.

There is a second way to see positivity. Integrating by parts once and neglecting the boundary term gives

$$\int dx\, \psi^\dagger(-\partial_x^2)\psi = \int dx\, (\partial_x\psi^\dagger)(\partial_x\psi). \qquad (9.19)$$

For fields that vanish suitably at infinity, or with periodic boundary conditions, the boundary contribution is zero. The result has the form of an absolute square and is nonnegative. It also displays a pattern common to field theory: the energy contains squares of spatial derivatives of fields.

9.5 A Local Contact Interaction

Let us now give the products of fields a physical job. Consider two species, which we will call an electron and a proton only to keep the picture familiar. For the moment they are deliberately fake: both are treated with bosonic operators, and their true Coulomb interaction is omitted. Their free Hamiltonian is

$$H_0 = \int dx \left[\psi_e^\dagger \left(-\frac{\partial_x^2}{2m_e} \right) \psi_e + \psi_p^\dagger \left(-\frac{\partial_x^2}{2m_p} \right) \psi_p \right]. \tag{9.20}$$

Add the model interaction

$$H_{\text{contact}} = g \int dx \, \psi_e^\dagger(x) \psi_p^\dagger(x) \psi_e(x) \psi_p(x). \tag{9.21}$$

Read it from right to left. It looks for an electron and a proton at the same point x, annihilates them, and recreates both at that point. If the two particles are not together, this local term gives zero. If their worldlines meet, it can change their individual momenta while preserving the total. Thus it acts as an idealized zero-range scattering potential.

Real electrons and protons interact at a distance through the Coulomb potential. The contact model

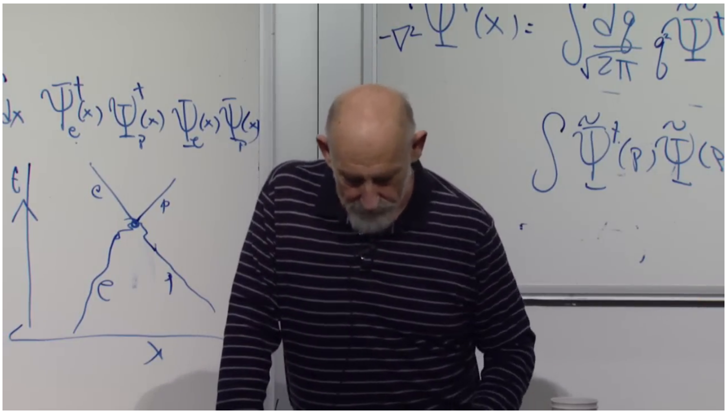

Figure 9.1: The contact interaction on the board. The four local field operators appear above the spacetime sketch; the meeting point is the only place where the idealized interaction acts.

Lecture frame: 00:35:36.

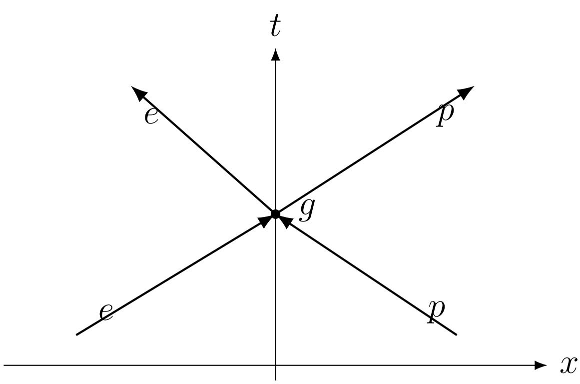

Figure 9.2: A clean reconstruction of the local scattering picture. The vertex is shorthand for the operator in Eq. (9.21), not a classical collision law.

is only the first short step: if experiment showed that two species scatter only when they overlap, Eq. (9.21) is the sort of term we would place in the Hamiltonian.

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[a] **Question from the audience.** Why is this scattering operator part of the Hamiltonian if the energy does not change?

Response. The Hamiltonian is both the energy operator and the generator that changes the state. A potential energy between the two particles produces scattering under time evolution. Conservation of the total energy does not mean that the state, or the particles' individual momenta, remains unchanged.

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[a]Lecture timestamp: 00:35:40.

This gives a practical rule. First learn from experiment what local processes particles can undergo. Then encode those processes in the operator that updates the state. Products of fields are not decorative algebra: their daggers and undaggered factors specify which species appear and disappear.

9.6 Decay, the Inverse Process, and Hermiticity

Suppose a bubble chamber reveals a track of species A ending at a point from which tracks B and C emerge. The local process is

$$A \longrightarrow B + C. \tag{9.22}$$

The operator that performs it is

$$V = g \int dx\, \psi_B^\dagger(x)\psi_C^\dagger(x)\psi_A(x). \tag{9.23}$$

The annihilation field removes A; the two creation fields produce B and C at the same point. Integrating with equal weight over every x expresses spatial translation invariance and immediately supplies

$$p_A = p_B + p_C. \tag{9.24}$$

A Hamiltonian must be Hermitian. This is what makes e^{-itH} unitary and preserves total probability. Equation (9.23) is not Hermitian by itself, so we must add its adjoint:

$$H_{\text{int}} = \int dx \left[g\, \psi_B^\dagger \psi_C^\dagger \psi_A + g^*\, \psi_A^\dagger \psi_C \psi_B \right](x). \tag{9.25}$$

For a real coupling g, the two coefficients are the same. Hermitian conjugation reverses operator order as well as exchanging daggers and undaggered fields. In this bosonic toy model, fields of distinct species commute, so that reversal causes no additional complication.

The second term performs the inverse process,

$$B + C \longrightarrow A. \tag{9.26}$$

Thus a closed, unitary Hamiltonian that permits the decay also contains the recombination amplitude. This does not promise that both directions occur for every chosen set of external momenta; energy and other conservation laws may make one direction kinematically inaccessible. It says that the operator and its adjoint both belong to the dynamics.

[a] **Question from the audience.** Can there be an exception to the rule that the inverse process is also allowed?

Response. Not at the level of a Hermitian closed-system Hamiltonian: the adjoint interaction is required. A particular reverse reaction can still be blocked by kinematics or another conserved quantum number, but the conjugate matrix element is present.

[a] Lecture timestamp: 00:42:48.

The number g is the coupling constant. If g is small, particles that meet interact only weakly; if it is large, the transition amplitude is larger. Couplings are usually parameters measured in experiments, although symmetries can relate several apparently different couplings.

The same grammar handles more elaborate reactions. In beta decay,

$$n \longrightarrow p + e^- + \bar{\nu}_e, \tag{9.27}$$

a schematic local term contains four fields: one annihilates the neutron and three create the proton, electron, and electron antineutrino, together with the Hermitian-conjugate process. This is enough to see how a small set of operator monomials begins to define a quantum field theory.

9.7 Why the Square of the Hamiltonian Matters

The short-time formula (9.5) kept only first order in ϵ. The full expansion is

$$e^{-i\epsilon H} = 1 - i\epsilon H - \frac{\epsilon^2}{2}H^2 + \frac{i\epsilon^3}{3!}H^3 + \cdots . \quad (9.28)$$

At second order the Hamiltonian acts twice. This is the algebraic origin of the first compound process pictures.

For the vertex $A \leftrightarrow B + C$, one action can combine $B + C$ into A, and a second can split that A back into $B + C$:

$$B + C \longrightarrow A \longrightarrow B + C. \quad (9.29)$$

The net contribution is a $B + C \to B + C$ scattering amplitude. The intermediate A is not an additional observed event; it labels the state summed over between two actions of the interaction.

---

[a] **Question from the audience.** Should the C-field in the written second-order product carry a dagger?

Response. Yes, the operator order must match the stated sequence. The first vertex annihilates B and C and creates A; the second annihilates A and recreates B and C. Writing the corresponding daggers correctly restores that interpretation.

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[a] Lecture timestamp: 00:49:54.

Now add a spectator B. One action lets A emit C and become B; a later action lets that C combine with the original B to form A. The external particles

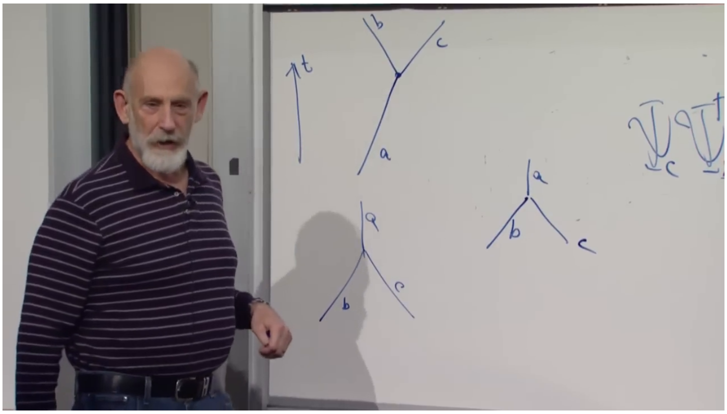


Figure 9.3: The two orientations of the elementary $A \leftrightarrow B + C$ vertex on the board. Juxtaposing them is the visual shorthand for the second-order process in Eq. (9.29).

Lecture frame: 00:48:56.

are again $A + B$, but their momenta can change. We describe this as A and B scattering by exchange of C.

Each local interaction point is called a vertex. Every vertex contributes a factor of its coupling, so a two-vertex contribution carries g^2, a three-vertex contribution g^3, and so forth. A single term in the Hamiltonian therefore correlates a large family of reactions. This economy is one reason quantum field theory is so powerful: a small number of local interactions organizes an enormous amount of scattering data.

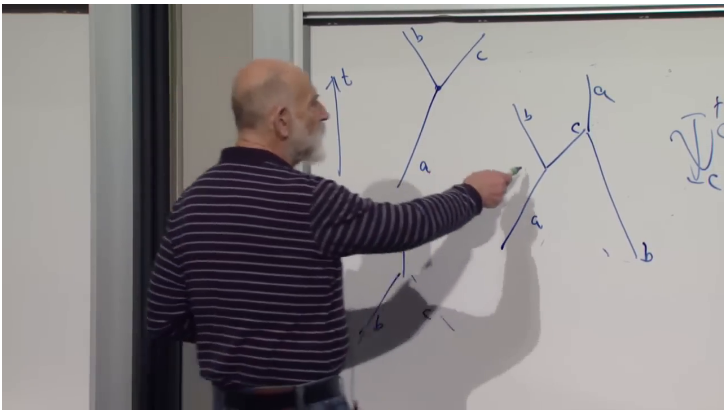
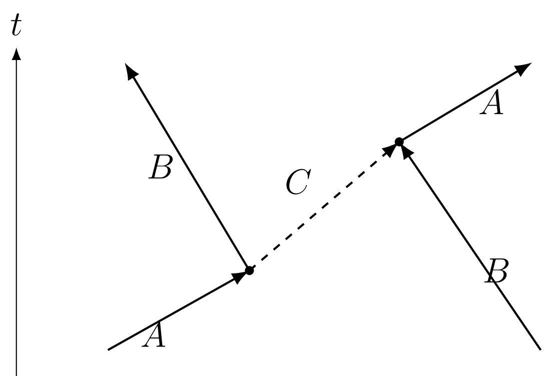


Figure 9.4: The exchange construction on the board. The right-hand diagram joins two elementary vertices with an internal C line while the external A and B particles scatter.

Lecture frame: 00:51:22.



$$A + B \to A + B$$

Figure 9.5: A clean reconstruction of C-exchange. Each dot contributes one factor of g, so this second-order term is proportional to g^2.

[a] **Question from the audience.** What do we call the process $A \to B + C \to A$?

Response. It is a self-energy contribution. A measured A is not merely the bare one-particle basis state; the interacting state also has an amplitude for nearby $B + C$ components. Scattering from it can reveal that dressed structure.

[a] Lecture timestamp: 00:53:14.

Quantum electrodynamics supplies the familiar example. A local electron–photon vertex can annihilate an electron, recreate an electron, and create a photon. At second order, the electron may emit and reabsorb a photon. It is tempting to narrate this as a tiny movie, but its disciplined meaning is a term in the perturbative expansion. The physical electron is a dressed state with photon-containing components.

The classroom estimate that this component is "about one percent" conveys the scale set by the fine-structure constant, $\alpha_{\text{em}} \simeq 1/137$. A precise probability for a photon Fock component depends on energy resolution and on how the interacting state is defined; it is not a universal one-percent property of an isolated electron. The robust statement is that interaction changes the electron's measurable structure.

9.8 What a Diagram Does and Does Not Say

The diagrams are calculational shorthand for powers of the interaction Hamiltonian. They are not reliable classical histories in which a particle definitely performs one event and then another. Looking inside to decide which intermediate alternative “really happened” changes the experiment and destroys the coherent sum, just as in ordinary quantum interference.

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[a] **Question from the audience.** Are both energy and momentum conserved at the first vertex?

Response. In the equal-time Hamiltonian construction performed here, the spatial integral immediately produces a momentum delta function. No time integral has yet produced an energy delta function for that isolated insertion. The complete time-translation-invariant scattering amplitude conserves total energy, while intermediate lines need not obey the on-shell relation of observed particles. In covariant diagram language four-momentum is conserved at each vertex, but internal particles may be off shell.

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[a] Lecture timestamp: 00:56:52.

The apparent missing energy in a naive vertex picture is another warning not to discard the interaction energy. The total Hamiltonian is more than the sum of free-particle energies. If confusion arises, return to Eq. (9.28): the picture stands for a term in the expansion of unitary time evolution. Full Feynman-diagram technology supplies systematic time ordering, propagators, integrations, and symmetry factors, but the basic physics begins here.

An extended question presses the diagram too literally by comparing the emitted photon with the electric field surrounding a slowly moving electron.

---

[a] **Question from the audience.** Is the electron's electric field at a distance literally produced by radiated photons? How does that fit with interference and the Aharonov–Bohm effect?

Response. A static or induction field is not a stream of real radiated photons. Quantum electrodynamics describes both radiation and the electron's electromagnetic dressing with the same field, but the corresponding quantum states and kinematics differ. For an ultrarelativistic electron, some field components can usefully behave like a flux of nearly real photons; for a slow electron, a literal photon trajectory is the wrong model. Two-slit interference and the Aharonov–Bohm effect are also nonclassical field phenomena, but they do not turn this self-energy diagram into a mechanical movie.

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[a]Lecture timestamp: 00:58:26.

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[a] **Question from the audience.** Why exactly does a classical model of the diagram fail?

Response. Because the calculation coherently adds amplitudes from all allowed intermediate alternatives. Experience makes the pictures intuitive, but their literal content never exceeds the operator expansion that defines them. They are a compact way to calculate $H^2, H^3, \ldots$, not a record of observed intermediate events.

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[a]Lecture timestamp: 01:01:28.

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[a] **Question from the audience.** Can the ordinary Coulomb interaction emerge as the classical limit of photon exchange?

Response. Yes. Evaluating the appropriate photon-mediated amplitude and taking the nonrelativistic static limit yields the Coulomb potential. The derivation is more technical than the present operator sketch, and it still should not be read as a photon literally jumping between two classical particles.

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[a] Lecture timestamp: 01:03:04.

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[a] **Question from the audience.** Are coupling constants calculated from the field products?

Response. Usually their numerical values are experimental input. Symmetry can constrain them or relate couplings that would otherwise be independent. When such a relation predicts the relative strengths of distinct reactions and experiment confirms it, the agreement is especially informative.

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[a] Lecture timestamp: 01:03:44.

[a] **Question from the audience.** What is the role of an accidental symmetry compared with a gauge symmetry?

Response. Gauge symmetry directly constrains the allowed structure and relations among couplings. An accidental symmetry is not imposed at the fundamental level but emerges because all lower-order allowed interactions happen to respect it. It can still have important consequences, although its origin and robustness differ from those of a gauge symmetry.

[a] Lecture timestamp: 01:04:48.

That is enough of the nonrelativistic field theory for now. We promised fermions. Rather than second-quantize them immediately, we shall first understand the relativistic one-particle equation that their field operators must satisfy.

9.9 Why the Electron Requires Relativity

Begin in one spatial dimension. This is not a realistic restriction, but it removes the geometrical complexity of three-dimensional spin while preserving the essential algebra. The electron is light enough that relativity matters even in atoms. In hydrogen its characteristic speed is of order

$$v \sim \alpha_{\mathrm{em}} c \approx 0.007c, \tag{9.30}$$

roughly one percent of the speed of light. For inner electrons in a heavy atom, the rough scale becomes $v \sim Z\alpha_{\mathrm{em}} c$, so relativistic corrections can be substantial. Whether historical curiosity or atomic accuracy supplied the decisive motivation, a relativistic Schrödinger-like equation is physically necessary.

The usual quantization recipe starts from the classical relation between energy and momentum. Nonrelativistically,

$$E = \frac{p^2}{2m}, \tag{9.31}$$

and the substitutions

$$E \mapsto i\partial_t, \qquad p \mapsto -i\partial_x \tag{9.32}$$

produce

$$i\partial_t \varphi(x,t) = -\frac{1}{2m}\partial_x^2 \varphi(x,t). \tag{9.33}$$

Relativity instead gives, with $c = 1$,

$$E^2 = p^2 + m^2. \tag{9.34}$$

Applying the substitutions to the squared relation gives

$$-\partial_t^2 \varphi = \left(-\partial_x^2 + m^2\right)\varphi, \tag{9.35}$$

or equivalently

$$\left(\partial_t^2 - \partial_x^2 + m^2\right)\varphi = 0. \tag{9.36}$$

This is the Klein–Gordon equation. It is a perfectly meaningful relativistic field equation, especially for spin-zero fields. But the desired electron equation is to be first order in time,

$$i\partial_t \Psi = H\Psi, \tag{9.37}$$

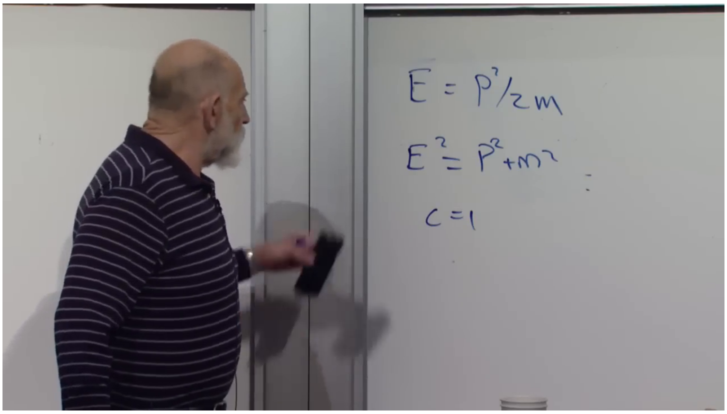

Figure 9.6: The board pivot from $E = p^2/2m$ to the relativistic relation $E^2 = p^2 + m^2$, with $c = 1$.

Lecture frame: 01:10:30.

so that H itself, rather than only H^2, generates evolution.

One could formally choose

$$H = \sqrt{-\partial_x^2 + m^2}. \tag{9.38}$$

The square root can be defined cleanly in momentum space, where it multiplies by $\sqrt{p^2 + m^2}$, but in position space it is a nonlocal pseudodifferential operator. It is technically meaningful and aesthetically unhelpful for the local equation we seek. We therefore start again.

9.10 A Massless Right-Moving Wave

Take first a massless particle moving to the right. Its positive energy and momentum satisfy

$$E = p \qquad (c = 1). \tag{9.39}$$

Quantization gives

$$i\partial_t\psi = -i\partial_x\psi, \tag{9.40}$$

or

$$(\partial_t + \partial_x)\psi = 0. \tag{9.41}$$

Every solution has the form

$$\psi(x,t) = f(x-t). \tag{9.42}$$

The profile is rigid and moves to the right at the speed of light.

Two difficulties appear immediately. First, if Eq. (9.41) is allowed to act on arbitrary Fourier modes, a negative momentum has energy $E = p < 0$, so the Hamiltonian has no lower bound. Restricting by hand to $p > 0$ would retain only right movers and is not yet the electron theory we want. Second, the equation is a one-way street: it contains no ordinary left-moving electron. We shall cure the left-right problem first and leave the negative energies exposed.

9.11 Two Components and the Alpha Matrix

Introduce two amplitudes,

$$\Psi(x,t) = \begin{pmatrix} \psi_1(x,t) \\ \psi_2(x,t) \end{pmatrix}, \tag{9.43}$$

where ψ_1 is right-moving and ψ_2 is left-moving in the massless limit:

$$(\partial_t + \partial_x)\psi_1 = 0, \qquad (\partial_t - \partial_x)\psi_2 = 0. \tag{9.44}$$

Quantum mechanics also allows arbitrary superpositions of the two. A particle can therefore have amplitudes for both directions rather than carrying a classical label chosen once and for all.

This extra two-valued degree of freedom resembles spin algebraically, but in one dimension it is not ordinary angular momentum. Define

$$\alpha = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \sigma_3. \tag{9.45}$$

Its $+1$ eigenvector selects the right-moving component and its -1 eigenvector the left-moving component. In the massless theory, the Hamiltonian is simply

$$H = \alpha p, \tag{9.46}$$

and the two equations combine into

$$i\partial_t \Psi = -i\alpha\partial_x \Psi. \tag{9.47}$$

---

[a] **Question from the audience.** What is the name of the diagonal matrix that distinguishes the two components?

Response. As a Pauli matrix it is σ_3, not σ_1. In Dirac's notation for the present construction we call it α. Its algebra is the familiar Pauli algebra even though the physical label here is direction, not spin angular momentum.

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[a]Lecture timestamp: 01:22:44.

[a] **Question from the audience.** Does the construction acquire a more geometrical meaning in three dimensions, where direction is a vector?

Response. Yes. The three-dimensional problem requires one matrix for each spatial direction and leads to a richer spinor structure. The one-dimensional model is nevertheless exact in its own setting and isolates the essential left-right algebra without that extra complexity.

[a]Lecture timestamp: 01:23:32.

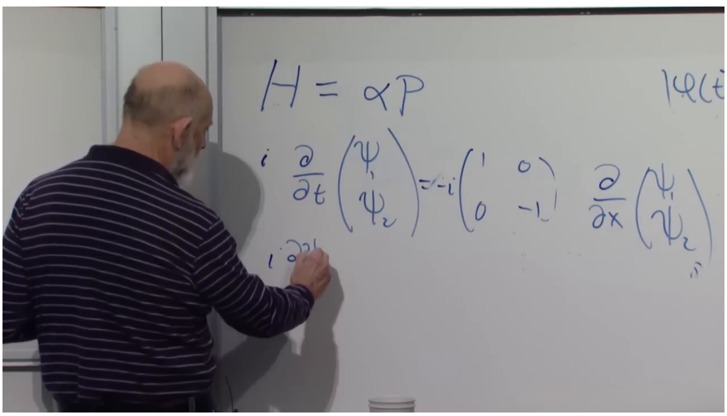

Figure 9.7: The massless two-component equation on the board: $H = \alpha P$, with the diagonal matrix assigning opposite propagation to ψ_1 and ψ_2.

Lecture frame: 01:26:00.

The one-way problem is gone. The negative-energy spectrum remains, and we have discovered a third defect: the two components are still massless and propagate at light speed. The next step must introduce a rest energy and couple the directions.

9.12 Adding Mass and Discovering the Dirac Algebra

Add the most general simple term proportional to the mass,

$$H = \alpha p + \beta m, \tag{9.48}$$

where β acts only on the two-component index. It could have been the identity or some other matrix; we let the relativistic dispersion relation decide.

Since H^2 must equal $p^2 + m^2$, square Eq. (9.48):

$$\begin{aligned} H^2 &= (\alpha p + \beta m)^2 & (9.49)\\ &= \alpha^2 p^2 + \beta^2 m^2 + (\alpha\beta + \beta\alpha) pm. & (9.50) \end{aligned}$$

The matrices commute with p because they act on the internal two-component space while p acts on the spatial wavefunction. Matching Eq. (9.34) requires

$$\alpha^2 = \mathbb{1}, \qquad \beta^2 = \mathbb{1}, \qquad \{\alpha, \beta\} = \alpha\beta + \beta\alpha = 0. \tag{9.51}$$

The first condition already holds for $\alpha = \sigma_3$. Any different Pauli matrix anticommutes with it and squares to the identity. Choose

$$\beta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \sigma_1. \tag{9.52}$$

The choice of σ_1 rather than σ_2 is a basis convention; both realize the same algebra after a unitary change of basis.

[a] **Question from the audience.** Did Dirac simply recognize that Pauli matrices solve these algebraic conditions?

Response. The historical derivation was carried out directly in three dimensions and introduced matrices satisfying the required relations. Once the conditions are written, the Pauli algebra makes the one-dimensional solution immediate: choose two different Pauli matrices.

[a] Lecture timestamp: 01:33:00.

[a] **Question from the audience.** Is this β the relativistic velocity parameter v/c?

Response. No. It is a matrix acting on the spinor components. The reused letter is only Dirac's notation; it has nothing to do with the kinematic quantity v/c.

[a] Lecture timestamp: 01:34:24.

In three spatial dimensions, 2×2 matrices are insufficient for the full set of required anticommuting matrices. That enlargement is postponed. In one dimension, Eqs. (9.45) and (9.52) are enough.

9.13 The Coupled One-Dimensional Dirac Equation

Substituting $p = -i\partial_x$ gives the one-dimensional Dirac equation in Hamiltonian form:

$$i\partial_t \Psi = (-i\alpha\partial_x + \beta m)\,\Psi. \tag{9.53}$$

Because α is diagonal, its derivative term keeps the two components separate. Because β is off diagonal, the mass term exchanges them. Written component by component,

$$i\partial_t\psi_1 = -i\partial_x\psi_1 + m\psi_2, \tag{9.54}$$

$$i\partial_t\psi_2 = +i\partial_x\psi_2 + m\psi_1. \tag{9.55}$$

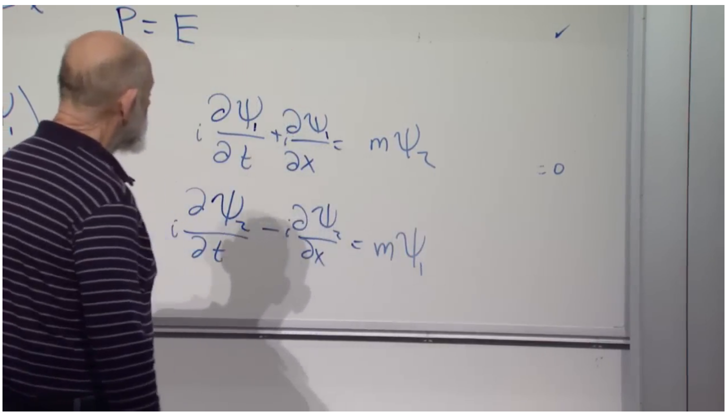

Figure 9.8: The coupled component equations on the board. The opposite derivative signs encode right and left motion; the off-diagonal mass term swaps ψ_1 and ψ_2.

Lecture frame: 01:37:44.

The components cease to evolve independently as soon as $m \neq 0$. Squaring the first-order equation recovers the Klein–Gordon operator on each component,

$$\left(\partial_t^2 - \partial_x^2 + m^2\right)\psi_a = 0, \qquad a = 1, 2, \tag{9.56}$$

and plane waves satisfy $E^2 = p^2 + m^2$. Their group velocity is

$$v_g = \frac{dE}{dp} = \frac{p}{E}, \tag{9.57}$$

whose magnitude is below one for nonzero mass.

The relation between mass and left-right mixing is not accidental. A massless right mover stays a right mover under every proper Lorentz boost; no subluminal observer can overtake light and reverse its direction. A massive particle can be brought to rest and then viewed from a faster frame moving in the same direction, where its velocity appears reversed. A massive relativistic theory therefore cannot keep right and left sectors permanently isolated. The off-diagonal mass term supplies precisely the required coupling.

---

[a] **Question from the audience.** Is the swapping of the two components a chiral oscillation?

Response. In this one-dimensional basis, α distinguishes the two massless chiral directions. The mass term is off diagonal and mixes them, so chirality is no longer conserved. In that sense the mass term is the source of chiral oscillation.

---

[a]Lecture timestamp: 01:40:01.

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[a] **Question from the audience.** Why should a fermion equation be the square root of a boson equation?

Response. It should not be understood that way. The dispersion relation $E^2 = p^2 + m^2$ applies to every free relativistic particle, regardless of statistics. The Dirac operator happens to provide a matrix factorization of the Klein–Gordon operator, but that algebraic relation does not derive fermions from bosons.

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[a]Lecture timestamp: 01:40:46.

[a] **Question from the audience.** Can the equation be reduced to a simple scalar factorization into separate left- and right-moving factors?

Response. Only in the massless limit do the two chiral equations decouple into independent scalar factors. For $m \neq 0$, the useful factorization is matrix-valued: the anticommuting α and β matrices are what remove the unwanted cross term.

[a]Lecture timestamp: 01:41:31.

[a] **Question from the audience.** Do these two coupled equations already constitute the Dirac equation?

Response. Yes. Together they are the Dirac equation in one spatial dimension, equivalently written in the compact form of Eq. (9.53). A more symmetric covariant notation can be obtained by multiplying by β, but that form is deferred.

[a]Lecture timestamp: 01:42:04.

Two tasks remain deliberately unfinished. The construction has not removed negative energies; it has supplied negative-energy solutions in both directional sectors, so that problem is now more visible rather than less. And the theory still lives in one spatial dimension. Interpreting the negative-energy solutions and enlarging the matrix algebra to $3 + 1$ dimensions are the next steps.

CHAPTER 10

FERMIONS, DIRAC MATRICES, AND THE DIRAC SEA

The Dirac sea is our destination, but the route matters. Before facing negative energy in the relativistic electron theory, we need two pieces of machinery. First comes the operator algebra that distinguishes fermions from bosons. Then comes a review of the one-dimensional Dirac equation, where every enlargement of the wavefunction can be seen without the geometry of three spatial dimensions. Once those pieces are in place, the three-dimensional equation will force its own matrix structure, and the opening discussion of filled and empty fermion states will return as the key to positrons.

We use $c = \hbar = 1$. Capital P denotes a momentum operator, lower-case p a momentum eigenvalue, and bold symbols their three-dimensional versions.

10.1 Exchange Becomes Operator Algebra

Consider two identical particles, one at x and one at y. The labels x and y denote complete positions; they are not two Cartesian coordinates of a single particle. Acting on the vacuum with two creation operators gives

$$|x, y\rangle = \psi^\dagger(x)\psi^\dagger(y)|0\rangle. \tag{10.1}$$

For bosons, exchanging the two slots leaves the state unchanged:

$$|x, y\rangle = |y, x\rangle. \tag{10.2}$$

The operator statement that guarantees this symmetry is

$$\psi^\dagger(x)\psi^\dagger(y) = \psi^\dagger(y)\psi^\dagger(x). \tag{10.3}$$

Two bosonic creation operators commute, so we may reverse their order without changing the state they create.

For fermions, exchange changes the sign:

$$|x, y\rangle = -|y, x\rangle. \tag{10.4}$$

The overall sign of an isolated state vector is not observable, but relative signs between alternatives are observable, so this minus sign must be carried consistently. Equation (10.4) requires

$$\psi^\dagger(x)\psi^\dagger(y) = -\,\psi^\dagger(y)\psi^\dagger(x). \tag{10.5}$$

For any two operators A and B, define the anti-commutator by

$$\{A, B\} := AB + BA. \tag{10.6}$$

The exchange rule is therefore

$$\{\psi^\dagger(x), \psi^\dagger(y)\} = 0. \tag{10.7}$$

The corresponding fermionic field algebra replaces the bosonic commutators by anti-commutators. One must keep two uses of the word distinct: field operators anti-commute because of Fermi statistics,

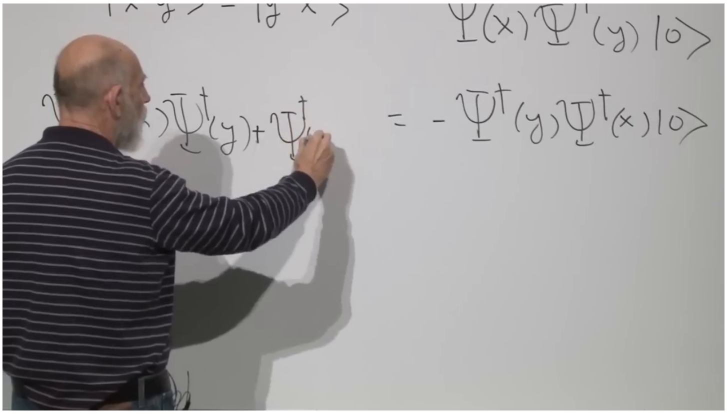

Figure 10.1: The board relation implementing fermionic exchange. Interchanging the creation operators reverses the sign of the two-particle state.
Lecture frame: 00:05:50.

while matrices will later anti-commute in order to cancel cross terms in a relativistic dispersion relation.

Now set $y = x$ in Eq. (10.7). Both products become the same:

$$0 = \{\psi^\dagger(x), \psi^\dagger(x)\} \tag{10.8}$$
$$= 2\,\psi^\dagger(x)\psi^\dagger(x). \tag{10.9}$$

Hence

$$\left(\psi^\dagger(x)\right)^2 = 0. \tag{10.10}$$

Trying to create two identical fermions in the same one-particle state gives the zero vector. That is the exclusion principle in operator form.

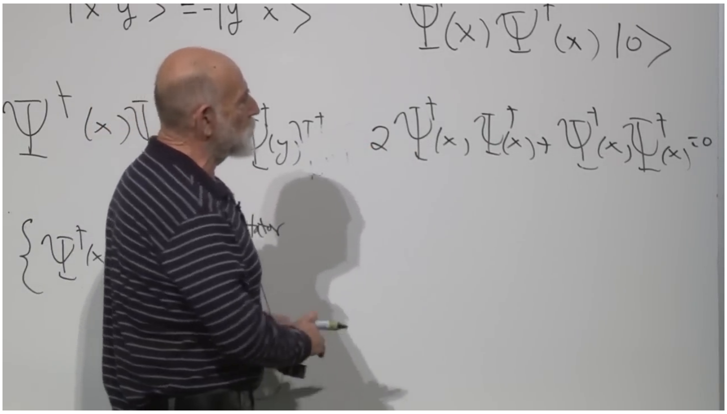

Figure 10.2: The equal-state specialization on the board. The repeated creation-operator product appears twice and therefore must vanish.

Lecture frame: 00:08:04.

$$\{\psi^\dagger(x), \psi^\dagger(x)\} = 0 \longrightarrow 2(\psi^\dagger(x))^2 = 0 \longrightarrow (\psi^\dagger(x))^2 = 0$$

Figure 10.3: The exclusion argument reconstructed without the partially erased intermediate board work.

10.2 Why Filled and Empty Are Symmetric for a Fermion

There is a second difference between bosonic and fermionic operators, subtler than exclusion and essential later. For a single bosonic mode,

$$[a, a^\dagger] = 1, \qquad n = 0, 1, 2, \ldots. \tag{10.11}$$

Interchanging a and $a^\dagger$ changes the sign of the commutator. Even if the operators were merely called A and B, the algebra would tell us which ordering raises the occupation number and which lowers it.

The bosonic occupation ladder also has an orientation. The annihilation operator eventually reaches $|0\rangle$ and gives zero, whereas the creation operator can continue indefinitely:

$$a|0\rangle = 0, \tag{10.12}$$

$$a^\dagger|n\rangle \propto |n+1\rangle, \qquad n = 0, 1, 2, \ldots. \tag{10.13}$$

Turning this ladder upside down does not reproduce the same spectrum. There is a bottom but no top.

For one fermionic mode,

$$\{a, a^\dagger\} = 1, \qquad a^2 = (a^\dagger)^2 = 0, \qquad n = 0, 1. \tag{10.14}$$

The creation operator takes the empty state to the filled state and then stops:

$$a^\dagger|0\rangle = |1\rangle, \qquad a^\dagger|1\rangle = 0. \tag{10.15}$$

The annihilation operator performs the reverse:

$$a|1\rangle = |0\rangle, \qquad a|0\rangle = 0. \tag{10.16}$$

There are only two states, so the diagram is symmetric under interchanging filled with empty and a with $a^\dagger$. The operators still have different physical meanings, but their algebra no longer singles out one as creation and the other as annihilation. This formal symmetry is the seed of the particle-hole reinterpretation.

---

[a] **Question from the audience.** If $(a^\dagger)^2 = 0$, is that a contradiction, or does it imply that $a^\dagger = 0$?

Response. Neither. A nonzero operator can have a vanishing square. Here $a^\dagger$ takes $|0\rangle$ to $|1\rangle$, but there is no allowed state above $|1\rangle$, so applying it twice gives zero. The same reasoning gives $a^2 = 0$.

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[a] Lecture timestamp: 00:13:54.

Keep this filled-empty symmetry in reserve. When a negative-energy electron is removed, the annihilation operator will be relabeled as a positron creation operator. That move is not an arbitrary change of notation; it uses exactly the two-state fermion algebra established here.

10.3 The One-Dimensional Dirac Construction

Return now to one particle in one spatial dimension. The simplest first-order relativistic wave equation is

$$i\partial_t\psi = -i\partial_x\psi. \tag{10.17}$$

With $P = -i\partial_x$, this is

$$H = P. \tag{10.18}$$

A momentum eigenstate has $E = p$, and its group velocity is

$$v_g = \frac{dE}{dp} = 1. \tag{10.19}$$

Positive or negative momentum, and therefore positive or negative energy, still gives a wave moving rigidly to the right. The sign of p changes the phase pattern, not the propagation direction.

A massive particle must be able to move in either direction. Introduce a second component satisfying

$$H\psi_1 = P\psi_1, \qquad H\psi_2 = -P\psi_2. \tag{10.20}$$

The first component moves right; the second moves left. Put them into a column,

$$\Psi = \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}, \tag{10.21}$$

and define

$$\alpha = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{10.22}$$

Both equations become

$$H\Psi = \alpha P \Psi. \tag{10.23}$$

The cost of including both directions is a new two-valued internal degree of freedom.

Direction alone does not produce mass. Introduce a second matrix,

$$\beta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \tag{10.24}$$

and try

$$H = \alpha P + \beta m. \tag{10.25}$$

The matrices satisfy

$$\alpha^2 = \beta^2 = \mathbb{1}, \qquad \{\alpha, \beta\} = 0. \tag{10.26}$$

Squaring the Hamiltonian gives

$$\begin{aligned} H^2 &= (\alpha P + \beta m)^2 \qquad (10.27)\\ &= \alpha^2 P^2 + \beta^2 m^2 + (\alpha\beta + \beta\alpha) P m \qquad (10.28)\\ &= P^2 + m^2. \qquad (10.29) \end{aligned}$$

The anti-commutator in Eq. (10.26) removes the unwanted term linear in both P and m.

The off-diagonal matrix β couples the two directional components. It is useful to picture the mass term as continually mixing right-moving and left-moving amplitudes. The picture should not be literalized into a particle bouncing back and forth along a microscopic track, but it captures why a massive wavepacket can have an average speed below light and can possess a rest frame, whereas an isolated massless chiral component cannot.

[a] **Question from the audience.** Is there a deeper physical interpretation of introducing both directional components?

Response. There is, but it emerges more clearly in three dimensions. At this stage they are simply the two components required to describe opposite propagation and to admit an off-diagonal mass coupling. Their later interpretation is handedness, or chirality.

[a]Lecture timestamp: 00:23:37.

10.4 Three Dimensions Force the Pauli Matrices

In three spatial dimensions momentum is a vector. A Hamiltonian linear in momentum and rotationally scalar therefore suggests another vector, now a vector whose components are matrices:

$$H = \boldsymbol{\alpha} \cdot \mathbf{P} = \alpha_x P_x + \alpha_y P_y + \alpha_z P_z. \tag{10.30}$$

Begin again with the massless case. We require

$$H^2 = \mathbf{P}^2 = P_x^2 + P_y^2 + P_z^2. \tag{10.31}$$

Expanding the square of Eq. (10.30),

$$\begin{align} H^2 = \alpha_x^2 P_x^2 + \alpha_y^2 P_y^2 \tag{10.32} \\ + \alpha_z^2 P_z^2 \tag{10.33} \\ + \{\alpha_x, \alpha_y\} P_x P_y \tag{10.34} \\ + \{\alpha_x, \alpha_z\} P_x P_z \tag{10.35} \\ + \{\alpha_y, \alpha_z\} P_y P_z. \tag{10.36} \end{align}$$

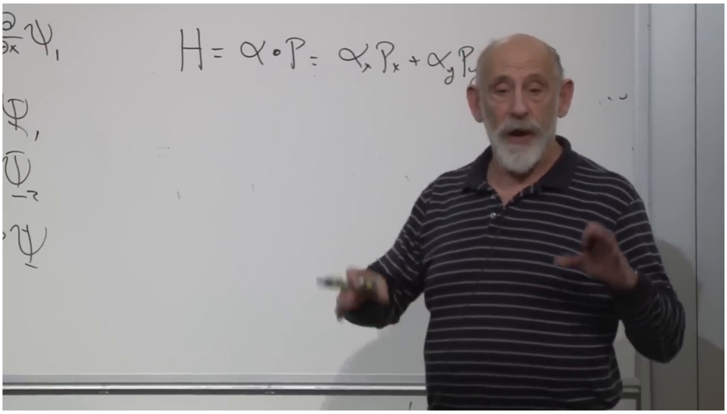

Figure 10.4: The three-dimensional massless ansatz $H = \boldsymbol{\alpha} \cdot \mathbf{P}$ at the point where one matrix becomes a vector of three matrices.

Lecture frame: 00:26:00.

The diagonal terms are correct if

$$\alpha_x^2 = \alpha_y^2 = \alpha_z^2 = \mathbb{1}, \tag{10.37}$$

and all cross terms disappear if

$$\{\alpha_i, \alpha_j\} = 0, \qquad i \neq j. \tag{10.38}$$

Three familiar 2×2 matrices do exactly this:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{10.39}$$

Thus one solution is

$$\alpha_i = \sigma_i, \qquad H = \boldsymbol{\sigma} \cdot \mathbf{P}. \tag{10.40}$$

For a momentum eigenstate,

$$\boldsymbol{\sigma} \cdot \mathbf{p} = |\mathbf{p}|\, \boldsymbol{\sigma} \cdot \widehat{\mathbf{p}}. \tag{10.41}$$

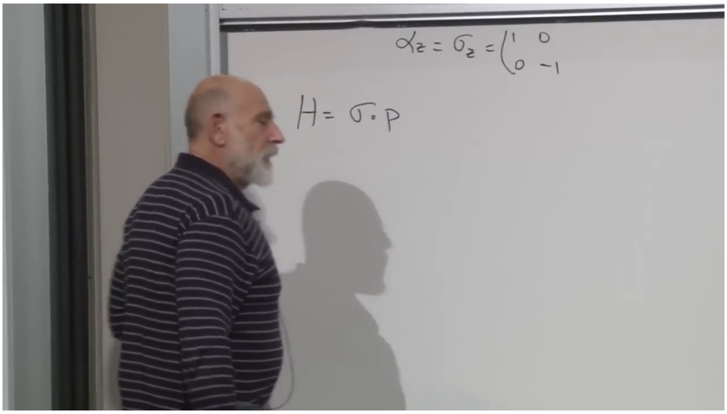

Figure 10.5: The compact massless Hamiltonian $H = \boldsymbol{\sigma} \cdot \mathbf{P}$, with $\alpha_z = \sigma_z$ still visible above it.

Lecture frame: 00:28:16.

The operator $\boldsymbol{\sigma} \cdot \widehat{\mathbf{p}}$ has eigenvalues $+1$ and -1, so

$$E = \pm|\mathbf{p}|. \tag{10.42}$$

In this two-component theory, positive energy has the Pauli degree of freedom aligned with momentum, while negative energy has it anti-aligned. The alignment is locked; it is not an independently adjustable spin direction for a fixed energy sign.

[a] **Question from the audience.** Does spin drop out of the Dirac equation, or was it inserted because the electron is already known to have a magnetic moment?

Response. Here "drop out" means "emerge as a consequence," not "disappear." The starting demand was a Hamiltonian linear in momentum whose square is $\mathbf{P}^2$. Ordinary numbers cannot satisfy the required algebra, but the Pauli matrices can. Their two-component structure therefore arrives with the linearization itself. This does not prove that every particle must have spin; it says that particles described by this Dirac-type equation carry the corresponding spinor degree of freedom.

[a] Lecture timestamp: 00:30:10.

This is one of the central lessons of the construction. The matrices are not decorative labels added after the relativistic equation is found. The attempt to take a local square root of the relativistic dispersion relation produces them.

10.5 Why Mass Requires Four Components

Try to repeat the one-dimensional mass construction without enlarging the 2×2 theory:

$$H = \boldsymbol{\sigma} \cdot \mathbf{P} + \beta m. \tag{10.43}$$

Assume $\beta^2 = \mathbb{1}$. The square contains the desired terms $\mathbf{P}^2 + m^2$, but it also contains

$$m \sum_i \{\sigma_i, \beta\} P_i. \tag{10.44}$$

To remove this term, one matrix β would have to satisfy

$$\{\sigma_x, \beta\} = \{\sigma_y, \beta\} = \{\sigma_z, \beta\} = 0. \tag{10.45}$$

No 2×2 matrix does so. The identity commutes with all three Pauli matrices, and every traceless 2×2 matrix is a linear combination of the Pauli matrices. Any such combination fails to anti-commute with the component parallel to it. The 2×2 algebra has no fourth independent matrix of the required kind.

The remedy is to enlarge the spinor. Four mutually anti-commuting matrices whose squares are the identity first appear in a 4×4 complex representation. This matrix size is not chosen because spacetime has four dimensions; it is the smallest matrix space capable of realizing the required algebra.

There are many equivalent representations. A unitary change of spinor basis changes the entries of all the matrices without changing any physical prediction. What matters is the abstract set of relations

$$\{\alpha_i, \alpha_j\} = 2\delta_{ij}\mathbb{1}, \qquad \{\alpha_i, \beta\} = 0, \qquad \beta^2 = \mathbb{1}. \tag{10.46}$$

[a] **Question from the audience.** Can β be regarded as a time component α_t?

Response. Not directly. The three α_i do not combine with β as the four components of one Lorentz vector. In covariant notation one instead defines $\beta = \gamma^0$ and $\alpha_i = \gamma^0\gamma^i$; the four γ^μ matrices satisfy the spacetime Clifford algebra.

[a] Lecture timestamp: 00:37:38.

A convenient block representation is

$$\alpha_i = \begin{pmatrix} \sigma_i & 0 \\ 0 & -\sigma_i \end{pmatrix}, \qquad \beta = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}. \tag{10.47}$$

The Pauli matrices in the diagonal blocks guarantee that different α_i anti-commute. The off-diagonal identity blocks in β guarantee that β anti-commutes with every α_i.

The massive Hamiltonian is now

$$H = \boldsymbol{\alpha} \cdot \mathbf{P} + \beta m = \begin{pmatrix} \boldsymbol{\sigma} \cdot \mathbf{P} & mI_2 \\ mI_2 & -\boldsymbol{\sigma} \cdot \mathbf{P} \end{pmatrix}. \tag{10.48}$$

Using Eq. (10.46),

$$H^2 = \mathbf{P}^2 + m^2. \tag{10.49}$$

The four-component structure is therefore not an optional embellishment. It is the smallest structure that permits both three-dimensional momentum and a mass term while keeping the equation first order.

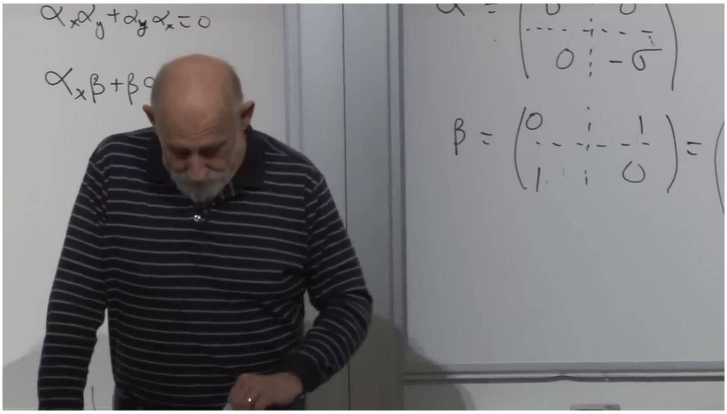

Figure 10.6: The board's 4×4 block representation. The upper equations record the required anti-commutators; the full matrices are typeset in Eq. (10.47).

Lecture frame: 00:41:10.

10.6 Chirality and the Meaning of the Extra Components

What physical distinction did the enlargement add? Start with the massless upper block,

$$H_R = \boldsymbol{\sigma} \cdot \mathbf{P}, \tag{10.50}$$

and the massless lower block,

$$H_L = -\boldsymbol{\sigma} \cdot \mathbf{P}. \tag{10.51}$$

For positive-energy states, the first has spin aligned with momentum and the second has spin anti-aligned. These are opposite helicities. In a massless theory helicity is invariant under proper Lorentz transformations, because no observer can overtake a lightlike particle and reverse its momentum while leaving its spin unchanged. The two blocks therefore describe opposite chiral sectors.

---

[a] **Question from the audience.** In one dimension an extra binary degree of freedom was needed to introduce mass. What is the corresponding physical distinction in three dimensions?

Response. It is the handedness of the spinor. The two 2×2 blocks carry opposite relations between spin and momentum. One is right-handed and the other left-handed. The 4×4 theory contains both.

---

[a]Lecture timestamp: 00:41:28.

The mass term in Eq. (10.48) is off diagonal. It sends an upper-block amplitude into the lower block and vice versa:

$$\beta m \begin{pmatrix} \Psi_R \\ \Psi_L \end{pmatrix} = m \begin{pmatrix} \Psi_L \\ \Psi_R \end{pmatrix}. \tag{10.52}$$

Mass is therefore a coupling between opposite chiralities.

---

[a] **Question from the audience.** Can a massless particle change chirality?

Response. Not in the free massless Hamiltonian displayed here. The two blocks are decoupled, so a state that begins right-handed or left-handed remains so. The off-diagonal mass term is precisely what mixes them when $m \neq 0$.

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[a]Lecture timestamp: 00:43:58.

This language is exact for the block representation, but the image of a particle mechanically flipping between two arrows should not be pushed too far. The physical statement is the matrix coupling in the wave equation.

---

[a] **Question from the audience.** Are these matrices built into the quaternions?

Response. The Pauli matrices are closely related to quaternion units: the matrices $-i\sigma_i$ square to $-\mathbb{1}$ and multiply with the quaternionic pattern, up to orientation conventions. The full Dirac matrices require the larger Clifford algebra, and the quaternion connection is not needed for the argument that follows.

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[a]Lecture timestamp: 00:44:50.

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[a] **Question from the audience.** If spin is angular momentum, is there a separate energy term associated with it here?

Response. No independent free-particle term is being added. The spinor structure already enters the kinetic Hamiltonian through the matrices. Coupling spin to an external magnetic field would introduce additional interaction terms, but that is a different problem.

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[a]Lecture timestamp: 00:46:34.

10.7 The Alpha Matrices Are Velocity, Not Spin

Each α_i has two $+1$ eigenvalues and two -1 eigenvalues. Different components do not commute, so they cannot all be measured sharply at once. To identify what one component represents, use the Heisenberg equation. For an operator L without explicit time dependence,

$$\dot{L} = i[H, L]. \tag{10.53}$$

The sign convention is fixed here by the Schrödinger evolution e^{-iHt}.

Apply Eq. (10.53) to the position operator x. The matrices act on spinor indices and commute with position, while only P_x fails to commute with x:

$$\begin{aligned} \dot{x} &= i[\boldsymbol{\alpha} \cdot \mathbf{P} + \beta m, x] && (10.54) \\ &= i\alpha_x [P_x, x] && (10.55) \\ &= \alpha_x, && (10.56) \end{aligned}$$

because $[P_x, x] = -i$. Thus α_x is the x-component of the velocity operator; similarly,

$$\dot{\mathbf{x}} = \boldsymbol{\alpha}. \tag{10.57}$$

---

[a] **Question from the audience.** Does a velocity-component measurement really give only $+1$ or -1?

Response. Yes, for the instantaneous operator α_i in units $c = 1$. A general state need not be an eigenstate, so its expectation value can lie anywhere between those values. The smoothly transported wavepacket has an average velocity below light when the particle is massive.

---

[a] Lecture timestamp: 00:51:50.

The velocity is not conserved:

$$\dot{\alpha}_x = i[H, \alpha_x] \neq 0. \tag{10.58}$$

The terms containing α_y, α_z, and β fail to commute with α_x. A state containing both positive- and negative-energy components consequently displays a rapid oscillatory contribution to its position. This is *zitterbewegung*, literally trembling motion.

After averaging over times long compared with the rapid oscillation, the expected relativistic velocity is

$$\overline{\mathbf{v}} = \frac{\mathbf{p}}{E}, \tag{10.59}$$

which reduces to $\mathbf{p}/m$ for a slow electron.

---

[a] **Question from the audience.** Does a free electron trembling without an external force violate Newton's law?

Response. No. Translational invariance conserves momentum, so $\dot{\mathbf{P}} = 0$. The Dirac velocity operator is not simply $\mathbf{P}/m$, and its rapid oscillatory part can change while momentum remains fixed. The time-averaged motion obeys the expected free-particle relation.

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[a]Lecture timestamp: 00:54:40.

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[a] **Question from the audience.** Is the velocity always c?

Response. An instantaneous measurement of one α_i has eigenvalues $\pm c$, but a massive wavepacket is generally a superposition and has a subluminal expectation value. Its mean group velocity is $\mathbf{p}/E$.

---

[a]Lecture timestamp: 00:55:22.

[a] **Question from the audience.** Can the oscillation be pictured as the mass varying the velocity?

Response. No. The mass is a fixed parameter. The oscillation comes from noncommuting velocity and Hamiltonian operators, or equivalently from interference between energy sectors, not from a fluctuating mass.

[a]Lecture timestamp: 00:55:30.

[a] **Question from the audience.** Could an instantaneous measurement find the velocity pointing one way even while the average motion points another way?

Response. Yes. The component operator has the two sharp eigenvalues, while repeated or time-averaged measurements recover the smoother drift.

[a]Lecture timestamp: 00:56:22.

The word spin must also be used carefully. The α_i transform as components of velocity, an ordinary vector. The genuine spin operator in this representation is

$$\mathbf{S} = \frac{1}{2}\boldsymbol{\Sigma}, \qquad \Sigma_i = \begin{pmatrix} \sigma_i & 0 \\ 0 & \sigma_i \end{pmatrix}. \tag{10.60}$$

It differs from α_i by the sign of the lower block. The matrices are closely related, but velocity and spin are not the same observable.

10.8 Negative Energy and the Lowest-Energy Vacuum

The negative-energy problem is already present before any of this matrix machinery. Return to the one-dimensional right mover,

$$H = P. \tag{10.61}$$

Since p ranges over the whole real line, $E = p$ has no lower bound.

The vacuum cannot merely mean a state with no particles. It must be the state from which no further energy can be removed. If an empty state allows us to add a negative-energy particle, lower the energy, add another, and continue indefinitely, then that empty state is not a stable vacuum.

Fermi statistics supplies a historical way out. Place one electron in every negative-energy state. Each added electron lowers the total energy, but exclusion allows only one electron in each one-particle state. Once every negative-energy state is filled, no additional electron can fall into that sector. Dirac identified this filled configuration as the vacuum.

The same construction would fail for bosons. Arbitrarily many bosons can occupy one state, so a negative-energy bosonic mode could absorb particles without limit. The fermionic bound $n = 0, 1$, introduced at the beginning of the lecture, is exactly what stops the descent.

The sea picture is historical language for a correct operator rearrangement. Its literal infinite charge and energy should not be mistaken for directly observable quantities; the modern reformulation will remove the need to imagine material electrons filling space.

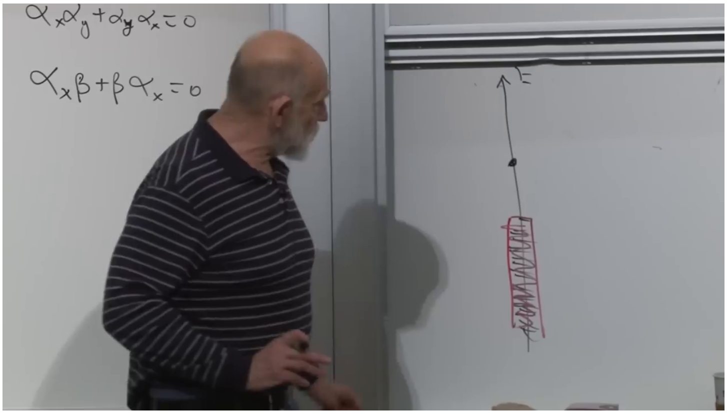

Figure 10.7: The historical Dirac-sea picture on the board: a ladder of negative-energy levels filled up to the dividing line, with positive-energy states above.
Lecture frame: 01:02:45.

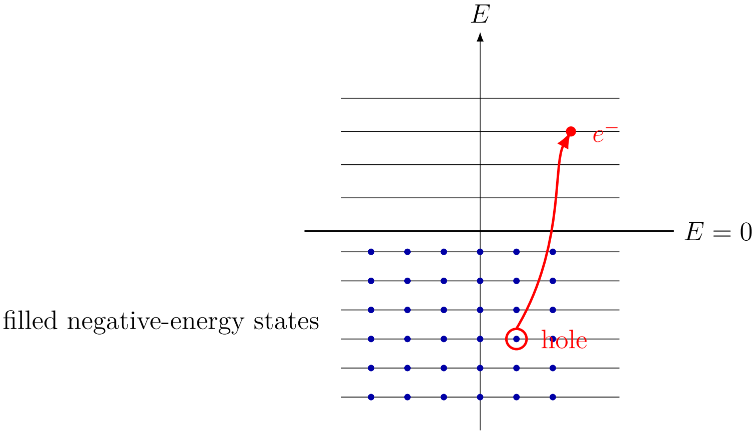

Figure 10.8: A clean reconstruction of excitation above a filled sea: promoting one negative-energy electron leaves a positive-energy hole.

10.9 A Hole Is a Positron

With all negative-energy states occupied, two kinds of excitation are possible. We may add a positive-energy electron above the sea. Or we may remove one negative-energy electron and leave a hole.

A hole has the opposite charge of the missing electron. Removing charge $-e$ changes the charge by $+e$. It also has positive excitation energy. If the removed electron had $E_- < 0$, then the energy measured relative to the filled vacuum changes by

$$E_{\text{hole}} = -E_- > 0. \tag{10.62}$$

The hole is a positive-energy, positively charged excitation: the positron.

A photon can promote a sea electron into a positive-energy state. Relative to the vacuum, the final state contains an electron and a positron. In a time-ordered diagram it is conventional to draw electron-number flow upward for the electron and the positron line with the opposite arrow. The arrows track fermion-number or charge flow; they are not little classical velocity vectors.

[a] **Question from the audience.** Does this picture require equal numbers of electrons and positrons?

Response. Not for an arbitrary state. Starting from the vacuum and creating a neutral pair gives one of each, but a state may already contain excess electrons or excess positrons. Charge conservation constrains reactions; it does not require every allowed state to have equal populations.

[a]Lecture timestamp: 01:05:14.

Dirac did not initially know that the hole would be a particle identical in mass to the electron and opposite only in charge and other additive quantum numbers. The equation and the hole construction led to that prediction before the positron was incorporated into the familiar particle catalogue.

10.10 One Fermion Field Contains Electrons and Positrons

The field operator brings the historical picture into modern algebra. Suppress spinor indices, normalization factors, and time dependence for the moment. A one-dimensional field may be expanded schematically as

$$\psi(x) \sim \int_{-\infty}^{\infty} dp\, a(p) e^{ipx}. \tag{10.63}$$

For the right-moving dispersion $E = p$, positive p labels positive energy and negative p labels negative energy. Split the integral:

$$\psi(x) \sim \int_{0}^{\infty} dp\, a(p) e^{ipx} + \int_{-\infty}^{0} dp\, a(p) e^{ipx}. \tag{10.64}$$

In the second term set $p = -q$, with $q > 0$, and reinterpret annihilation of a filled negative-energy electron as creation of a positron:

$$b^{\dagger}(q) := a(-q). \tag{10.65}$$

Then

$$\psi(x) \sim \int_0^{\infty} dp\, \left[a(p)e^{ipx} + b^{\dagger}(p)e^{-ipx}\right]. \tag{10.66}$$

The exact signs of the plane-wave phases depend on the spacetime convention; the invariant content is that ψ contains an electron annihilator and a positron creator.

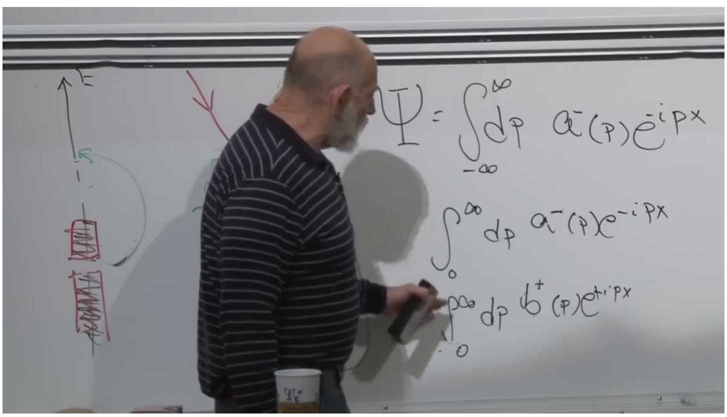

Figure 10.9: The field integral split into positive- and negative-energy sectors. The lower line relabels the negative-energy electron operator as a positron creation operator.

Lecture frame: 01:11:35.

This is where the filled-empty symmetry of Eq. (10.14) pays off. For fermions the operator algebra permits the relabeling without changing its anti-commutation structure.

A schematic electromagnetic interaction has the form

$$H_{\text{int}} \sim \int dx\, \psi^\dagger(x)\psi(x)A(x), \tag{10.67}$$

where A denotes the photon field. Because each fermion field contains electron and positron pieces, a single local operator contains several channels:

$$e^- \longrightarrow e^- + \gamma, \tag{10.68}$$

$$e^- + e^+ \longrightarrow \gamma, \tag{10.69}$$

$$\gamma \longrightarrow e^- + e^+, \tag{10.70}$$

together with their inverse and crossed processes. Energy-momentum conservation decides which channel is kinematically possible in a particular environment. For example, an isolated real photon in vacuum cannot create a pair without another participant to absorb recoil, even though the local vertex is present in the interaction.

One compact product of fields therefore organizes many apparently different reactions. Feynman later supplied the systematic diagrammatic rules for evaluating the amplitudes, but much of the operator structure is already visible here.

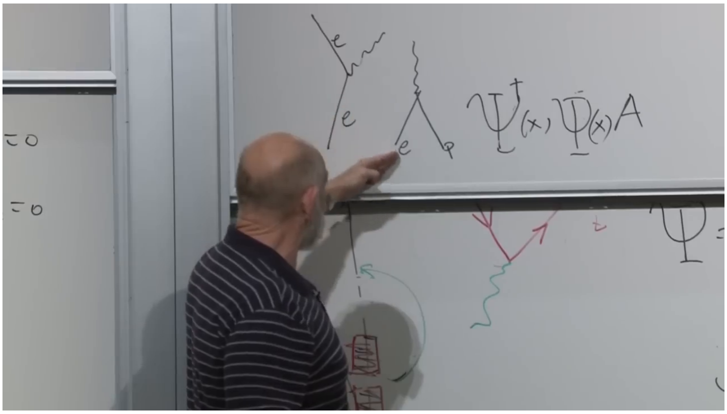

Figure 10.10: The board juxtaposes electron emission, pair annihilation, and the schematic interaction $\psi^\dagger\psi A$. Different selections from the same field operators generate the different channels.

Lecture frame: 01:13:30.

10.11 Modern Vacuum Language and the Fermi-Sea Analogy

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[a] **Question from the audience.** How do modern physicists describe this without treating the vacuum as a literal Dirac sea?

Response. They perform the operator reinterpretation directly. Negative-energy electron annihilation is renamed positive-energy positron creation, and the vacuum is defined as the state annihilated by all electron and positron annihilation operators. The resulting formulation is symmetric between particles and antiparticles and needs no material sea.

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[a] Lecture timestamp: 01:14:20.

The historical sea remains useful because its logic has a close condensed-matter analogue. In a crystal, a fixed number of electrons fill the lowest available one-particle states up to the Fermi level. The filled ground state is not empty. Promoting one electron above that level leaves an unoccupied state below it. Relative to the filled background, the hole carries positive excitation energy and can behave as a positive charge carrier.

There are important differences. A solid-state hole lives in a medium and its effective mass and charge response depend on a band structure. A positron is an elementary antiparticle in the relativistic quantum field. Even so, the common particle-hole algebra makes the analogy powerful.

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[a] **Question from the audience.** Does the Klein–Gordon theory for bosons suffer the same cascade into negative-energy states?

Response. No. A properly quantized complex Klein–Gordon field is organized into positive-energy particles and antiparticles without filling a bosonic sea. Its second-order equation and canonical structure differ from the Dirac Hamiltonian construction, so the historical fermionic sea argument is not transferred unchanged.

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[a] Lecture timestamp: 01:17:14.

[a] **Question from the audience.** Where does spin one for bosons come from?

Response. Not from the Pauli-matrix linearization used here. Spin-one fields have their own Lorentz transformation law and field equations. The present derivation explains the spinor structure of Dirac fermions, not all possible particle spins.

[a]Lecture timestamp: 01:17:30.

[a] **Question from the audience.** Why was Fermi statistics essential to the negative-energy solution?

Response. Because exclusion caps every state at one electron. Once all negative-energy states are occupied, no further electron can fall into them. Bosonic occupation has no such cap, so a literal negative-energy boson sea would still have no bottom.

[a]Lecture timestamp: 01:17:46.

[a] **Question from the audience.** Is electron versus positron simply another binary degree of freedom, like charge?

Response. As state counting, a massive Dirac field has two electron spin states and two positron spin states, so two binary labels are a useful mnemonic. But particle versus antiparticle is not an independently rotatable internal qubit of one electron: every electron has charge $-e$, every positron $+e$, and the distinction emerges from the positive- and negative-frequency sectors after quantization.

[a]Lecture timestamp: 01:20:02.

The infinite sea raises one final question.

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[a] **Question from the audience.** Would the infinite Dirac sea produce an infinite cosmological constant?

Response. It exposes the vacuum-energy problem. Bosonic zero-point modes contribute $+\frac{1}{2}\hbar\omega$, while fermionic modes contribute with the opposite sign. Formal cancellations are possible when spectra match, but the electron and photon spectra do not match, and an enormous remainder is expected. Without gravity, a constant shift of the Hamiltonian can be subtracted because only energy differences matter. With gravity, absolute vacuum energy gravitates, so the subtraction becomes a physical problem rather than mere bookkeeping.

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[a] Lecture timestamp: 01:20:54.

The last return is to the meaning of mass.

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[a] **Question from the audience.** Can we say more about how mass relates left- and right-handed components?

Response. The exact statement is already in the Hamiltonian: the off-diagonal term $m\beta$ maps one chiral block into the other. Any more mechanical picture of microscopic flipping is optional and inevitably incomplete.

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[a] Lecture timestamp: 01:22:56.